

Chapter 3: Magnetic Resonance Imaging

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3.1 Introduction to MRI

Magnetic Resonance Imaging (MRI)

- Developed in 1970s
 - 1976: First MR scan of a human body part
- Based on the magnetic resonance of atomic nuclei (mostly hydrogen)
- Requires a strong static magnetic field, and high-frequency electromagnetic pulses
- Typical image resolution:
Several mm³



Discovery of NMR and MRI

Nuclear spin manipulation through external fields:

- Static magnetic field: $B_0 \cdot \mathbf{e}_z$ $\Rightarrow$ spin polarization
 - Radiofreq. field: $B_1(t) \cdot \mathbf{e}_{xy}$ $\Rightarrow$ resonance signal
 - Gradient field: $(\mathbf{G}(t) \cdot \mathbf{r}) \cdot \mathbf{e}_z$ $\Rightarrow$ spatial frequencies
- $\rightarrow$ Imaging!**



Felix Bloch
(1905-1983)



Edward Purcell

1952 Nobel Prize in Physics

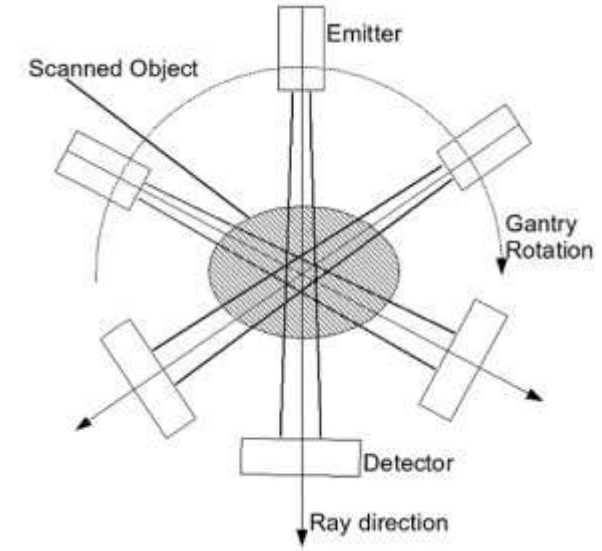


Paul Lauterbur and Sir Peter Mansfield

2003 Nobel Prize in Medicine

Computed Tomography (CT/CAT)

- Introduced in 1968
- Nobel Prize in 1979
- Based on measuring X-ray absorption from many different directions
 - Algorithmic reconstruction of a slice image
 - Multiple slices combined into a 3D volume
 - Often acquired simultaneously
- Can achieve sub-mm voxel size
 - e.g., inspect fine fractures



MRI vs. CT: Applications

- CT ideally suited for **skeletal structures**
- MRI provides a larger variety of contrasts for **soft tissue** (e.g., brain)
- Compared to MRI, CT is **more quantitative**
 - There are ways to make MRI quantitative; some of them will be treated in this lecture
- CT **cheaper** and **more widely available**



CT scan



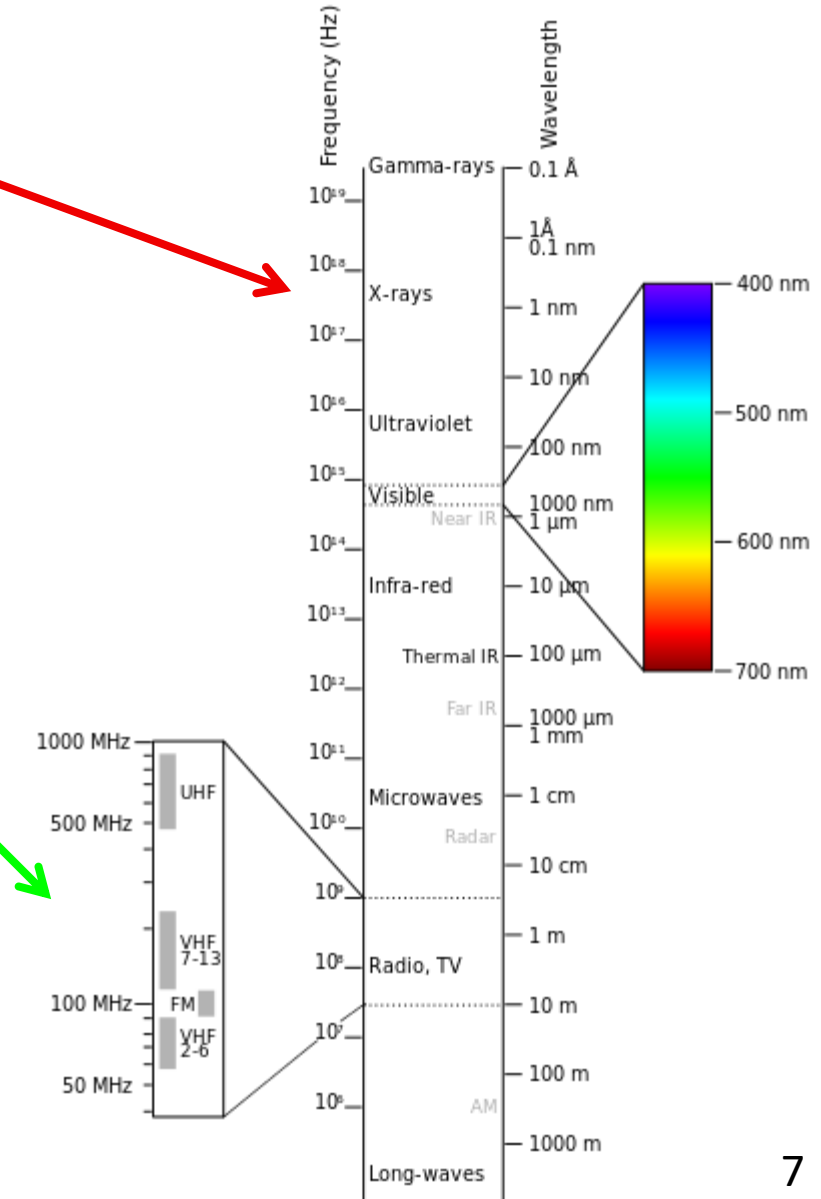
MRI scan

MRI vs. CT: Safety

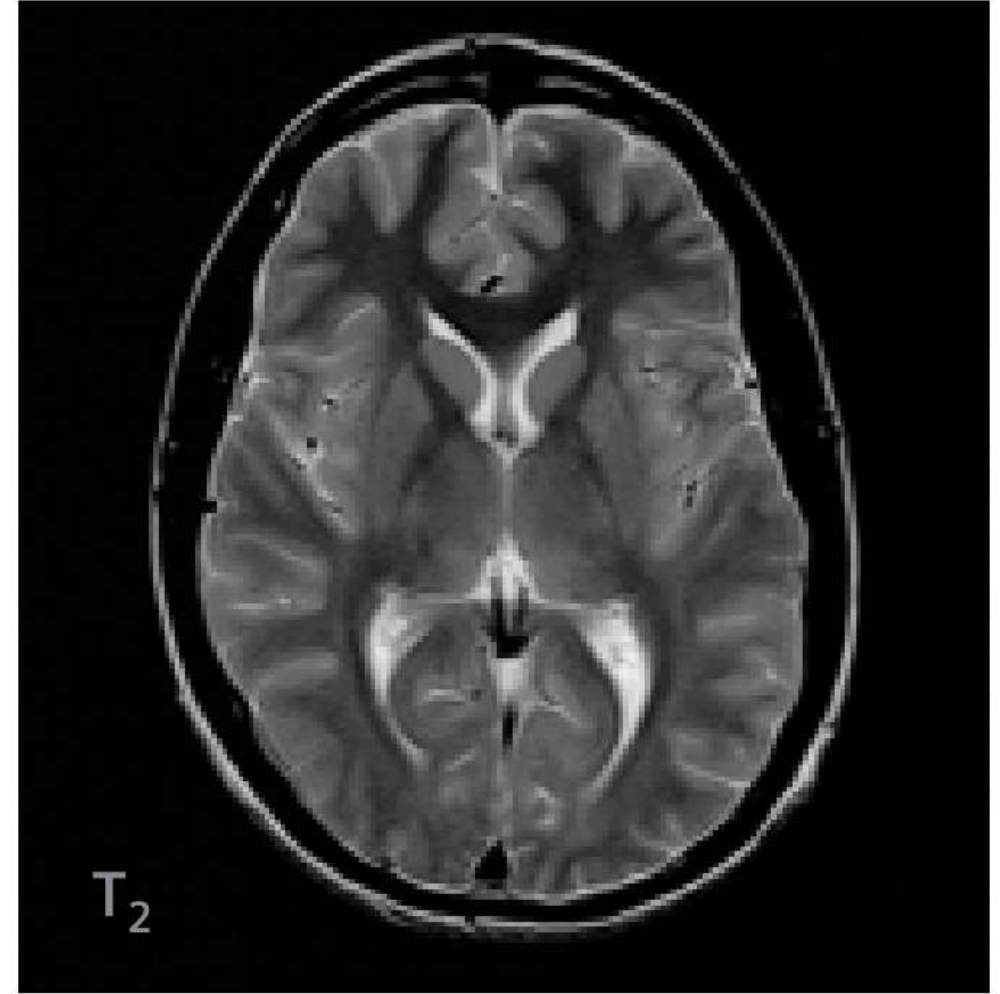
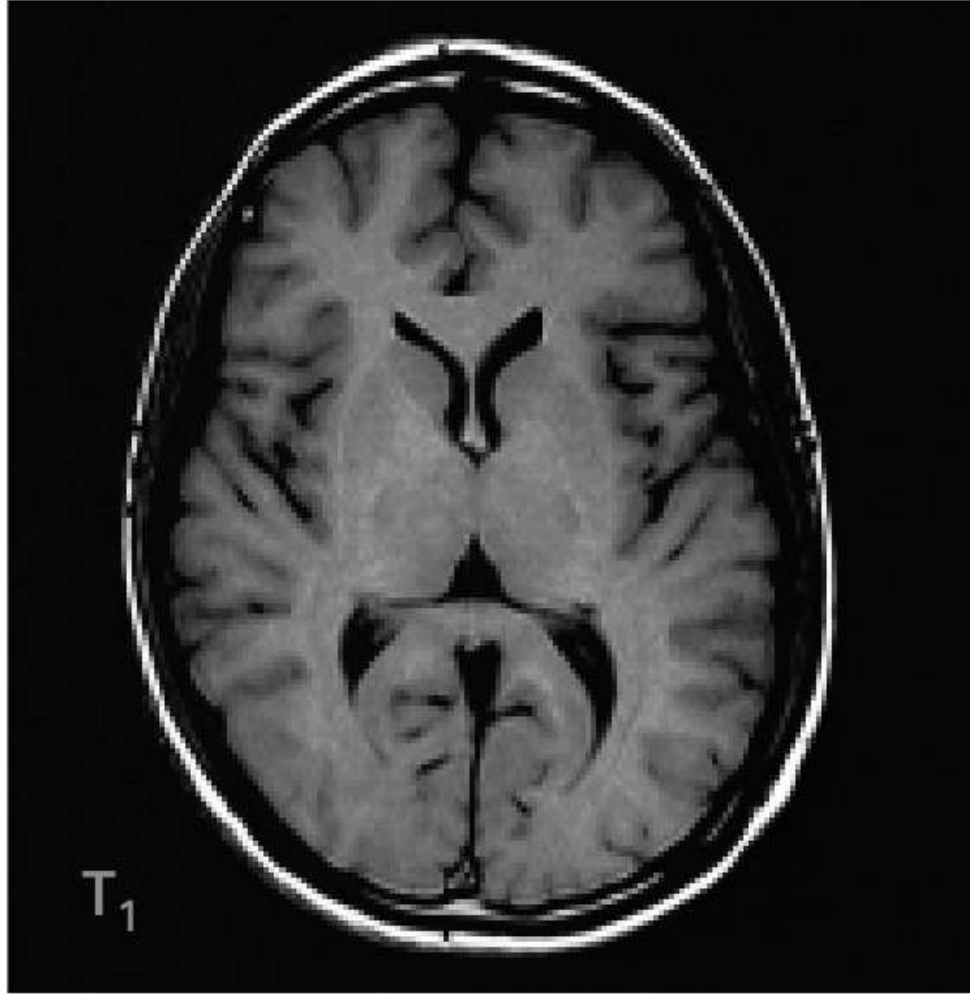
- X-ray radiation in CT is **ionizing** and potentially harmful to biological tissue
 - Rules on allowable dose
- MRI has **no known harmful effects**, safe to use on healthy volunteers
 - Footnote: Beware of implants, heating of certain tattoos, ...and office chairs



Image Sources: Victor Blacus / www.impactedenurse.com

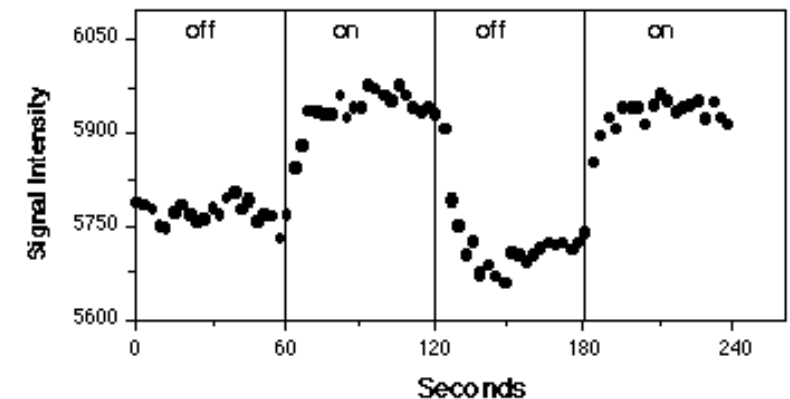
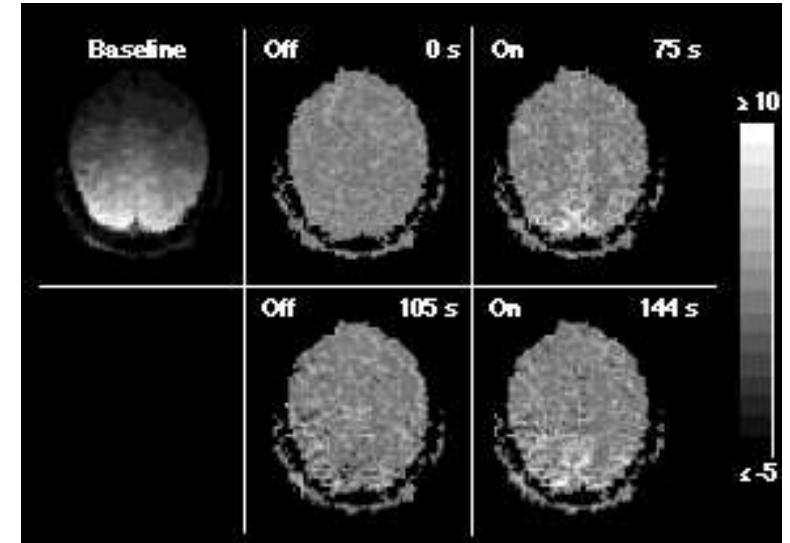


Examples: Multi-Contrast (T1/T2-weighting)



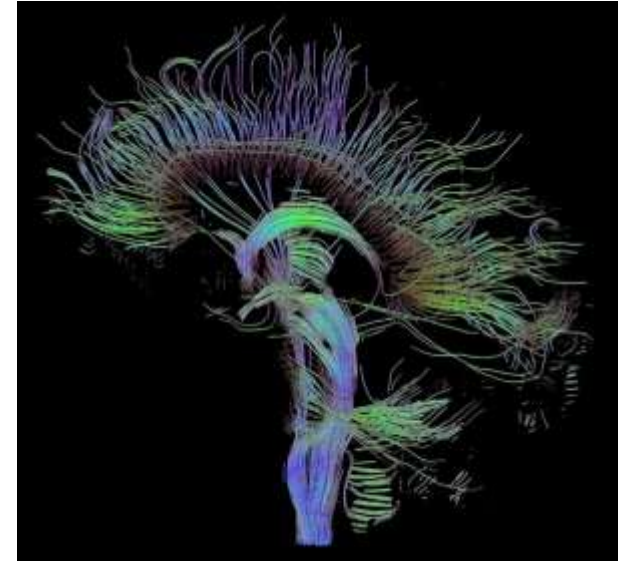
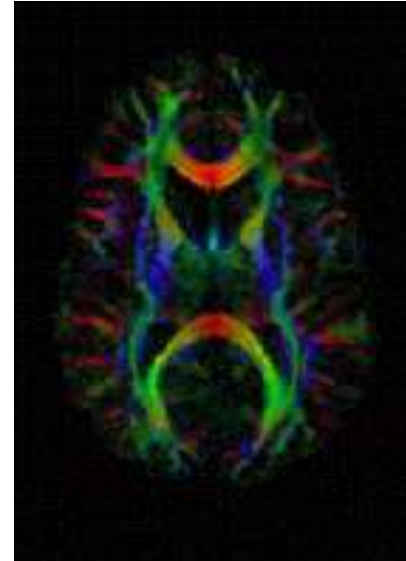
Functional MRI

- Functional MRI takes repeated images in quick succession
- The level of blood oxygenation leads to a slight variation in MR contrast
- „Watch the brain work“



Diffusion MRI

- Developed in mid-1980s, much refinement since then
- MR signal attenuated by heat motion (Brownian motion) of imaged nuclei
- Spatial resolution: 1-10 mm³
 - Probe of much smaller microstructure due to microscale interactions between molecules and tissue



Summary

- **Advantages of MRI** as an imaging modality:
 - **No ionizing radiation**
 - **Contrast agents** only required for specific use cases
 - **Well-suited for soft tissue** such as the brain
 - Fast enough for **functional imaging**
 - Flexible sequences offer **variety of contrasts**
 - different physical properties (T_1 , T_2 , diffusion, etc.)
 - Can be made **quantitative**
 - E.g., measuring T_1 as opposed to having a T_1 -weighted image in arbitrary units, in which brightness also depends on other factors
 - **Flexible orientation** of slices

3.2 Nuclear Magnetic Resonance

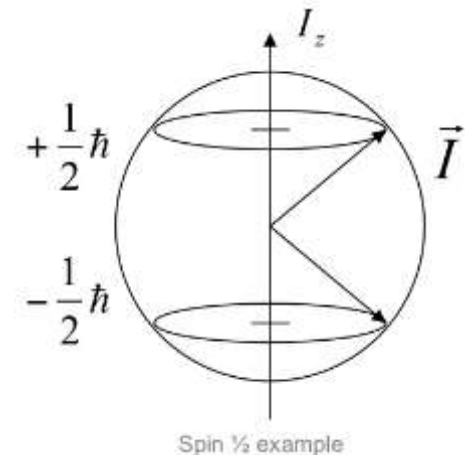
Partly based on slides by Prof. Tony Stöcker

Key Questions of this Section

- How does the **signal** arise that an MR scanner measures?
- Why do we need such a **strong magnetic field**?
- Why is it called “**Magnetic Resonance**”?
- How does MR differentiate between different **types of soft tissue**?

Quantum Mechanical Spin

- Human body consists of $\approx 70\%$ H_2O 
- Hydrogen nuclei (protons) have *nuclear spin*, a quantum mechanical property that is somewhat similar to classical **angular momentum** (e.g., rotating ball)
 - But it is **intrinsic**, cannot speed up or slow down
 - Protons have spin $\frac{1}{2}$, $I_z = \pm \frac{1}{2} \hbar$
 - Individual spins are in a *superposition* of parallel and antiparallel states with a certain phase



h = Planck constant $\approx 6.626068 \cdot 10^{-34}$ Js

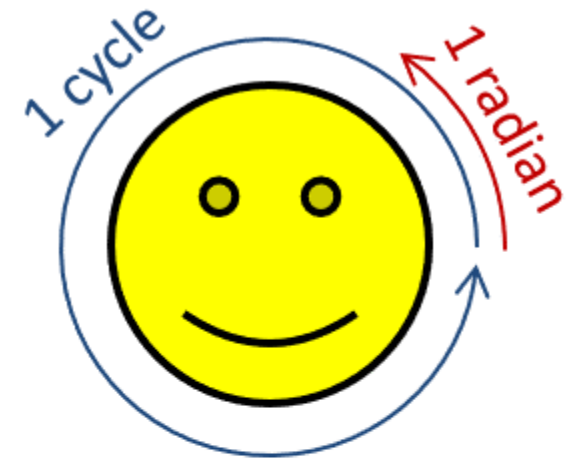
$$\hbar = \frac{h}{2\pi} \quad \text{“reduced Planck constant”}$$

Frequency vs. Angular Frequency

- **Rotational frequency** ν refers to the number of cycles per second (unit: Hz)
- **Angular frequency** ω refers to the change in angle (in radians) per second

$$\omega = 2\pi\nu$$

- Fourier basis with respect to
 - Rotational frequency: $e^{2\pi i\nu t}$
 - Angular frequency: $e^{i\omega t}$



Time (in seconds) = 0.00 s

Angle (in radians) = 0.00 rad

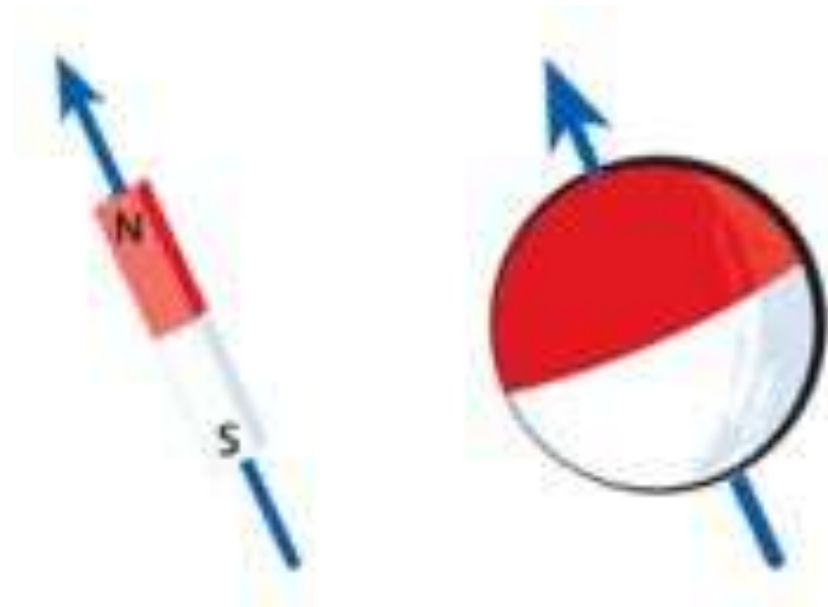
Rotation (in cycles) = 0.00 c

$$\omega = \frac{0.00 \text{ rad}}{0.00 \text{ s}} =$$

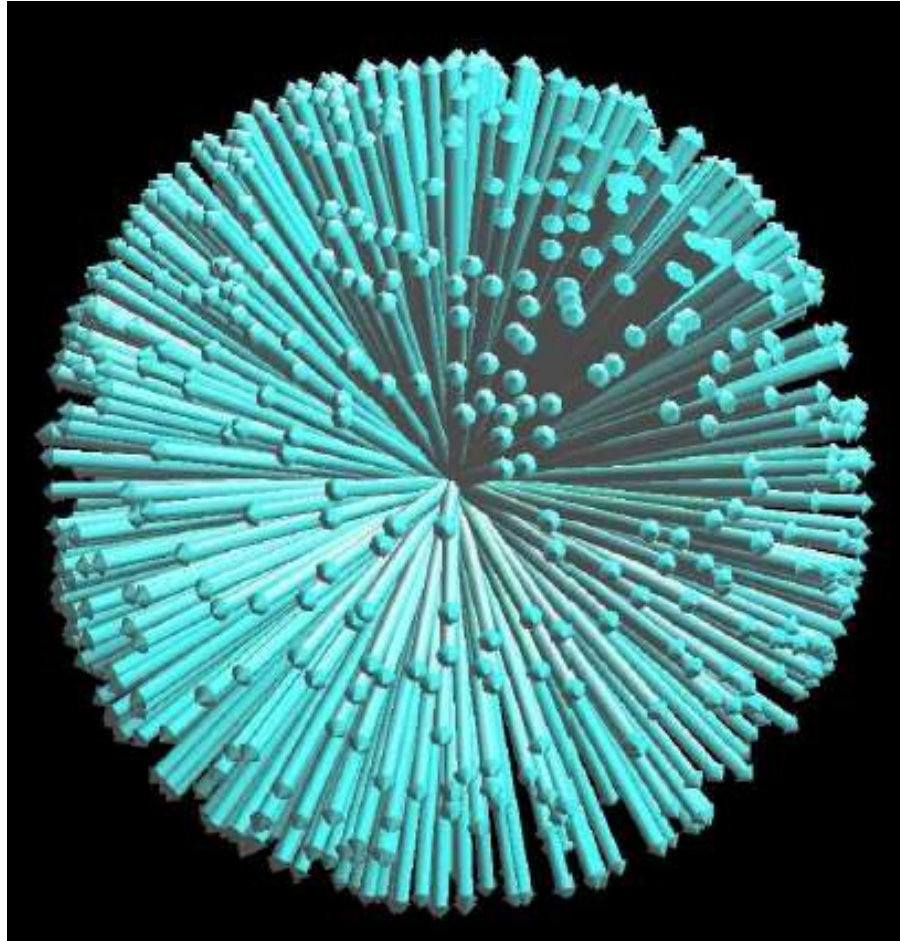
$$\nu = \frac{0.00 \text{ c}}{0.00 \text{ s}} =$$

Nuclear Magnetic Moment

- Every nuclear spin comes with a **nuclear magnetic moment**
 $\mu = \gamma I$
 - γ = nucleus-specific gyromagnetic ratio
 - For hydrogen, $\gamma = 267.513 \cdot 10^6$ rad/(s·T).

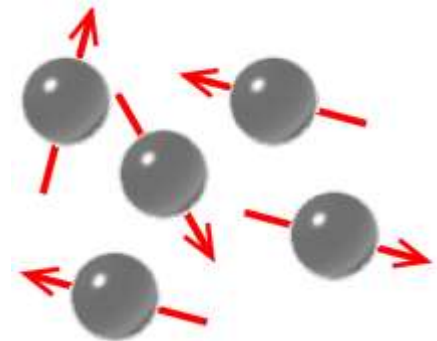


Spins at $B_0 = 0$



Hanson 2008

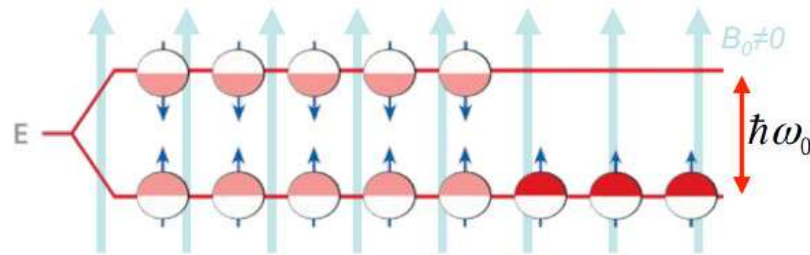
- Spin states can be represented as vectors whose
 - Polar angle indicates parallel / anti-parallel alignment
 - Azimuth angle indicates phase
- Without external magnetic field
 - Spins have random orientation
 - Vector sum over spins is zero
 - No net magnetisation



Nuclear Zeeman Effect

- When exposed to a static external magnetic field $\mathbf{B}_0$, the z-components of the nuclear magnetic moments align with $\mathbf{B}_0$.
 - For our purposes, the direction of $\mathbf{B}_0$ defines the z-axis.
- Two possible orientations of the individual magnetic moments: parallel and anti-parallel with energies

$$E_{\uparrow,\downarrow} = \mu_{\uparrow,\downarrow} B_0 = \mp \frac{1}{2} \hbar \gamma B_0$$

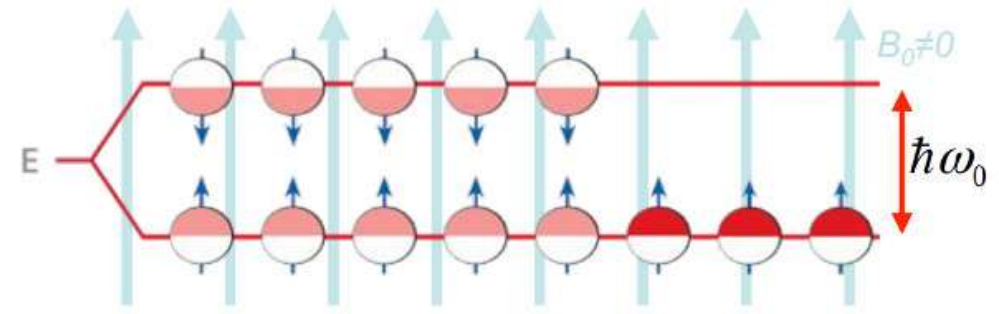


Zeeman splitting:

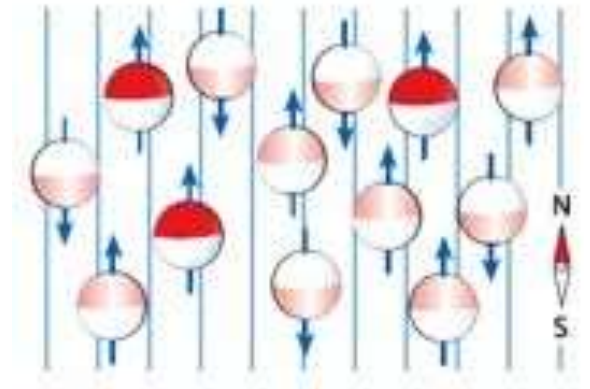
$$\Delta E = \hbar \gamma B_0$$

Equilibrium Magnetization

- At equilibrium, there is a tiny excess of spins in the lower energy state

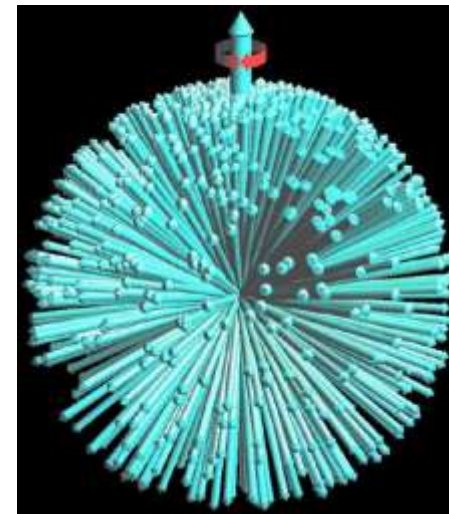
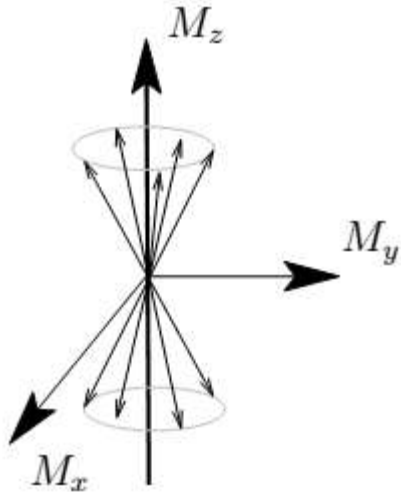


- Boltzmann statistics $\frac{n_{\downarrow}}{n_{\uparrow}} = e^{-\frac{\Delta E}{k_B T}}$
- If you plug in the Boltzmann constant $k_B \approx 1.381 \cdot 10^{-23} \text{ J/K}$, body temperature T , and ΔE at clinical magnetic field strength B_0 , you get a number very close to one
- The number of spins per voxel is large enough so that ensemble average produces a **macroscopic magnetization** $M_0 \sim B_0$



Spins in a Static Field: Larmor Precession

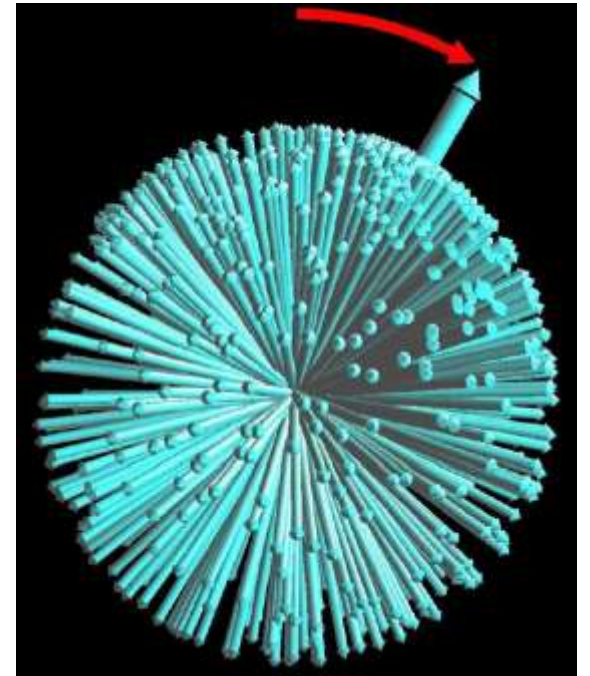
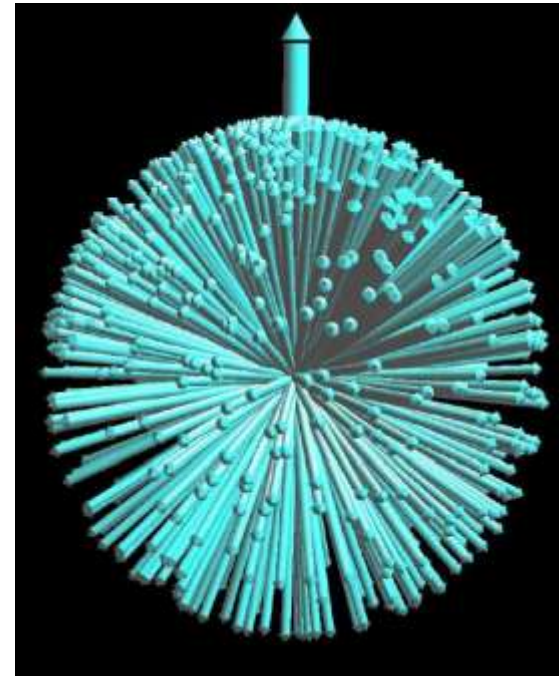
- Since the axis of rotation does not align with $\mathbf{B}_0$, precession around the direction of $\mathbf{B}_0$ takes place
 - Larmor frequency: $\omega_0 = \gamma B_0$
 - Spin distribution skewed slightly towards magnetic north pole, phases are random
 - Macroscopic magnetization is static and aligned with $\mathbf{B}_0$



[Hanson 2008]

Nuclear Magnetic Resonance

- **Nuclear Magnetic Resonance:**
 - Electromagnetic wave B_1 perpendicular to B_0 at ω_0 (**RF pulse**) *coherently rotates* the spin distribution, deflects the macroscopic magnetization
 - Net magnetization starts to precess with ω_0

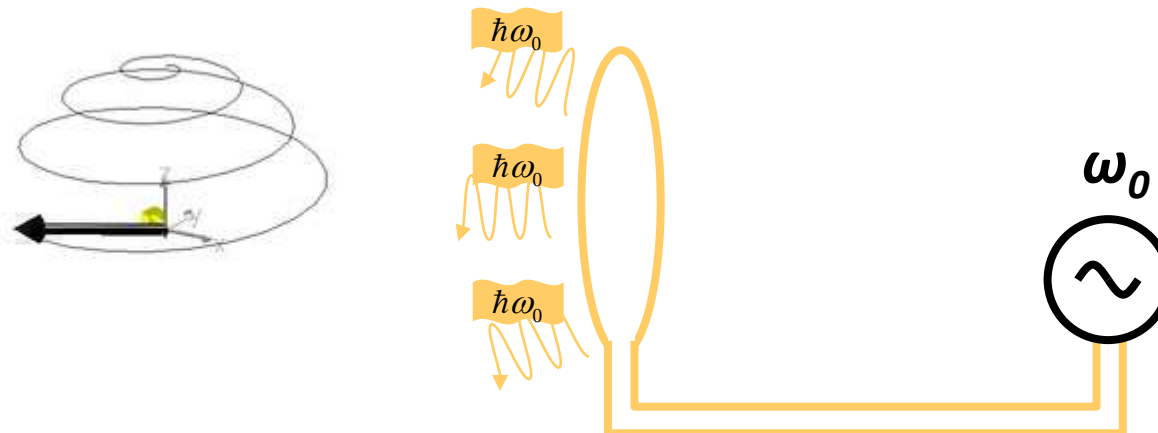


[Hanson 2008]

Excitation / Magnetic Resonance

→ Apply a **resonant radiofrequency (RF) pulse** to excite all spins, i.e. excite the macroscopic magnetisation from its equilibrium.

→ **Resonance** $\leftrightarrow$ frequency of RF pulse = Larmor frequency ω_0 .



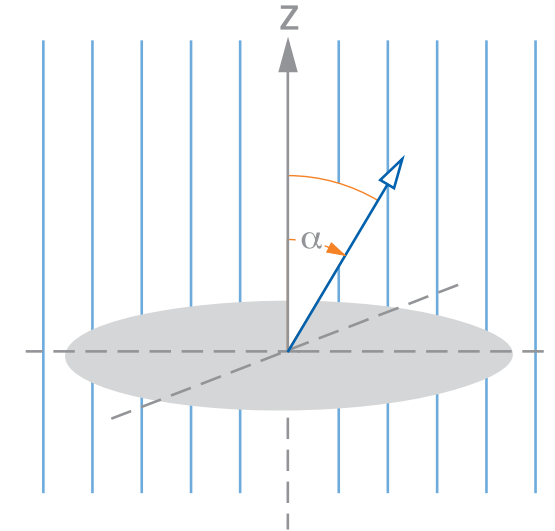
Excitation Flip Angle

- The **flip angle** by which we are deflecting the macroscopic magnetization from the z axis depends on the strength B_1 and duration τ of the transverse on-resonance RF field

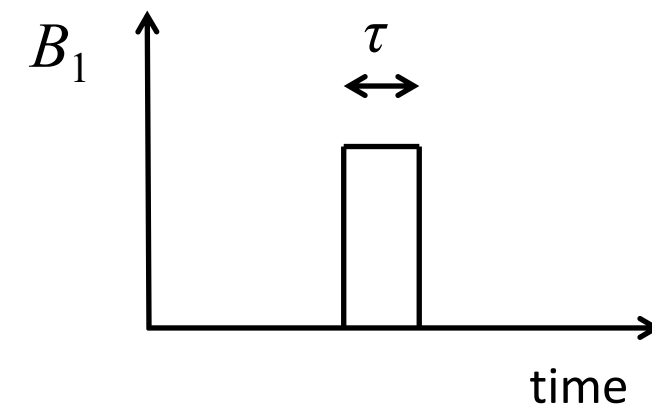
$$\alpha = \gamma \int_{-\tau/2}^{\tau/2} B_1(t) dt$$

- With constant amplitude and phase („rectangular pulse“):

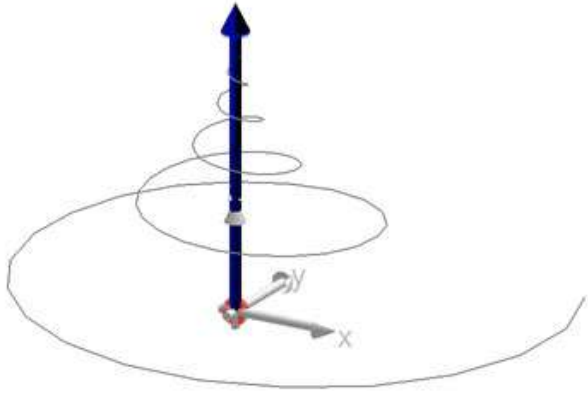
$$\alpha = \gamma B_1 \tau$$



Reminder: $\gamma = 267.513 \cdot 10^6 \text{ rad}/(\text{s} \cdot \text{T})$



Relaxation



- Following an excitation, the magnetization returns to its equilibrium state
- Longitudinal magnetization regains its equilibrium value M_0
- Transverse magnetisation decays to zero
- Both happen exponentially, transverse decay is faster

Longitudinal Relaxation

- Longitudinal relaxation due to exchange of energy with the surrounding tissue (lattice)
- Time evolution is given by

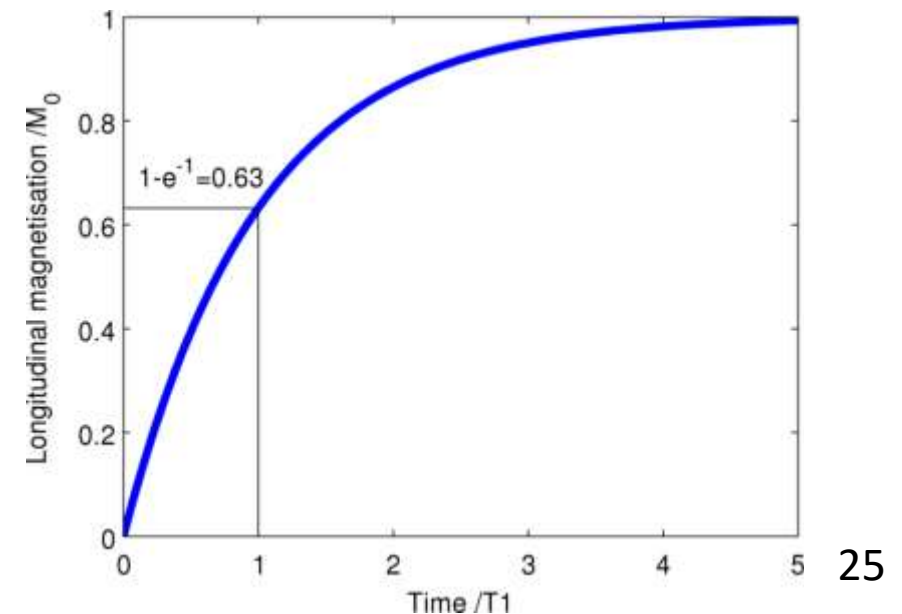
$$M_z(t) = M_0 + (M_z(0) - M_0)e^{-\frac{t}{T_1}}$$

- Assuming a 90° excitation pulse at t=0; $M_z(0)=0$:

$$M_z = M_0(1 - e^{-\frac{t}{T_1}})$$

M_0 = longitudinal equilibrium magnetization

T_1 = spin-lattice relaxation time constant

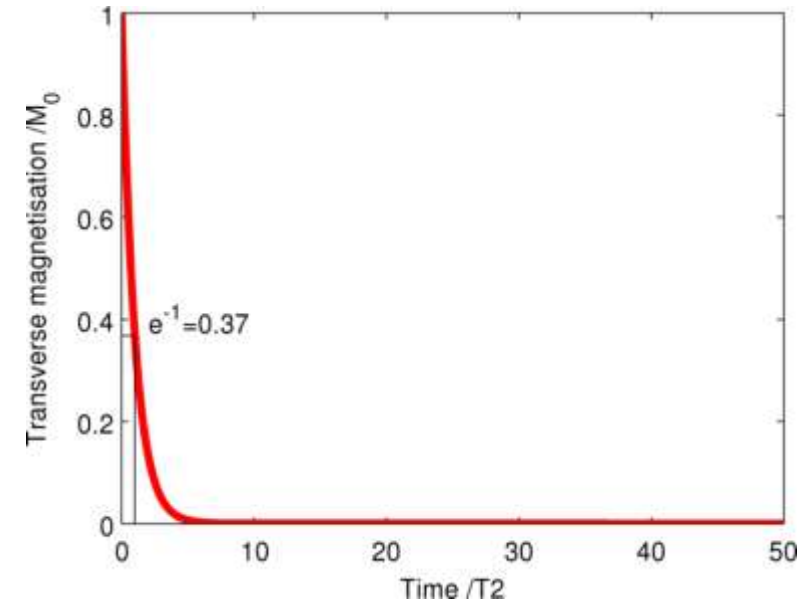


Transverse Relaxation

- Transient and random local field fluctuations occur on atomic level through spin-spin interactions
- Leads to *irreversible* loss of phase coherence between the spins
- Assuming a 90° pulse at $t=0$ and $M_{xy}(0)=M_0$ the solution is

$$M_{xy} = M_0 e^{-\frac{t}{T_2}}$$

T_2 is the spin-spin relaxation time constant.



Typical T1 and T2 Times in Tissue

- T1 times (in ms):

	0.2 T	1.0 T	3T
Fat		240	≈400
Muscle	370	730	≈1200
White Matter	388	680	≈900
Gray Matter	492	809	≈1500
CSF	1400	2500	≈4000

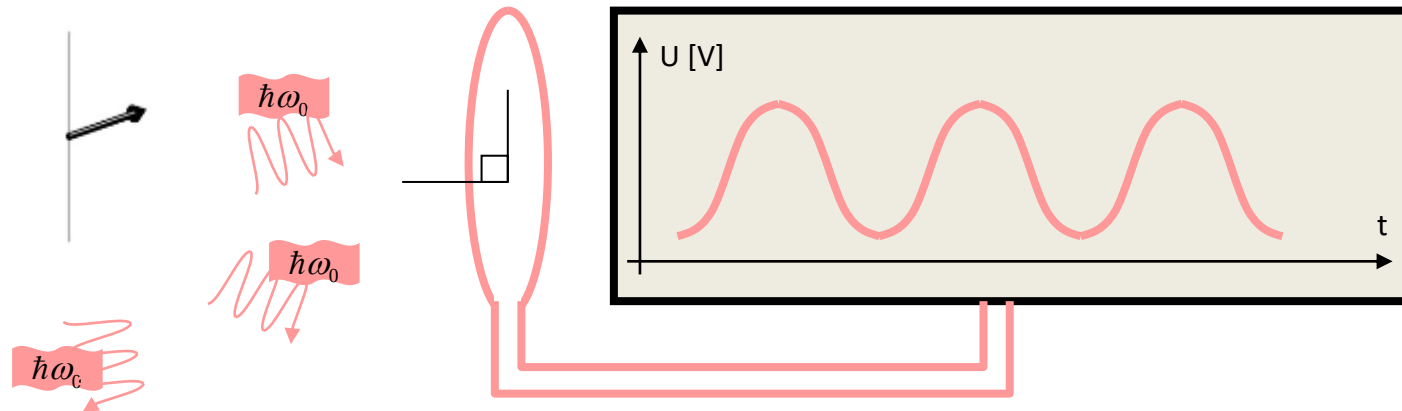
- T2 times (in ms):

Fat	84
Muscle	47
White Matter	92
Gray Matter	101
CSF	1400

Note: Measurements show high variability. (Bojorquez et al.,
What are normal relaxation times
of tissues at 3T? MRI 2017)

MR Signal Reception

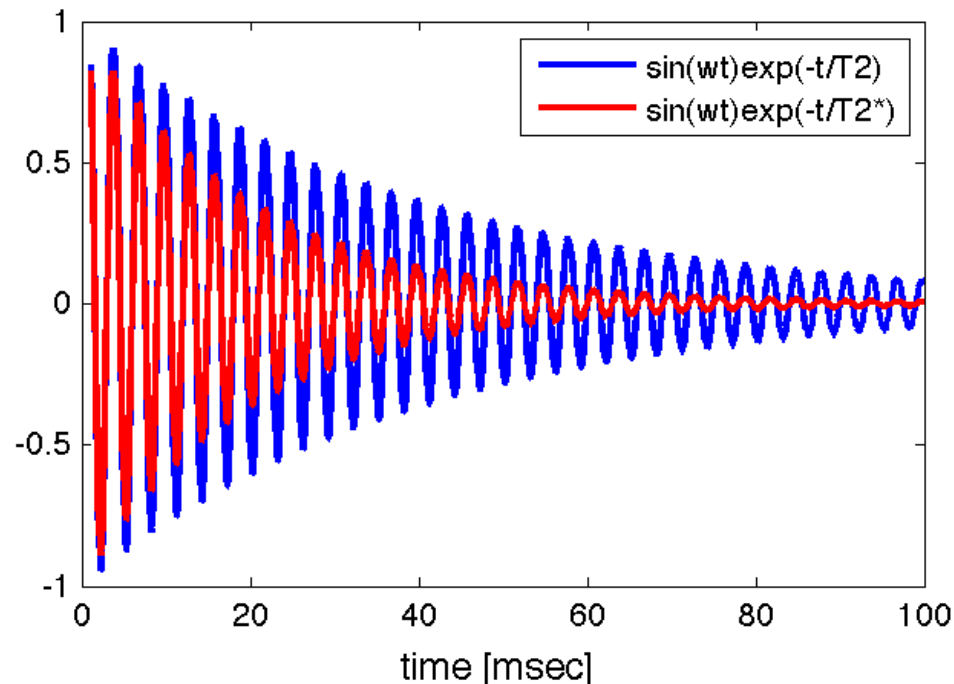
- Faraday's law: "A changing magnetic flux induces a voltage in a conductive loop."
(cf. bicycle dynamo)
- Maximize the area of the loop by orienting it perpendicular to the transverse plane



Free Induction Decay (FID) and T_2^* (effective transverse relaxation)

- Spin-spin interaction lead to irreversible loss of phase coherence between the spins (T_2 -decay)
- B_0 inhomogeneities and susceptibility effects cause additional shortening with time constant T_2'
- **Resulting decay constant: T_2^***

$$\frac{1}{T_2^*} = \frac{1}{T_2} + \frac{1}{T_2'}$$

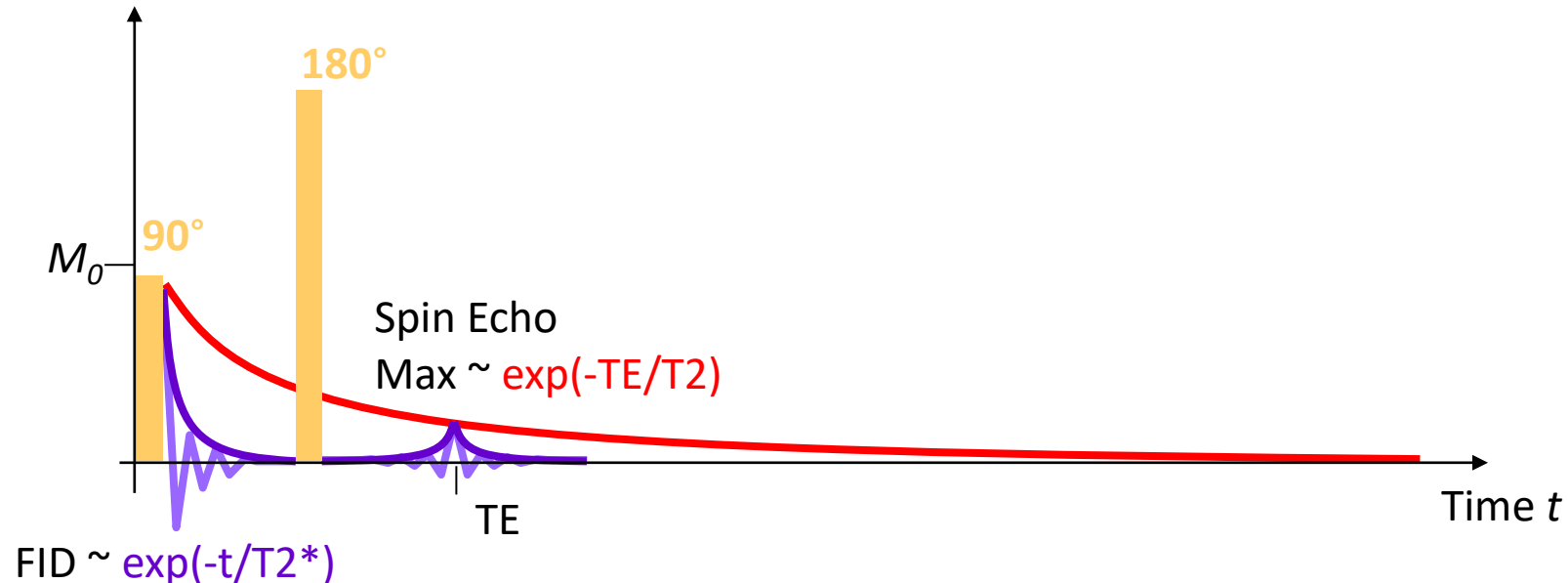


Key Questions of this Section

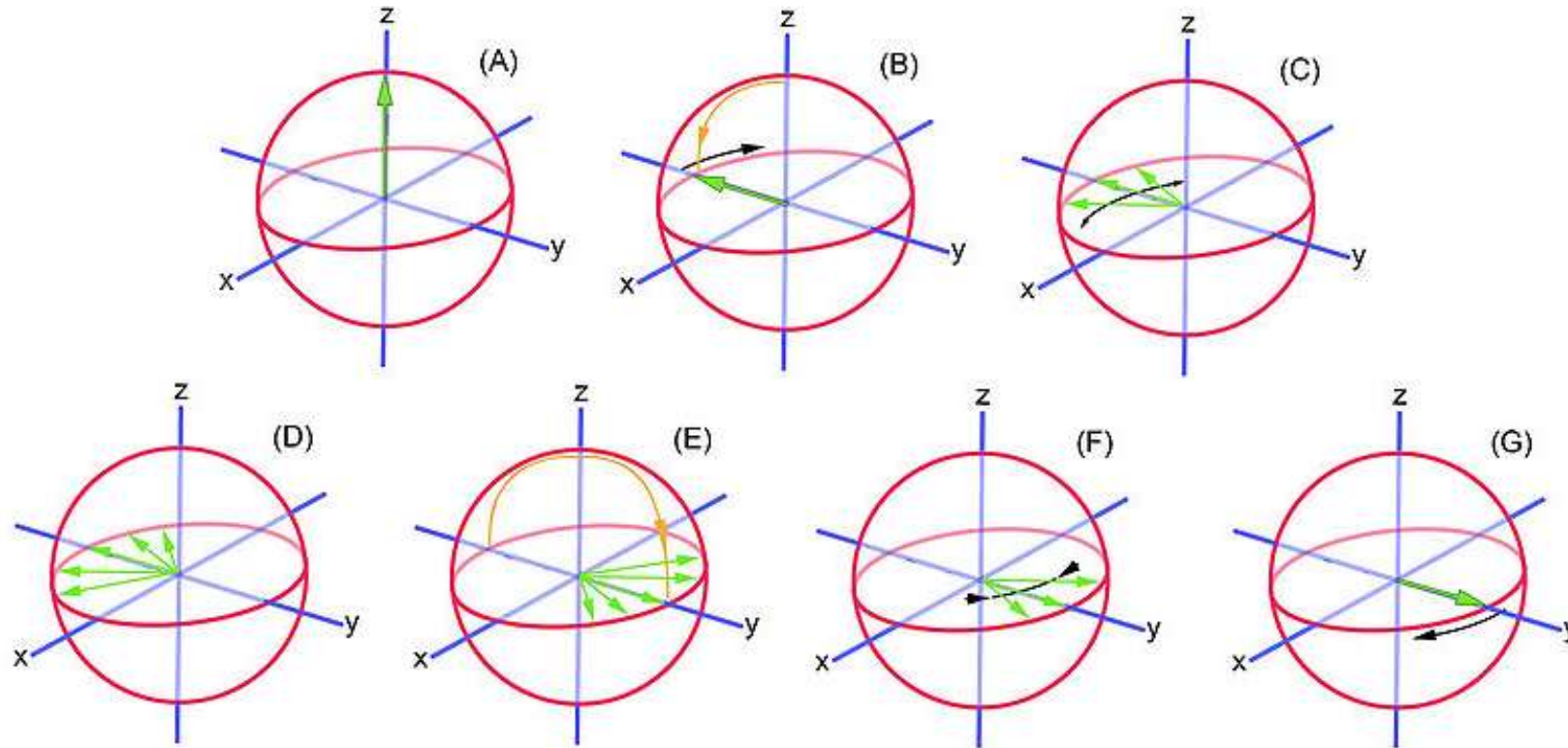
- How does the **signal** arise that an MR scanner measures?
 - Why do we need such a **strong magnetic field**?
 - Why is it called “**Magnetic Resonance**”?
 - How does MR differentiate between different **types of soft tissue**?
-
- What is the basic setup of an **echo measurement**?
 - What are the main **safety issues** related to MRI?

Spin Echo (SE)

- Signal losses due to B_0 inhomogeneities is reversible!
- Recall a ***spin echo*** by inversion of the spins using a 180° pulse.
- The individual spin's phase evolution due to the B_0 induced *frequency dispersion* (T_2') is “rewound”. But T_2 decay is *irreversible*.

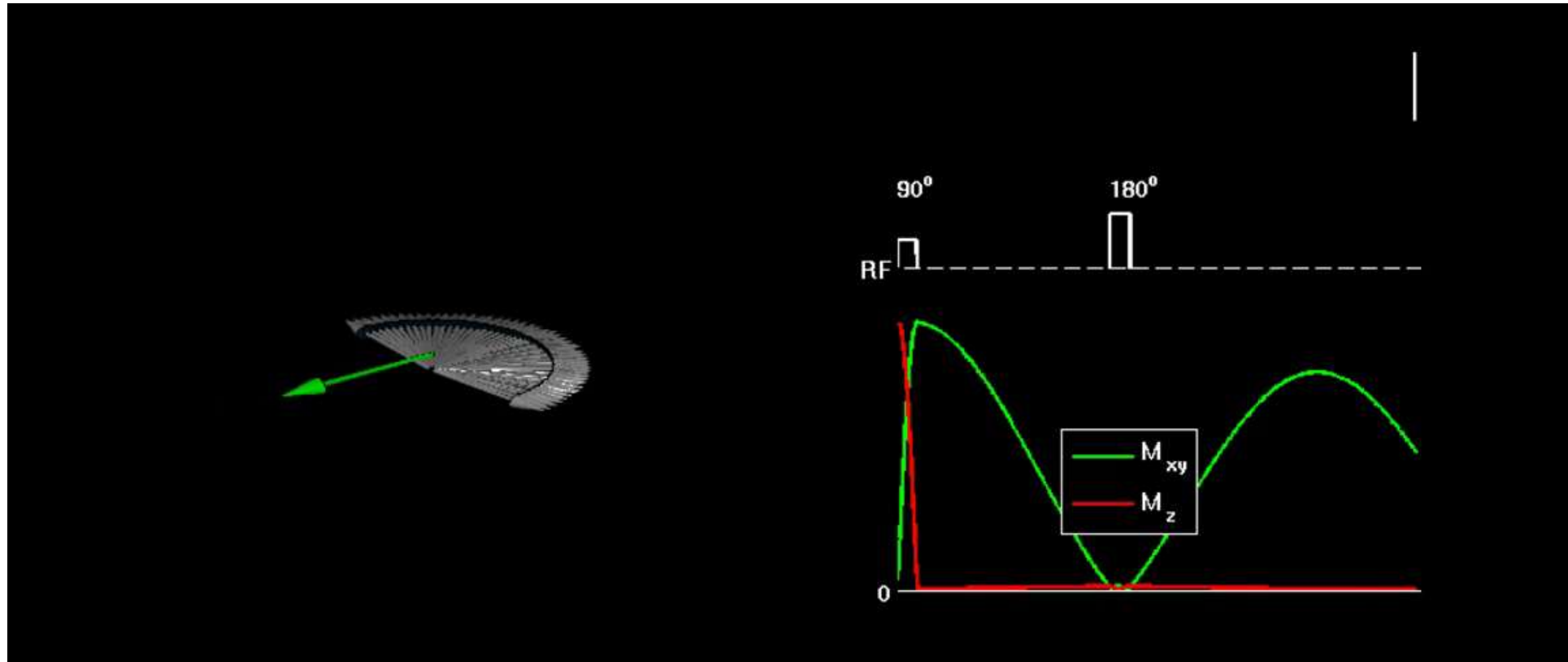


Spin Echo

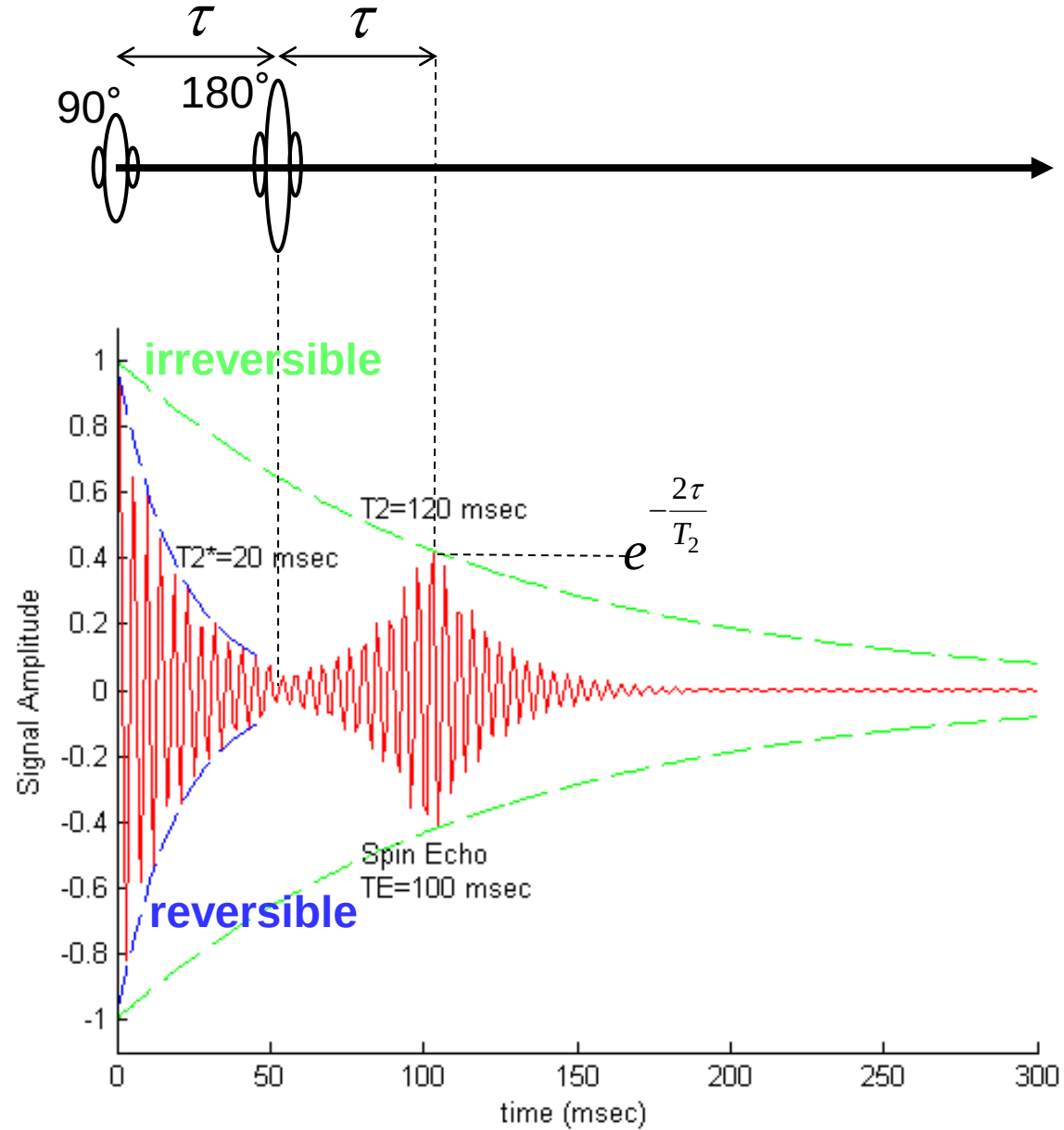


- A) Equilibrium
- B) 90 degree RF pulse (excitation)
- C)-D) de-phasing of spins
- E) 180 degree RF pulse (refocusing)
- F)-G) re-phasing of spins => Spin Echo!

Illustration: Spin Echo



Spin Echo



Gradient Recalled Echo (GRE)

- Recall a ***gradient echo*** by *dephasing* and subsequent *rephasing* in the opposite direction.
- Much faster and less RF than spin echo.
- But: the individual spin's phase evolution due to the B_0 induced *frequency dispersion* (T_2^*) is *NOT* “rewound”.

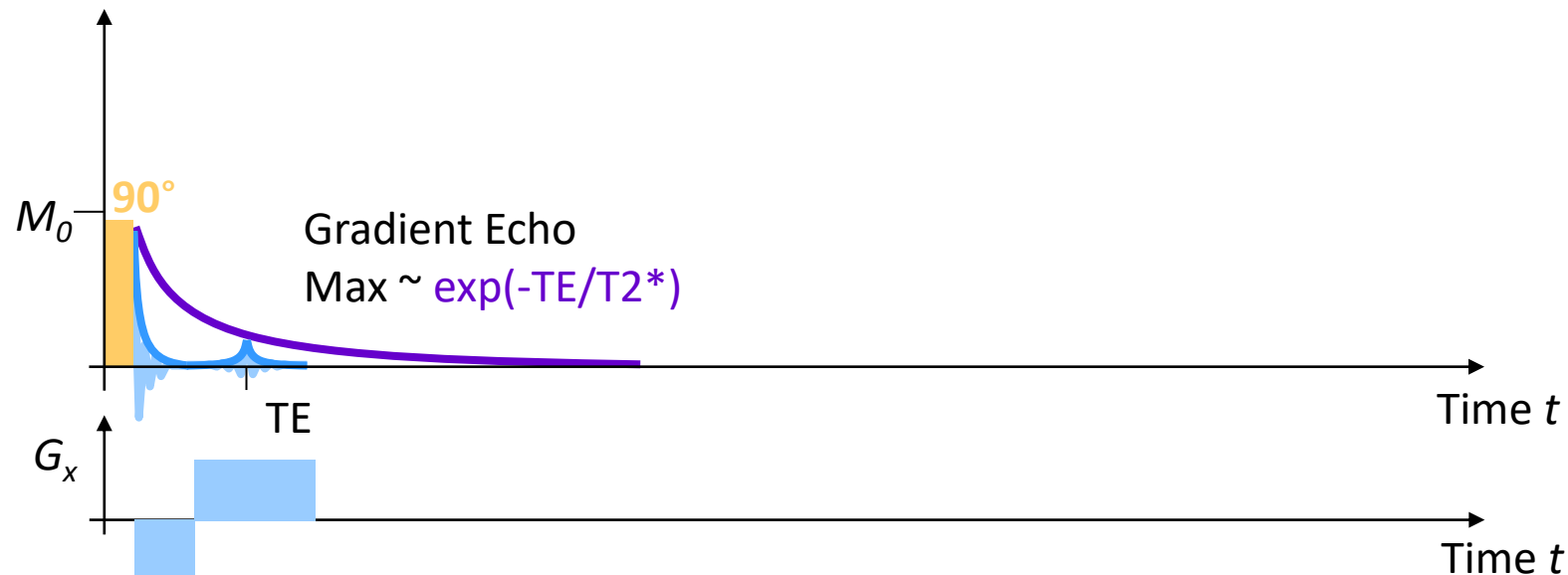
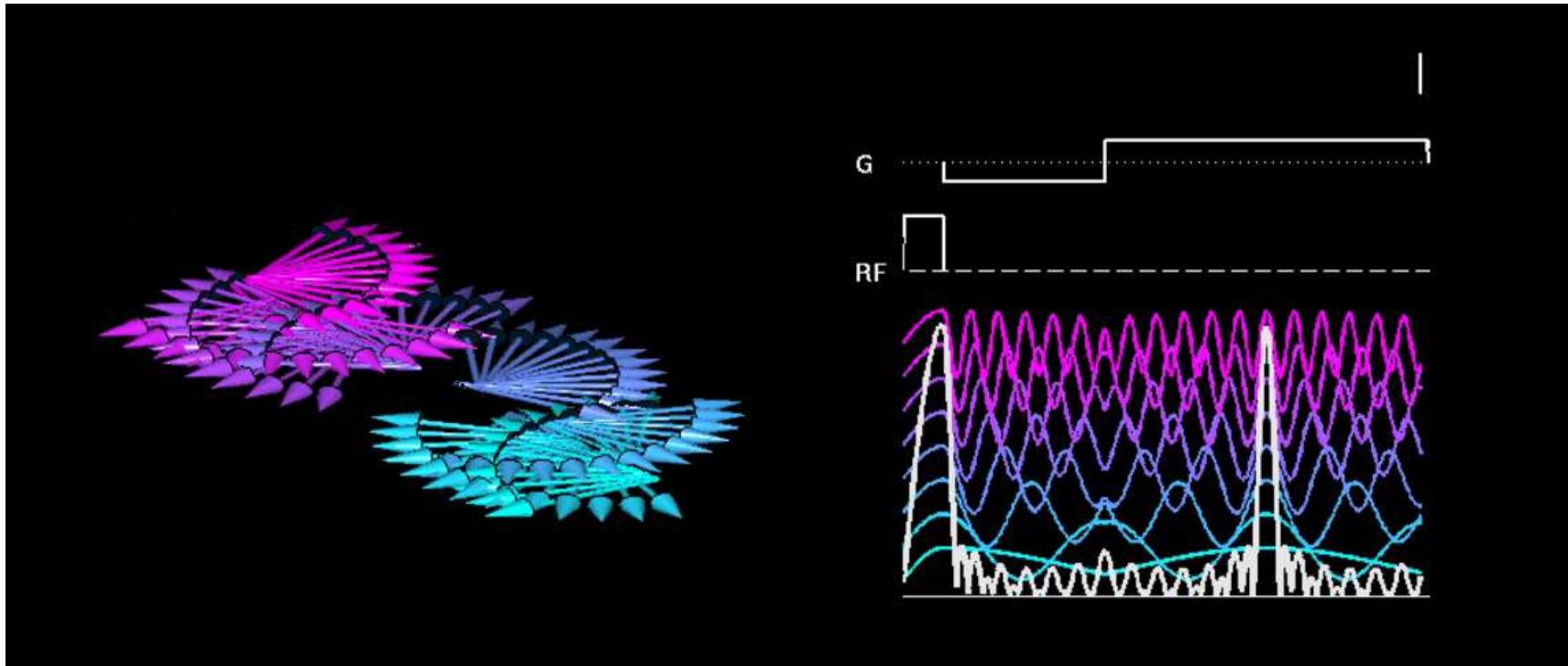


Illustration: Gradient Echo



Basic Safety Issues – The Main Field

- Permanently cooled-down (liquid Helium, $\sim 4\text{K}$) superconductor produces the high, static magnetic field: $B_0 \sim 1 - 3 \text{ Tesla}$ ($1\text{T} = 10000 \text{ Gauss}$).
 - Earth's magnetic field on its surface $\sim 0.5 \text{ Gauss}$.
- ➔ Huge force on ferromagnetic material growing rapidly when approaching the magnet.
- ➔ Objects like a normal fire extinguisher, tools, keys etc. can cause immense (live-threatening) damage.
- ➔ Credit cards, cell phones, pacemakers will not work anymore.
- ➔ Currents are induced in conducting material by changing magnetic flux, e.g. in organs when entering the scanner too quickly (from 0 to B_0). Can cause dizziness.

Basic Safety Issues – Field Gradients

- Imaging gradients are ramped up and down repetitively during an imaging sequence.
 - Max. Amplitude ~ 40 mT/m.
-
- ➔ Slew rate is limited to max. $200 \text{ mT}/(\text{m ms})$ in order not to stimulate the patient's peripheral nerves (muscle contraction in arms, back, ...)
 - ➔ Patients should not cross their hands in order not to create a conducting loop through their body.
 - ➔ Again, ferro-magnetic parts are not allowed, they would move and cause live-threatening damage.

Basic Safety Issues – RF fields

- RF fields can lead to hot spots at metallic edges, e.g. cables in contact with the skin or MR-unsafe implants.
 - Heating of make-up / tattoos with magnetic properties
- Higher B_0 fields require higher frequencies, wavelength of the RF fields approaches the extent of, e.g., the head.
 - High amplitudes due to standing-wave effects.
 - Amplitude and duration of RF must be controlled due to SAR (Specific Absorption Rate) constraints.

Summary

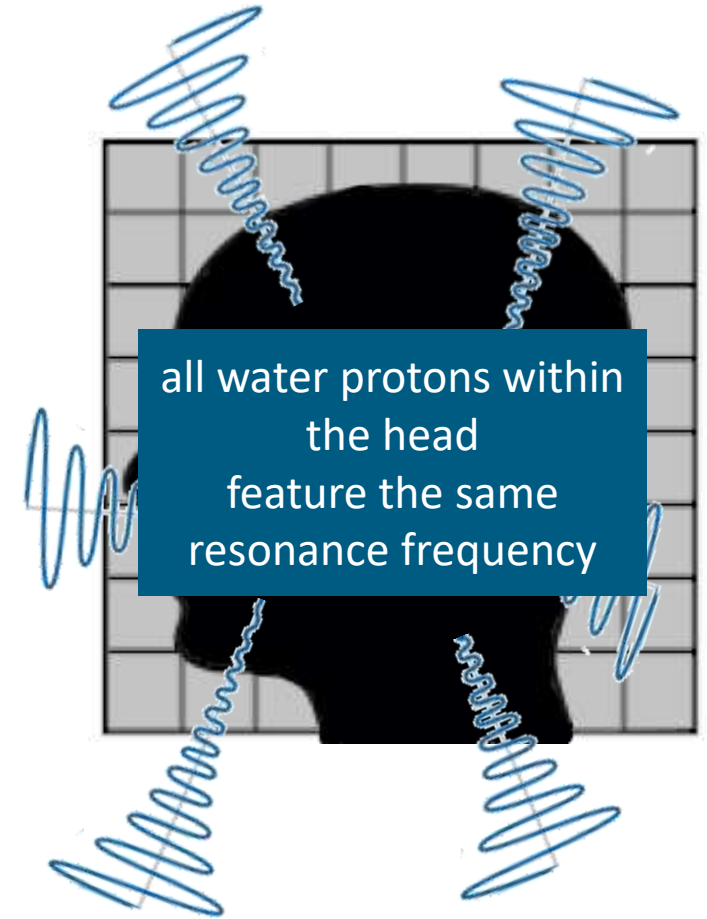
- MR places proton-spins in high B_0
 - Results in macroscopic magnetization (along B_0 in equilibrium).
- Spins excitable using *resonant RF* pulses.
 - Flips macroscopic magnetization vector
 - Rotating transverse part induces a current
- *Relaxation* described by T_1 , T_2 , T_2^* .
- MRI acquires echo-signals, usually *Spin echoes* or *Gradient recalled echoes*.

3.3 Magnetic Resonance Imaging

Partly based on slides by Prof. Tony Stöcker

Imaging ?

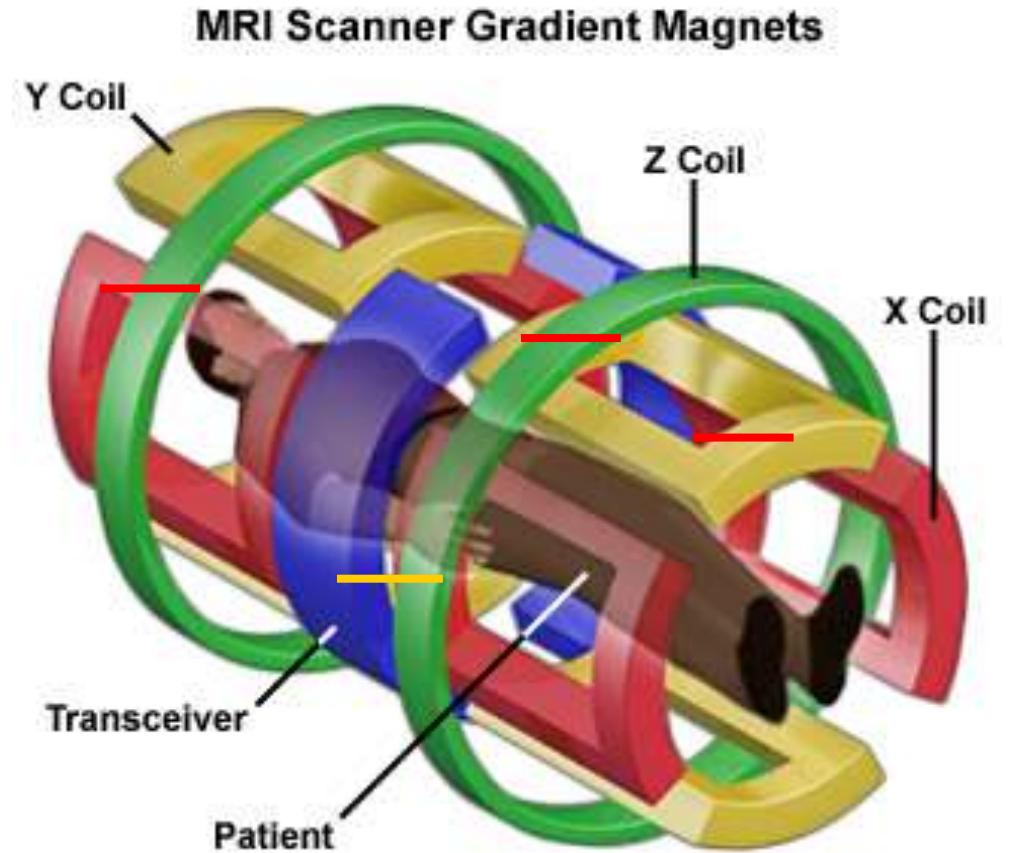
- So far, the MR signal is the sum of all nuclear magnetic resonances within the entire sample.
- Since signals are not localized spatially, we cannot use them to create an image.



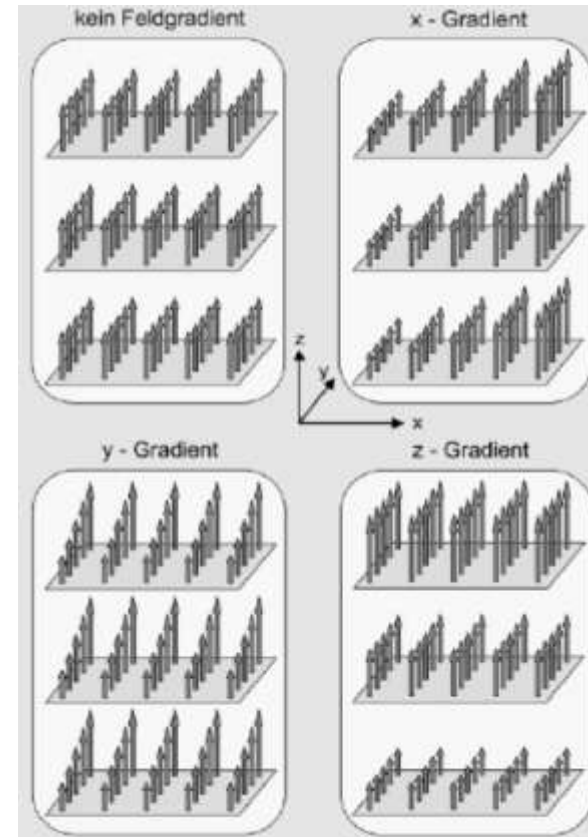
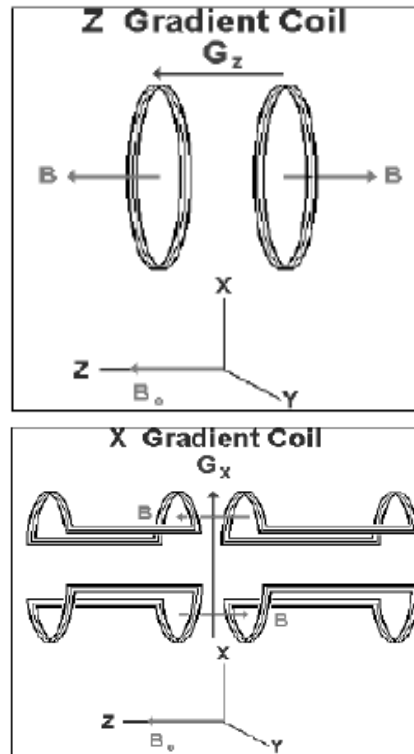
MR Scanner Components (Simplified)

Three types of magnetic fields:

1. Static field ($B_0 \parallel z$)
2. Radio-frequency ($B_1 \perp z$)
3. Dynamic “gradients” ($G_i = dB_z/dr_i$)



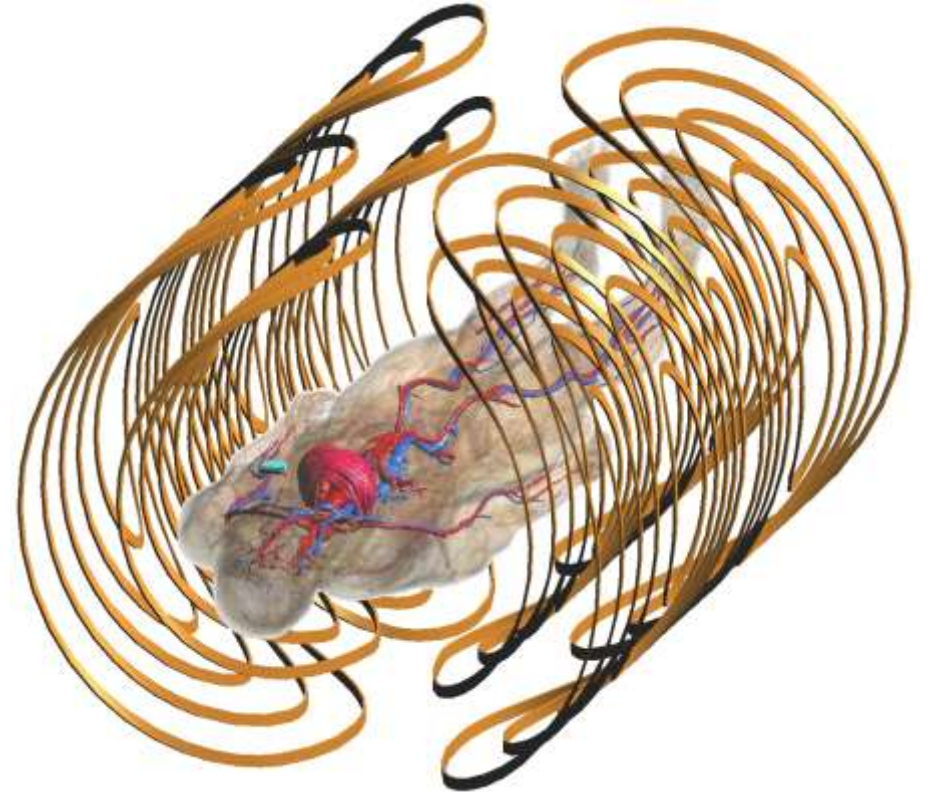
Gradient Coils: Linear Field Variation



$$\Rightarrow \mathbf{B}(t) = (B_0 + \mathbf{G}(t) \cdot \mathbf{r})\mathbf{e}_z$$

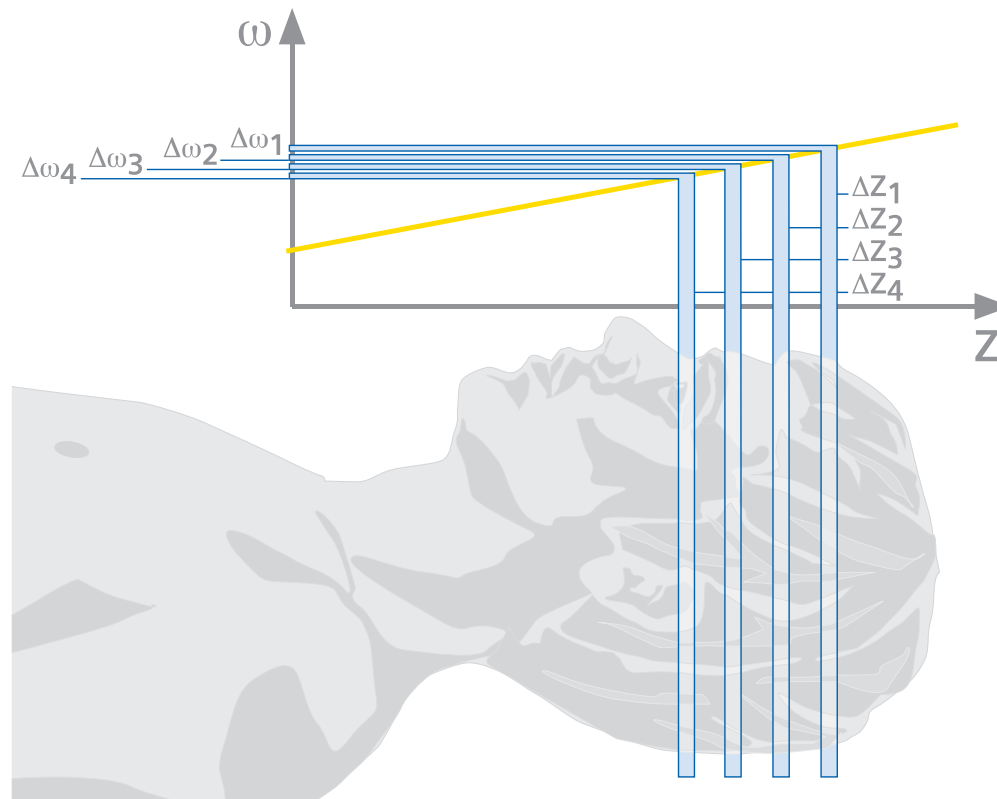
Some Facts About Gradient Coils

- typical range: $1\text{-}40 \times 10^{-3} \text{ T/m}$
e.g.: $x=30 \text{ cm}$, $B_0=1\text{T}$, $G=10 \text{ mT/m}$
→ $B = 0.9985 \text{ T} \dots 1.0015 \text{ T}$
- rise time: $200\text{-}600 \mu\text{s}$
- linearity: $40\text{-}60 \text{ cm}$
- power: e.g. 500 A at 2000 V in short time, power in MW range
- need liquid cooling
- “make the noise”



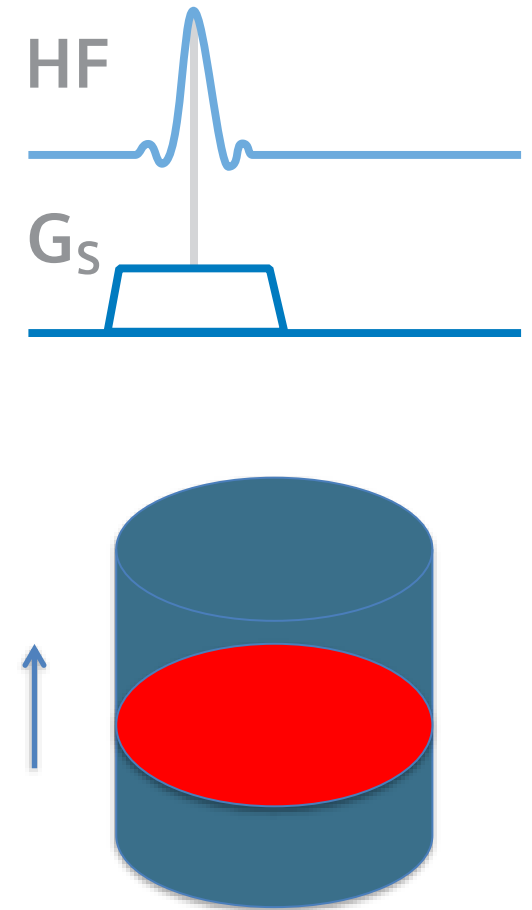
Slice Selection - Motivation

- Magnetic Resonance occurs only when RF pulse matches Larmor frequency
- Larmor frequency proportional to field strength



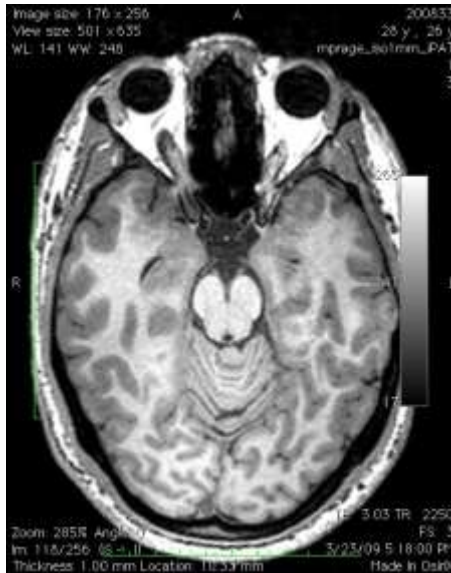
Slice Selection

- Simultaneous application of gradients and RF pulses
- Slices can be created in arbitrary orientations (*double oblique*) by simultaneous application of several gradients



Examples of Oblique Slices

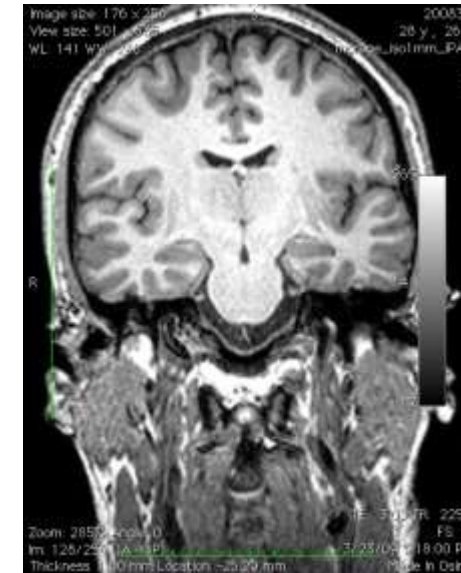
- Achieved through superposition of physical gradients G_x , G_y , G_z



axial slice

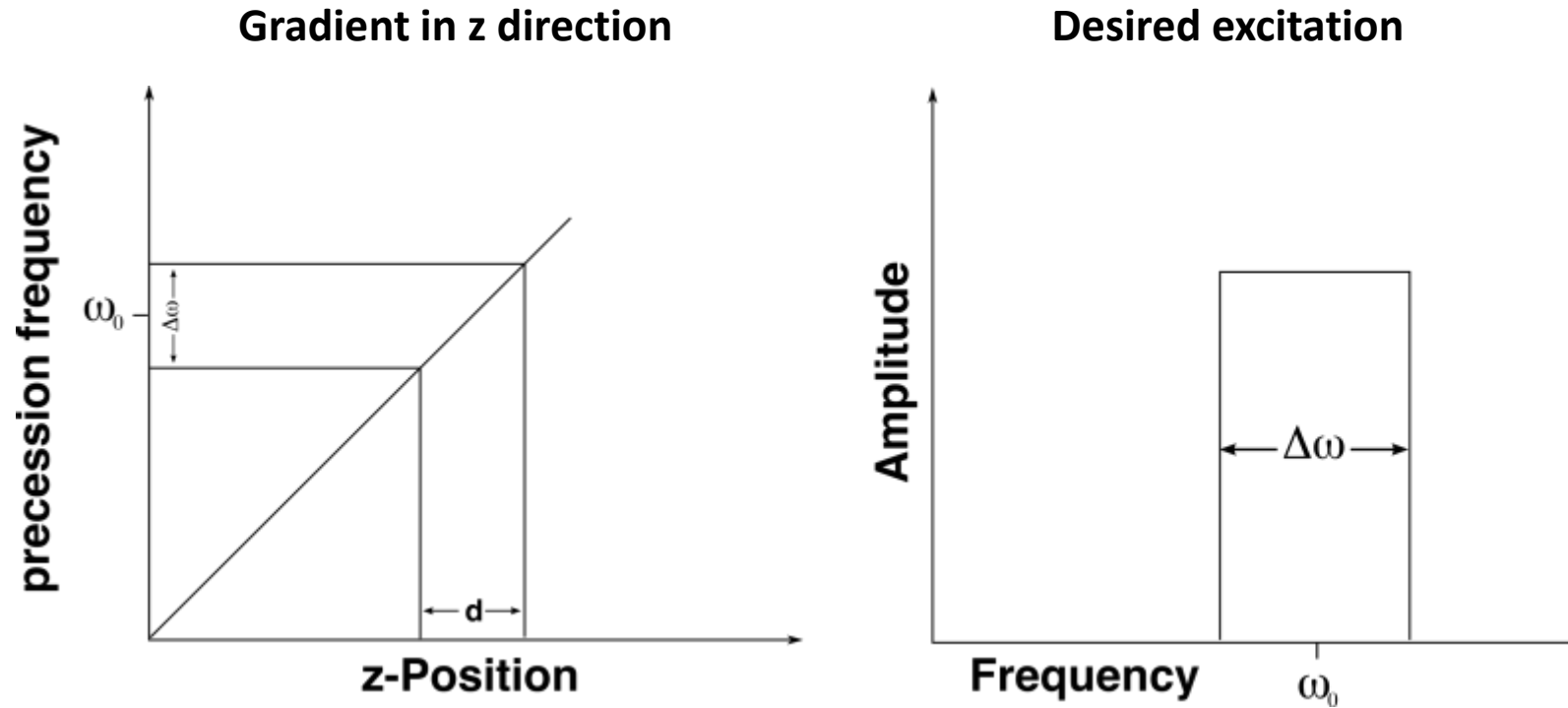


sagittal slice

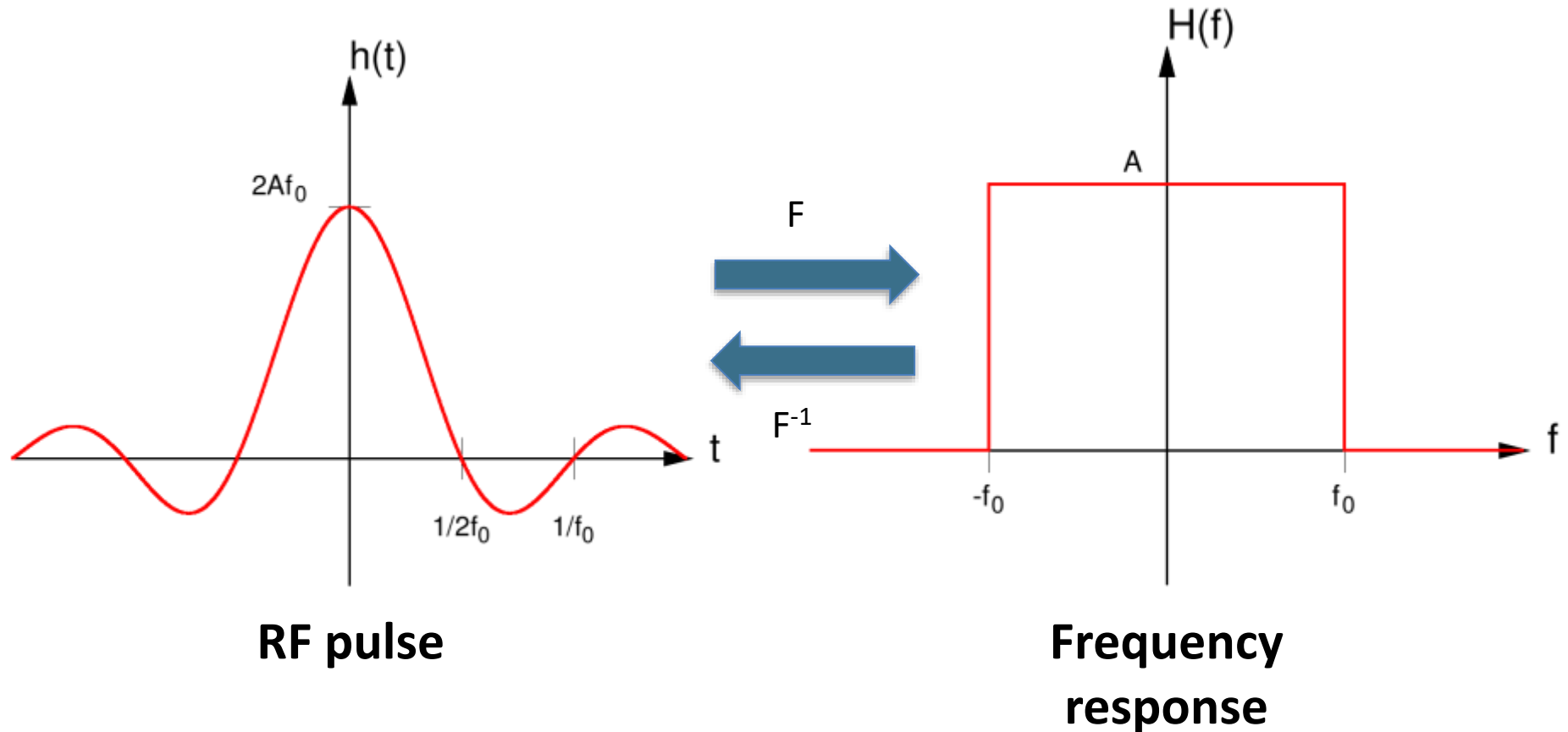


coronal slice

Slice Selection – Pictorial Description



Exciting a rectangular slice profile



Slice Selection – Slice Thickness

- The RF bandwidth Δf determines the *range of frequencies* contained in a pulse
- The slice thickness is determined by the frequency band of the excitation pulse (as a function of gradient amplitude)

$$\Delta z = \frac{2\pi\Delta f}{\gamma G_z}$$

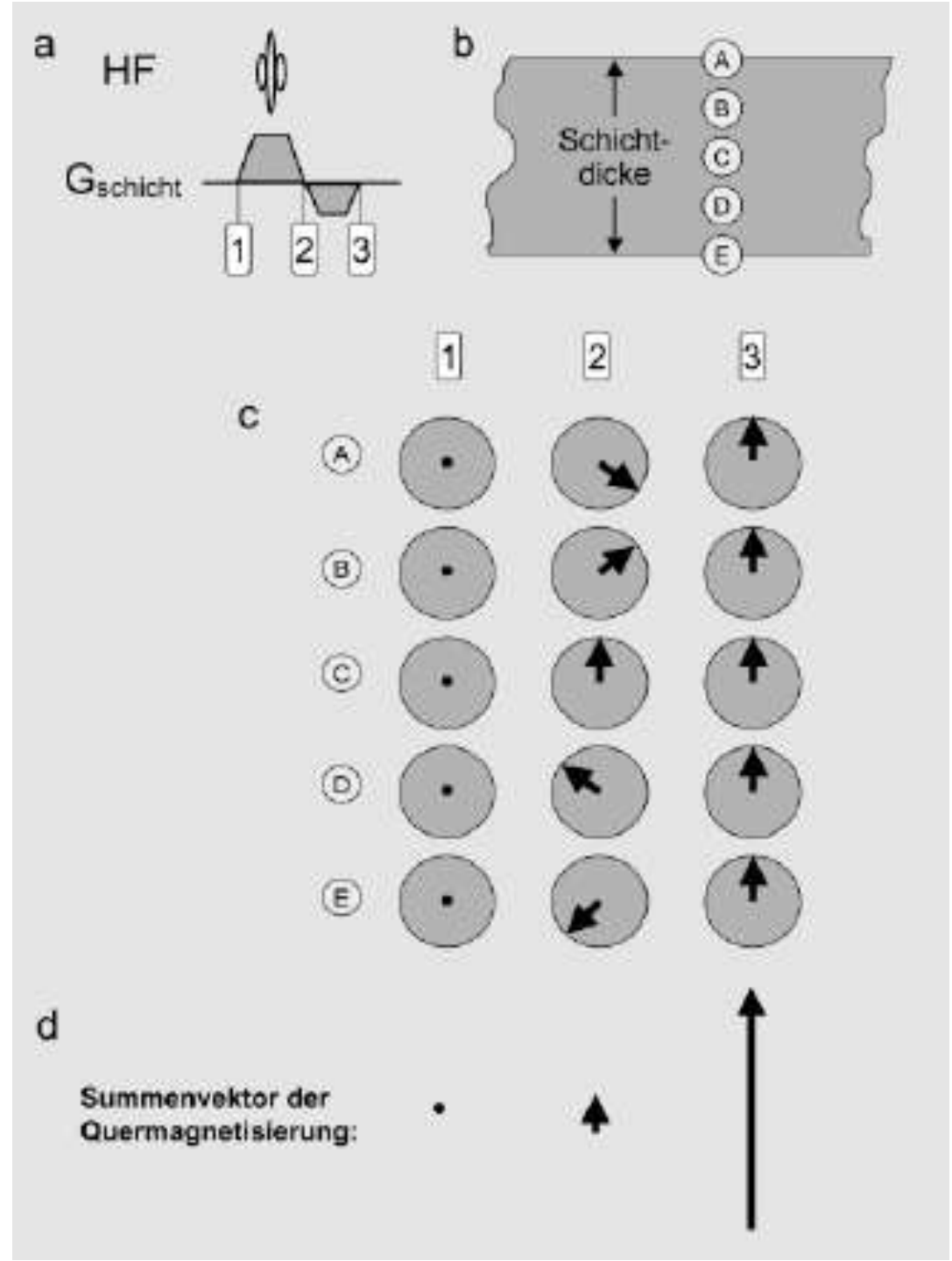
- Practical RF pulses do not have a rectangular freq. response; *Full-Width-Half-Maximum (FWHM)* is usually defined as bandwidth

Slice Refocusing

- Different Larmor frequencies within slice lead to phase dispersion by factor

$$\exp\left(\frac{i\gamma G_z z \tau}{2}\right)$$

- τ = duration of RF pulse
- Removed by application of opposite z-gradient (“refocusing lobe” or “slice rewinder”)

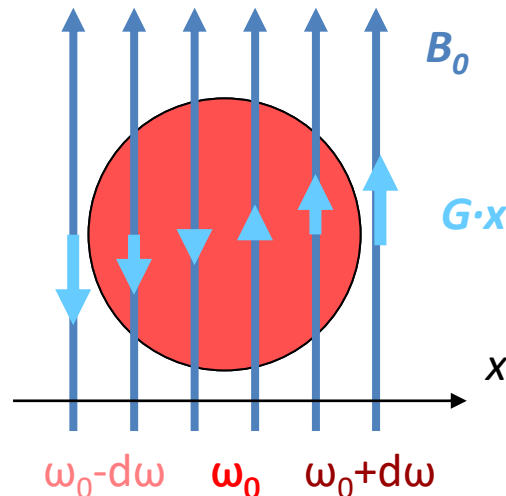


Spatial Encoding

- The basic idea of MRI:

Make the precessional frequency a function of space!

- The “spectrum” then reflects spatial distribution.
- Linear field gradients of the B-field in z-direction, e.g. $G_x = dB_z/dx$

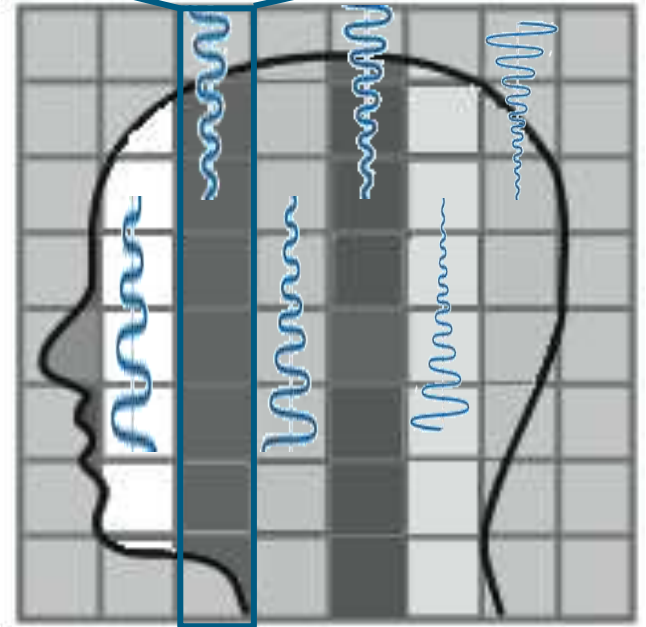


Frequency Encoding

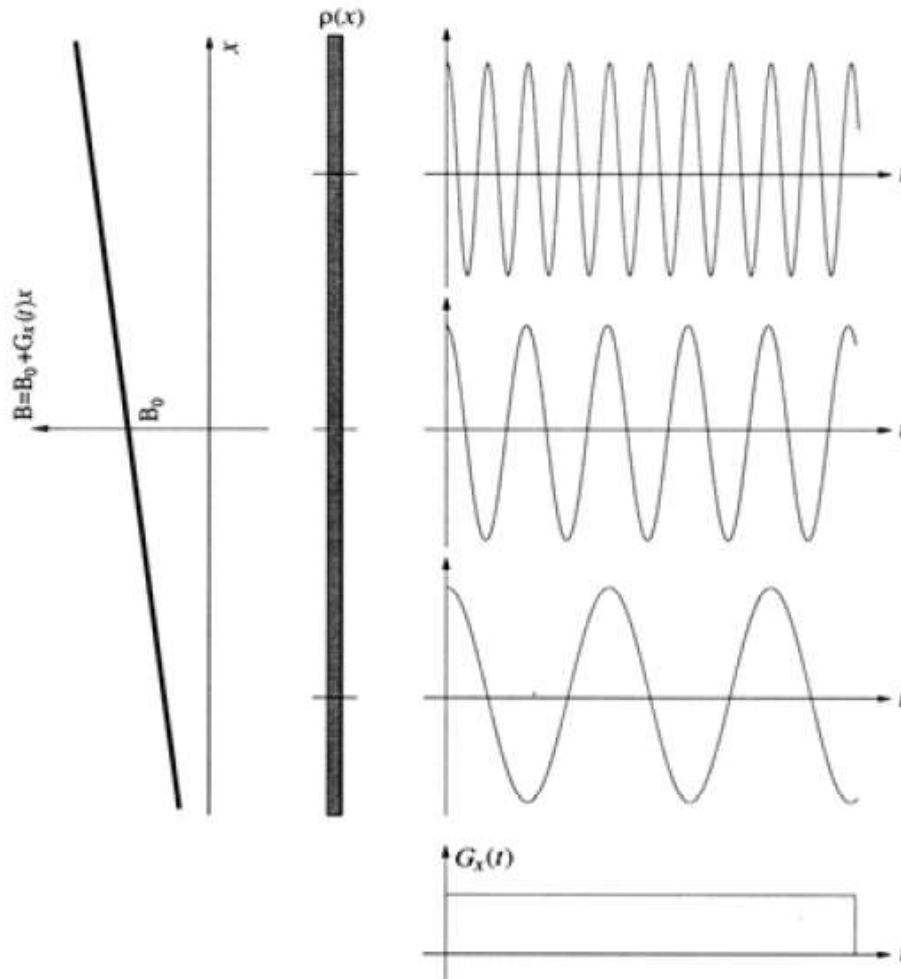
- Frequency encoding: apply gradient during data acquisition
 - “Readout gradient”

$$\Delta\omega = \gamma G_x x$$

$$M_{\perp}^n(t) = M_0 e^{-\frac{t}{T_2^*}} e^{-i(\omega_0 + \Delta\omega^n)t}$$



Frequency Encoding

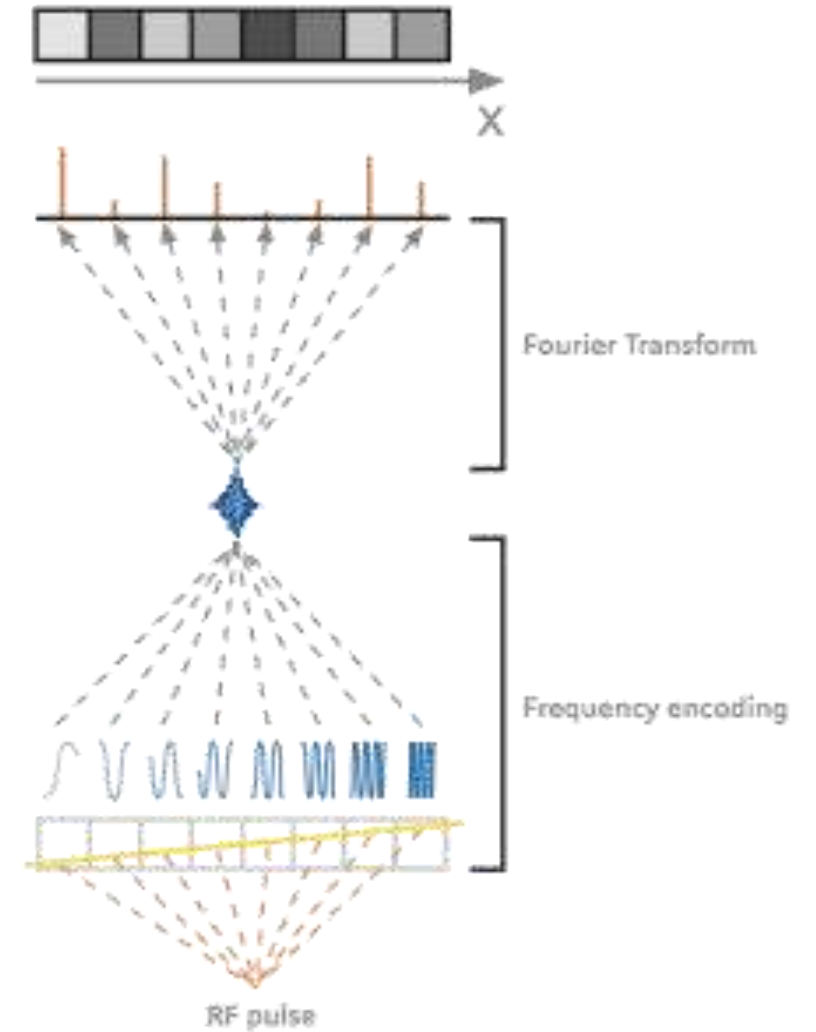


Larmor frequency
is a linear
function of the
spatial
coordinate:

$$\omega(x) = \gamma(B_0 + G_x \cdot x)$$

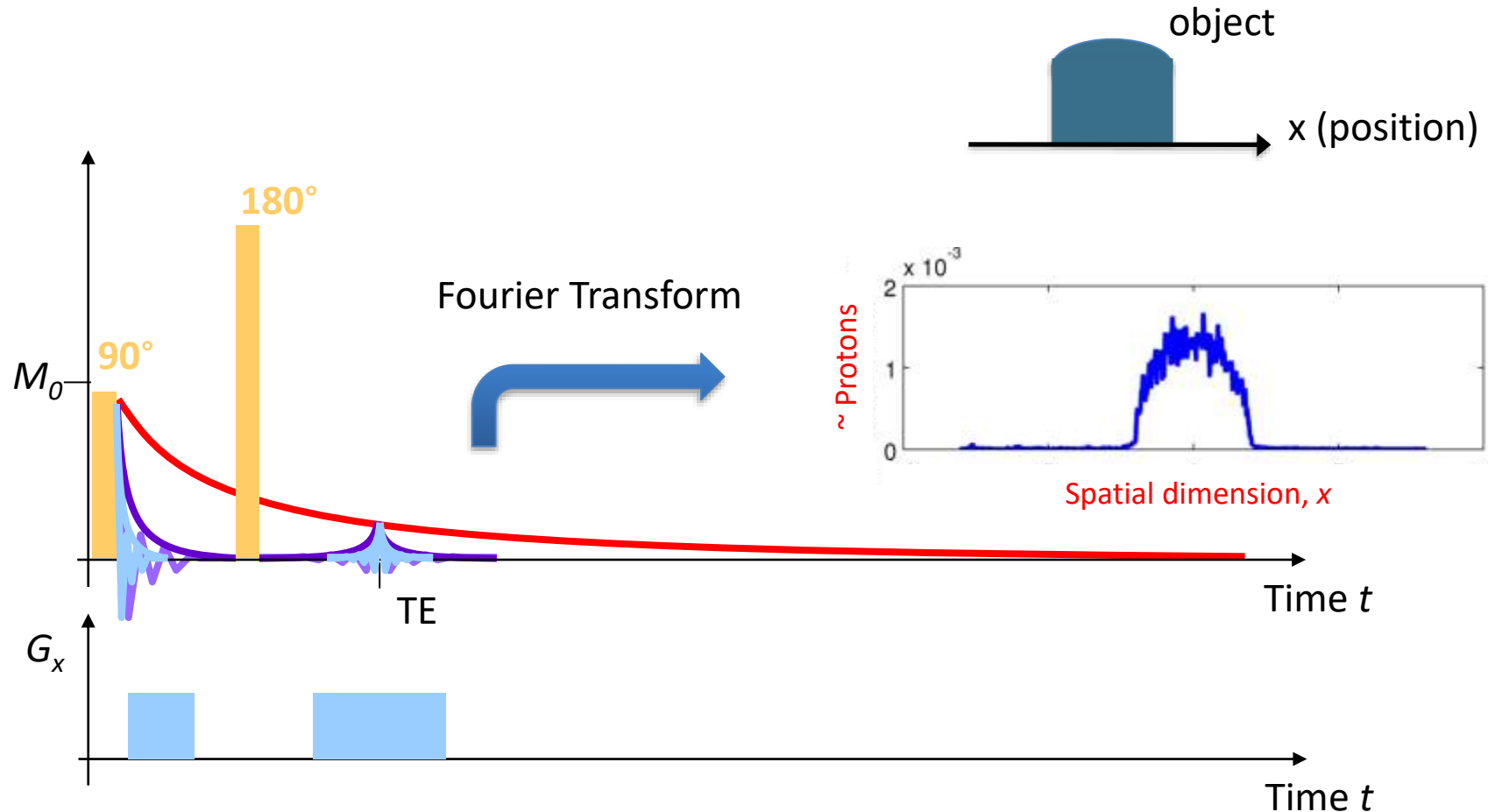
Frequency Encoding

- The Fourier Transformation allows us to transform that frequency information back to spatial location



Spin Echo with Frequency Encoding

Take the spin echo and introduce gradients along $x \Rightarrow$ 1D image !



Frequency Encoding: Sampling

- MR signal $S(t)$ is digitized by using an “analog to digital converter” ADC and a discrete timing interval Δt in a total acquisition window t_{aq}
- For the Fourier-analysis of the MR signal there are in total $N = t_{aq}/\Delta t$ measured data points: $S(\Delta t), S(2\Delta t), S(3\Delta t), \dots, S(N\Delta t)$
- Sampling theorem links Field of View (FOV; maximum object diameter) to sampling interval Δt :

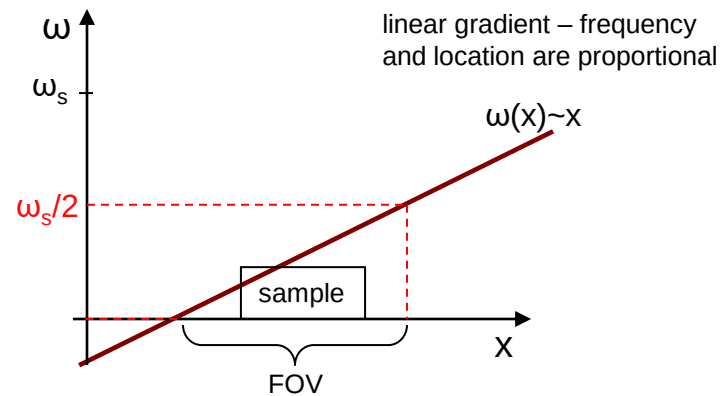
$$f_{\max} = \frac{\gamma G_x}{2\pi} \frac{\text{FOV}}{2} < \frac{1}{2\Delta t} \Rightarrow \text{FOV} = \frac{2\pi}{\gamma G_x \Delta t}$$

- Spatial resolution Δx results as $\Delta x = \text{FOV} / N$
- *Example:* with $N = 256$, $\Delta t = 30 \mu\text{s}$, $G_x = 1.566 \text{ mT/m}$ the spatial resolution in x-direction (pixel resolution in x) is:

$$\Delta x = 1.953 \text{ mm and } X = N \Delta x = 50 \text{ cm (= field-of-view FOV)}$$

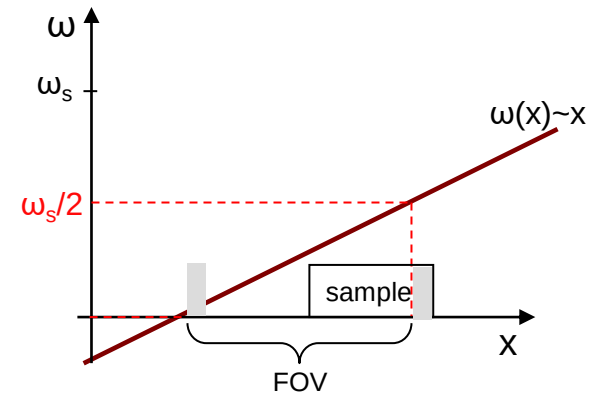
Aliasing in MRI

- When using frequency encoding, **aliasing manifests as a fold-over** of objects outside the FOV to incorrect locations.



$$\omega_{\max} < \frac{\omega_s}{2}$$

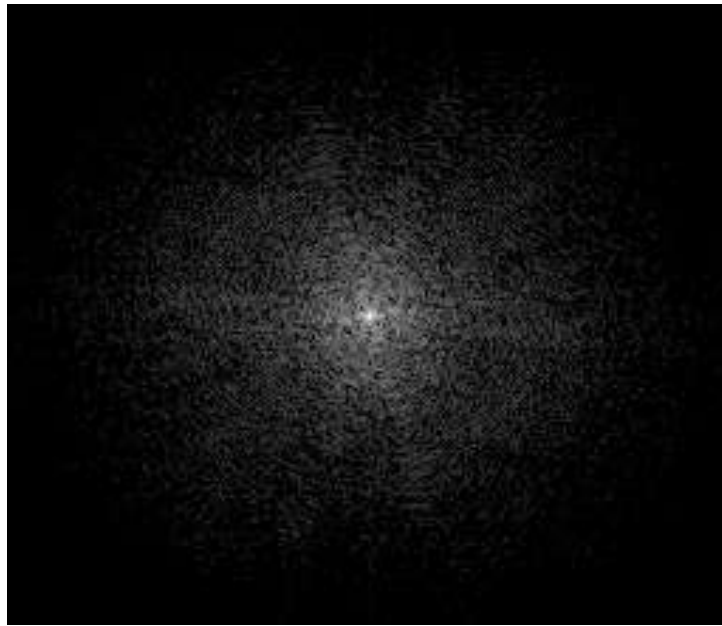
no aliasing



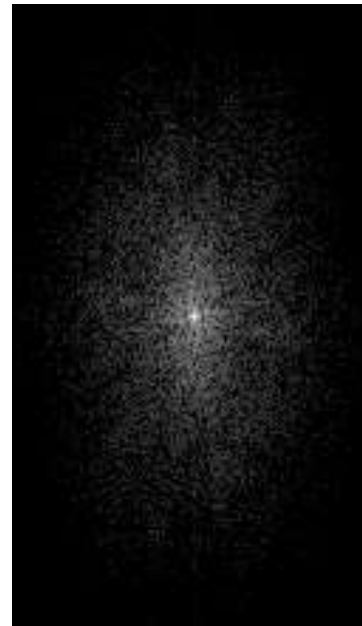
$$\omega_{\max} > \frac{\omega_s}{2}$$

aliasing

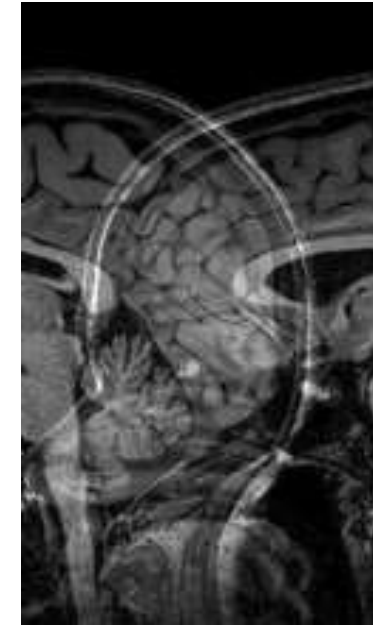
Aliasing - Illustration



**Original k space
representation**



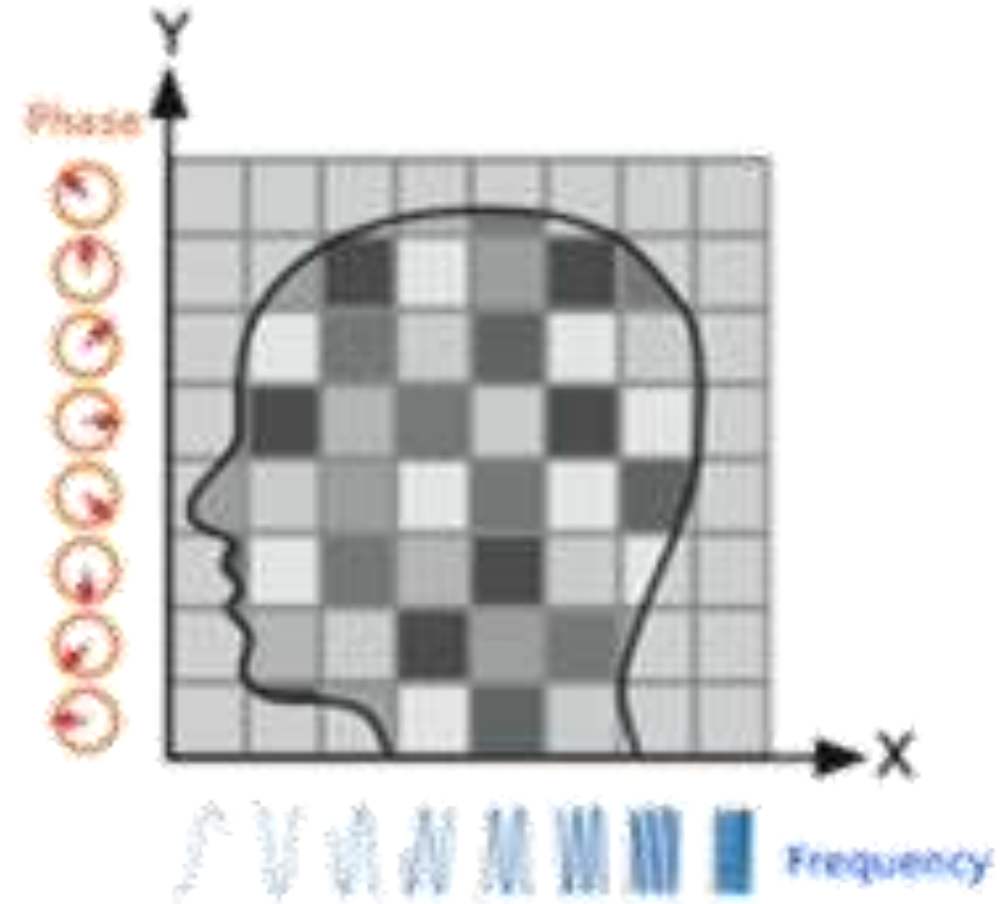
**Remove every
other column**



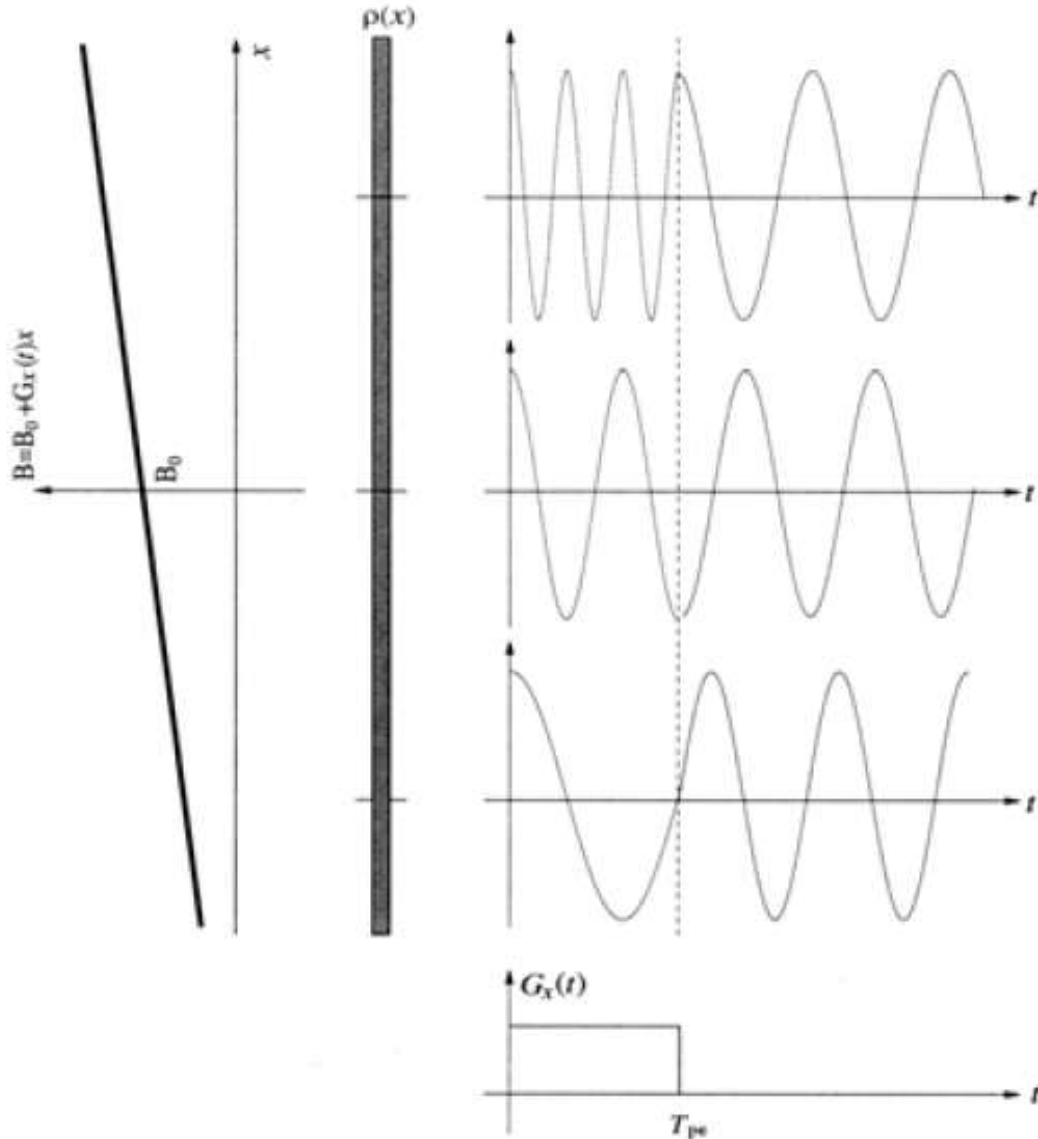
Aliased image

Phase Encoding

- Frequency encoding is applied to one spatial dimension, e.g. the x-axis.
How to encode the y-axis?
- Switch on the y-gradient for a short time *between excitation and readout*, in order to modulate the spins' phase in y direction.
- Repetition of the process with linearly varying phase again encodes a frequency!

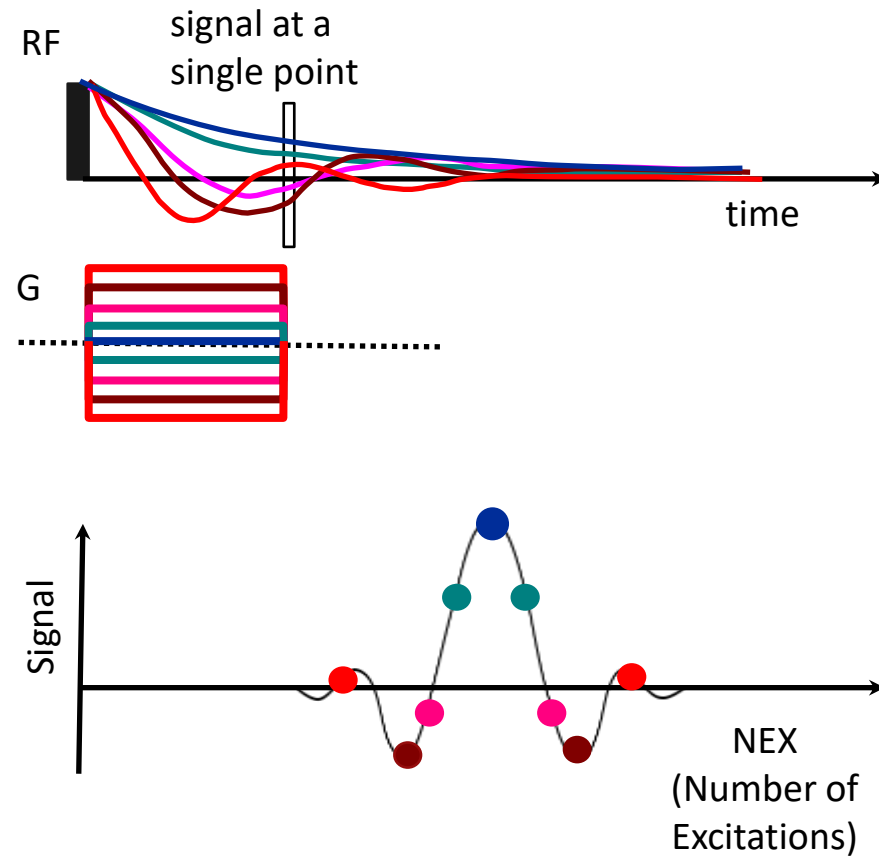
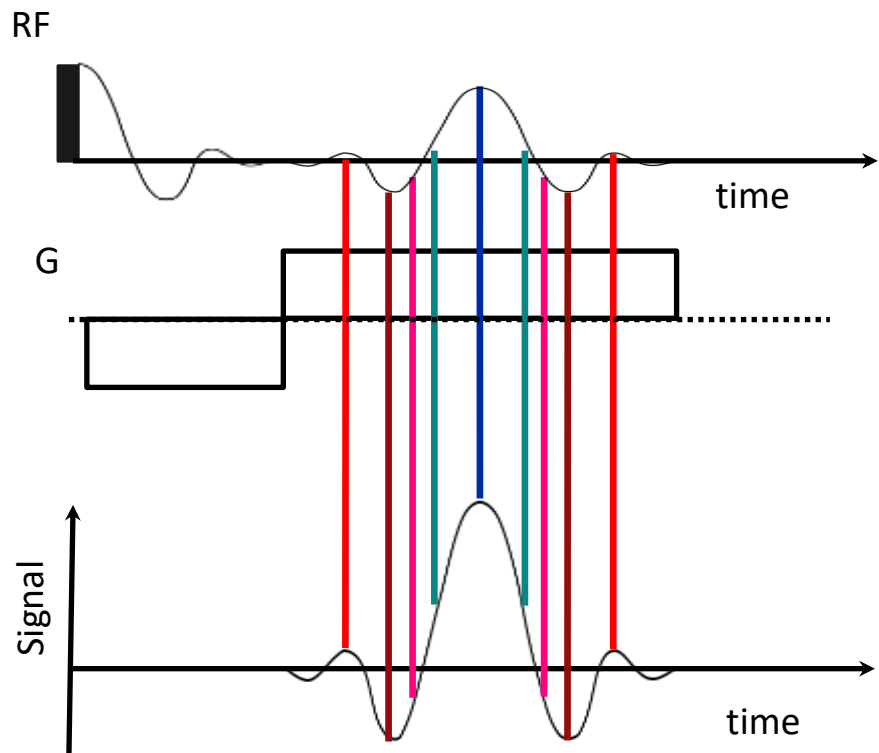


Phase Encoding

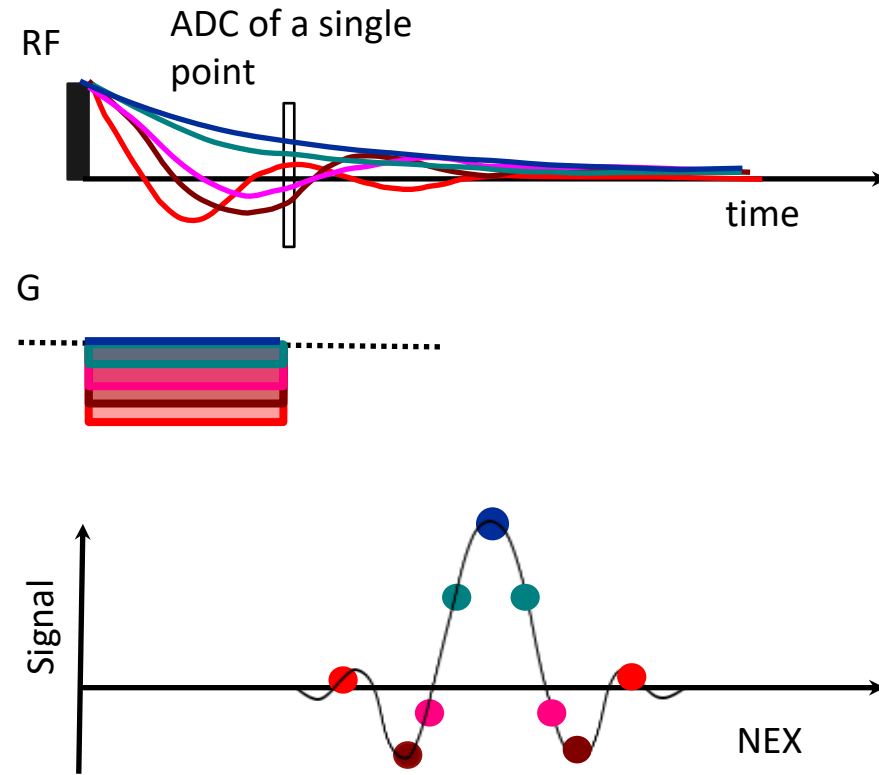
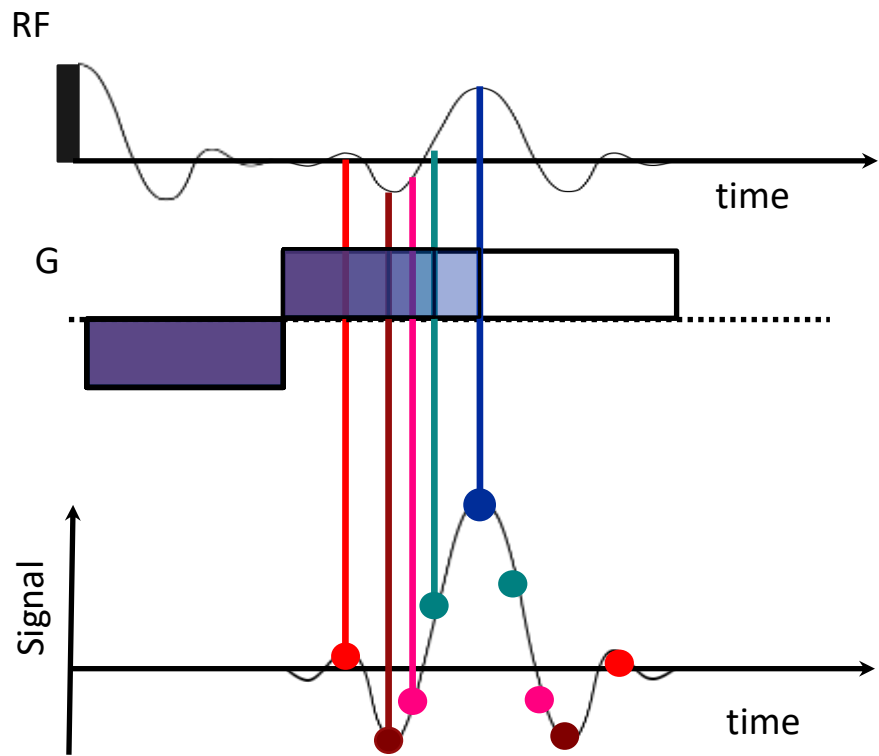


- Phase encoding stamps an initial phase angle onto the excited spins
- After switching off the phase encoding gradient, precession continues at a fixed frequency ω_0 , but with different phase
- Phase information of an activated MR-signal is linearly dependent on spatial coordinate

Frequency Encoding vs. Phase Encoding



Frequency Encoding vs. Phase Encoding



Signal Equation – Slice Selection

- The overall signal after a 90° pulse is given as:

$$s(t) = \iiint_{xyz} M_0(x, y, z) \exp\left(\cancel{\frac{-t}{T_2^*}}\right) \exp(-i\omega_0 t) dx dy dz$$

neglected!

equilibrium magnetisation ~ proton density
spin-spin relaxation
precession of the spins

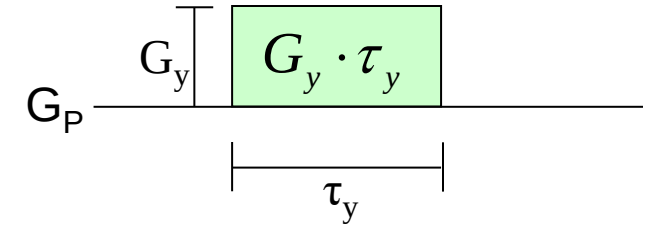
- 2D integral after slice selection:

$$s(t) = \iint_{xy} m(x, y) \exp(-i\omega_0 t) dx dy$$

$$\text{with } m(x, y) = \int_{z_0 - \frac{\Delta z}{2}}^{z_0 + \frac{\Delta z}{2}} M_0(x, y, z) dz$$

Signal Equation – Phase Encoding

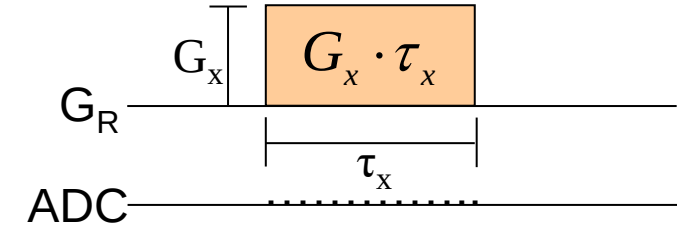
If we apply a linear gradient in y-direction between excitation and readout, it changes the signal equation as follows:



$$s(G_y, \tau_y, t) = \iint_{xy} m(x, y) \exp(-i\omega_0 t) \exp(-i\gamma G_y \tau_y y) dx dy$$

Signal Equation – Frequency Encoding

Similarly, a frequency encoding gradient in x-direction while readout changes the signal equation:



$$s(G_x, G_y, \tau_x, \tau_y, t) =$$

$$\iint_{xy} m(x, y) \exp(-i\omega_0 t) \exp(-i\gamma G_x \tau_x x) \exp(-i\gamma G_y \tau_y y) dx dy$$

Signal Equation - Demodulation

$$\iint_{xy} m(x, y) \exp(-i\omega_0 t) \exp(-i\gamma G_x \tau_x x) \exp(-i\gamma G_y \tau_y y) dx dy$$

demodulation in frequency by ω_0

(change from laboratory to rotating frame)

$$\iint_{xy} m(x, y) \cancel{\exp(-i\omega_0 t)} \exp(-i\gamma G_x \tau_x x) \exp(-i\gamma G_y \tau_y y) dx dy$$

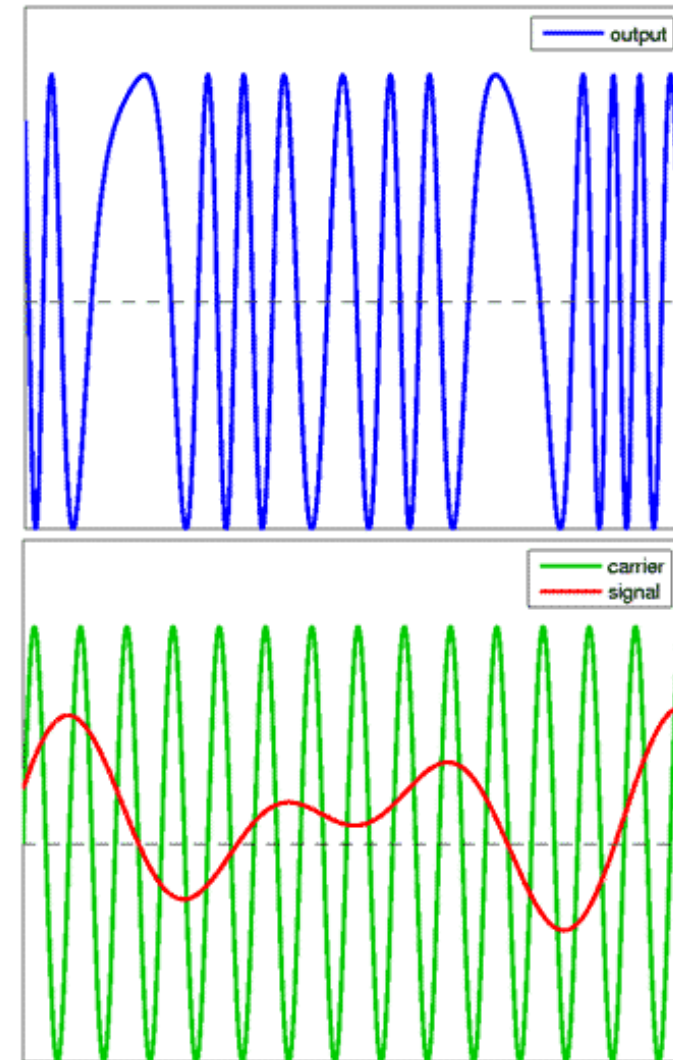


Figure from: http://en.wikipedia.org/wiki/Phase_modulation

Signal Equation – Fourier Interpretation

Thus, the demodulated signal from the slice selectively excited sample which was exposed to linear gradients in x- and y-direction is:

$$s(G_x, G_y, \tau_x, \tau_y) = \iint_{xy} m(x, y) \exp(-i\gamma G_x \tau_x x) \exp(-i\gamma G_y \tau_y y) dx dy$$

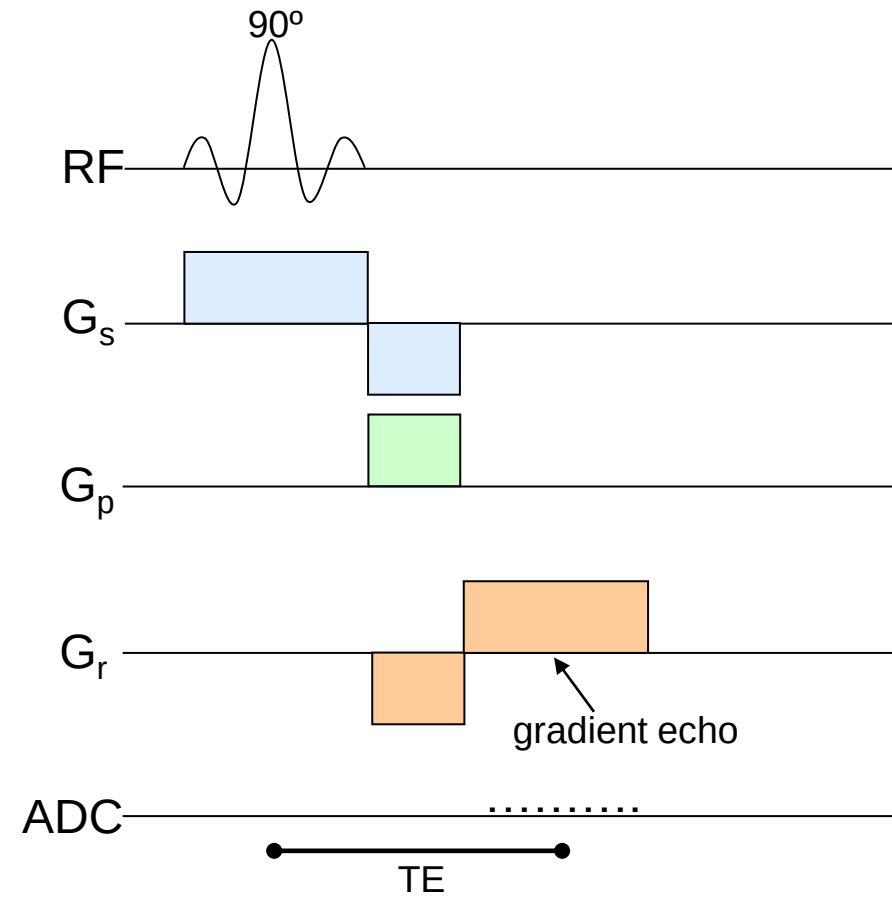
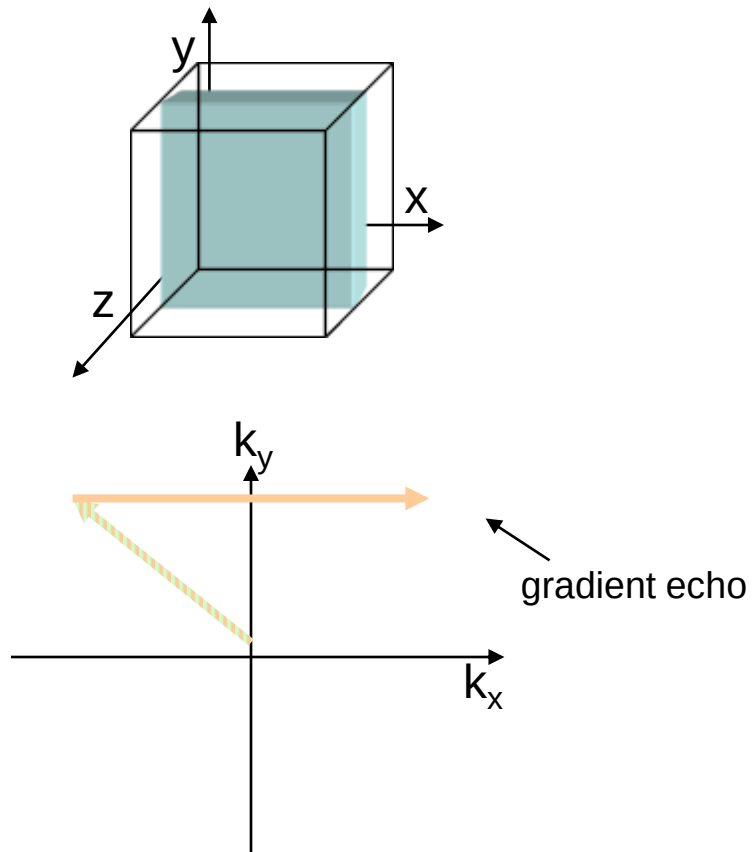
Our signal is the Fourier transform of the magnetisation!

Definition: $k_x(G_x, \tau_x) = \frac{\gamma}{2\pi} G_x \tau_x, \quad k_y(G_y, \tau_y) = \frac{\gamma}{2\pi} G_y \tau_y$



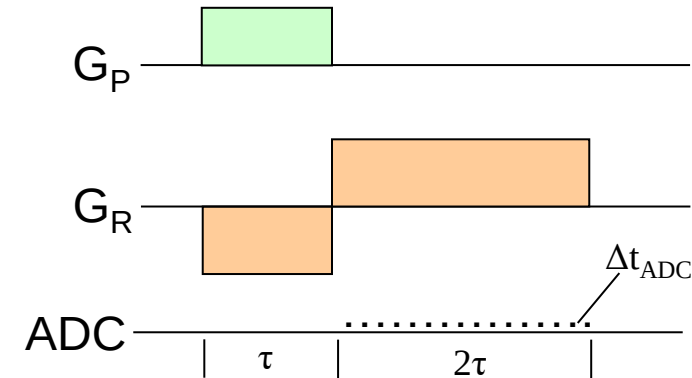
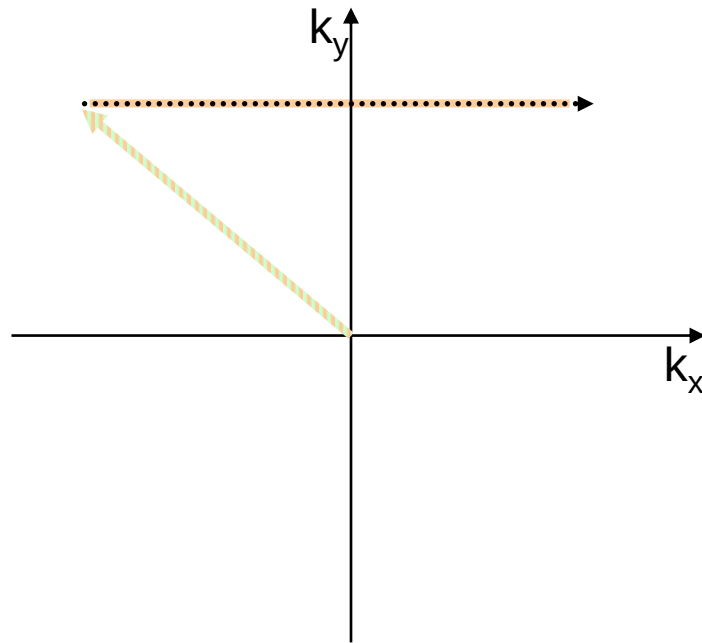
$$s(k_x, k_y) = \iint_{xy} m(x, y) \exp(-i2\pi k_x x) \exp(-i2\pi k_y y) dx dy$$

Gradient Echo MRI



k-space: Sampling

The combination of phase and frequency encoding we just learned about acquires **one line** of *k*-space:

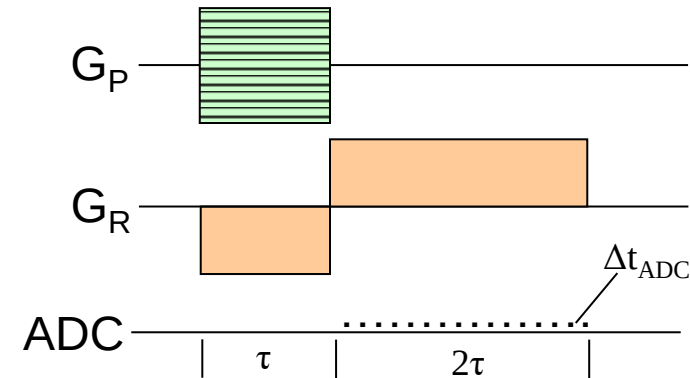
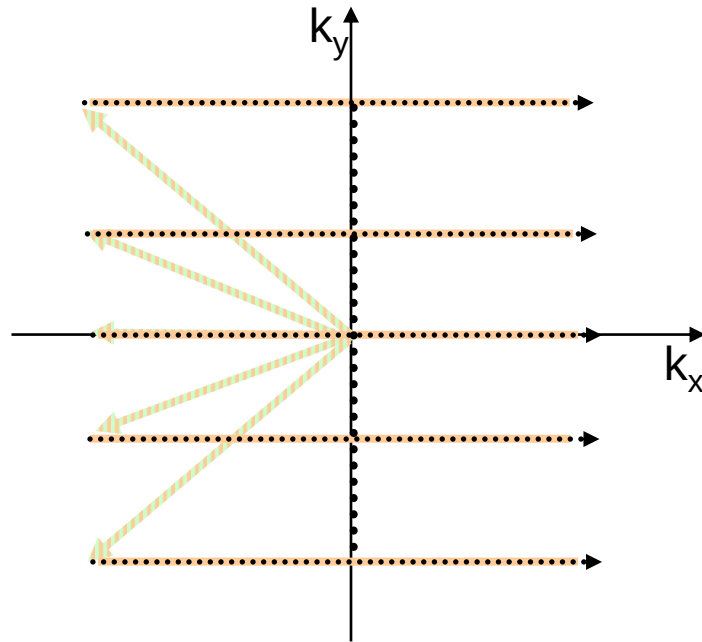


$$k_x = \frac{\gamma}{2\pi} (-G_x \tau + G_x \{0, \Delta t_{ADC}, 2\Delta t_{ADC}, \dots, 2\tau\})$$

$$k_y = \frac{\gamma}{2\pi} G_y \tau = \text{const.}$$

k-space: Sampling

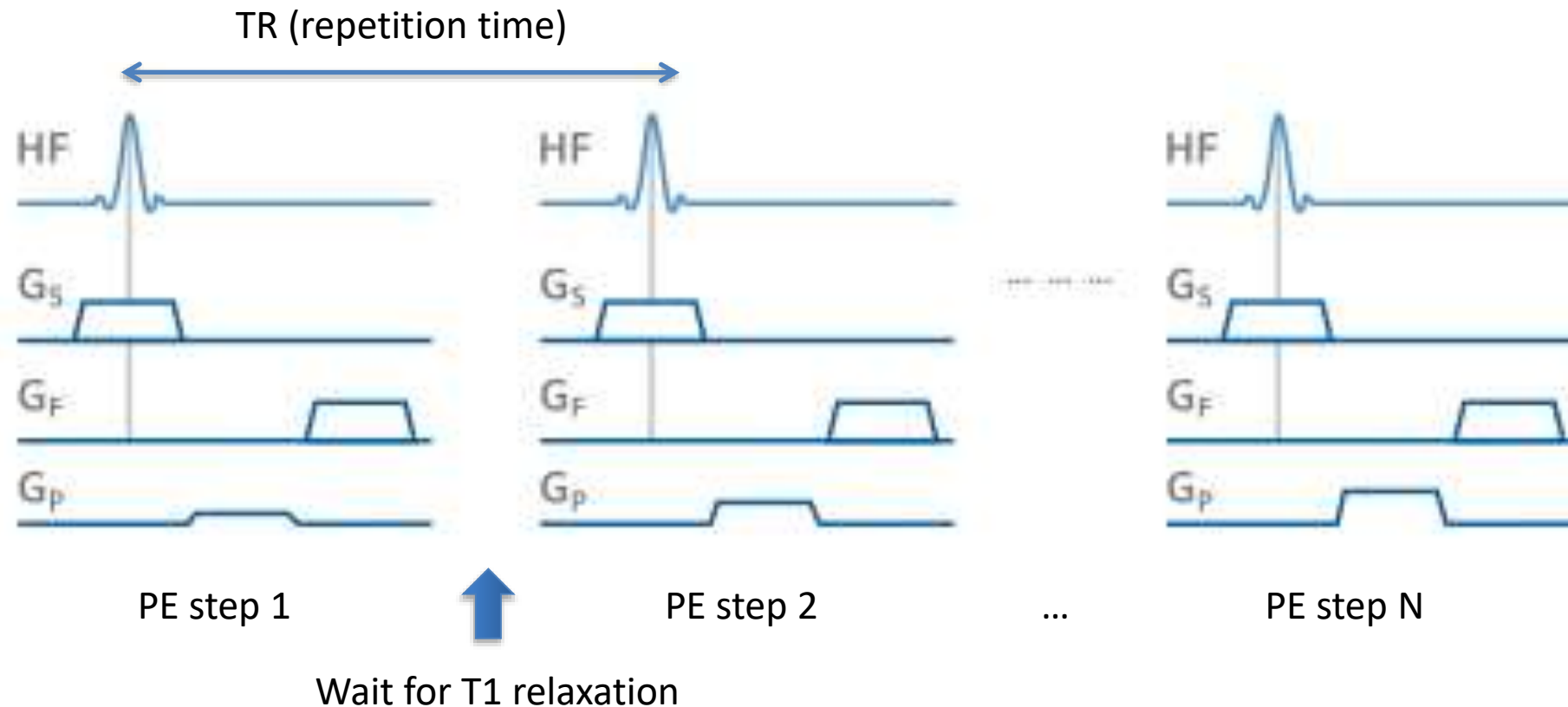
The sequence is repeated with different y-gradient amplitudes to acquire **a whole** 2-dimensional *k*-space.



$$k_x = \frac{\gamma}{2\pi} (-G_x \tau + G_x \{0, \Delta t_{ADC}, 2\Delta t_{ADC}, \dots, 2\tau\})$$

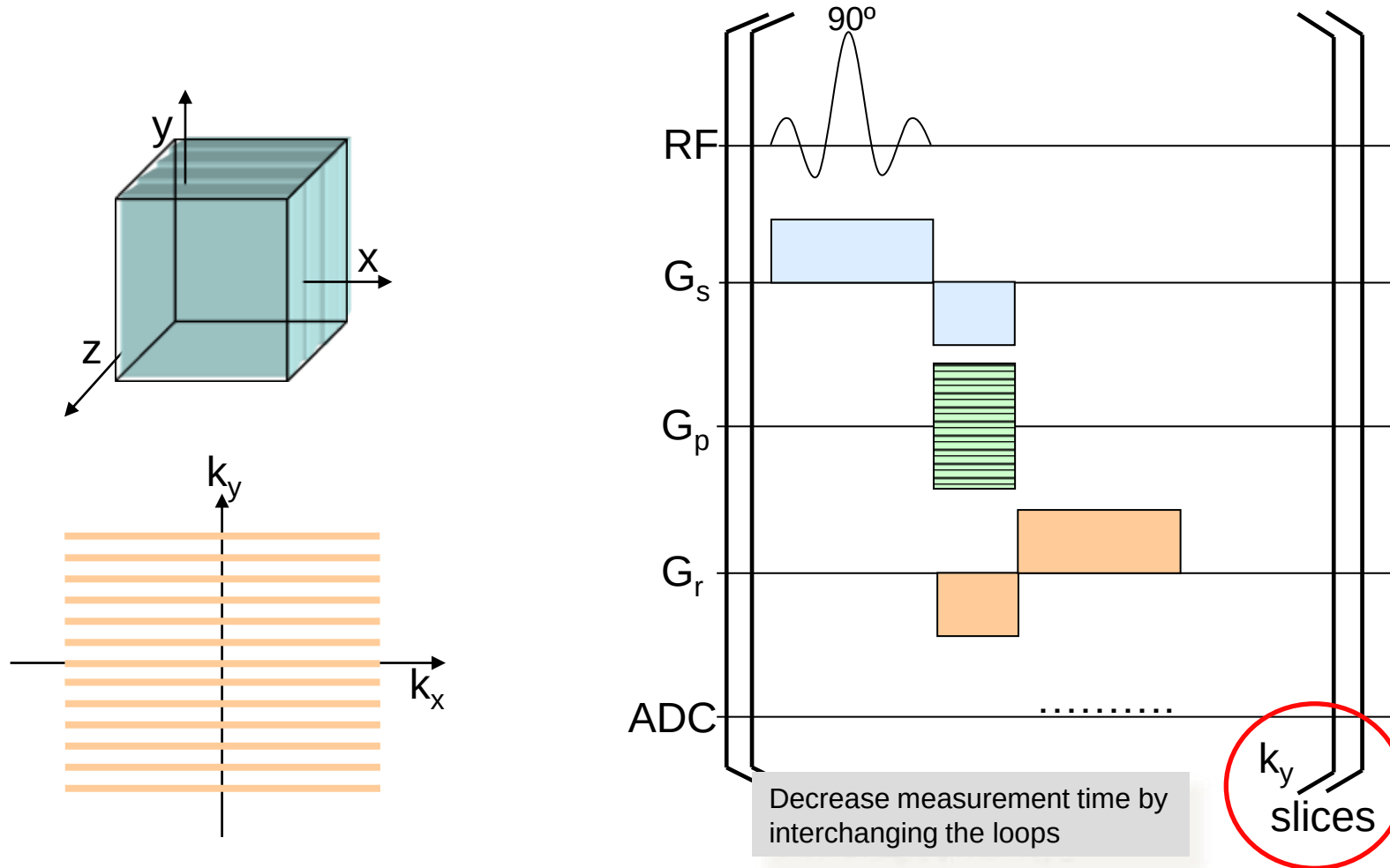
$$k_y = \frac{\gamma}{2\pi} \{G_{y,\max}, \dots, G_{y,\min}\} \tau$$

Timing for Phase and Frequency Encoding



The total acquisition time is given by the repetition time, TR, times the number of phase encode steps!

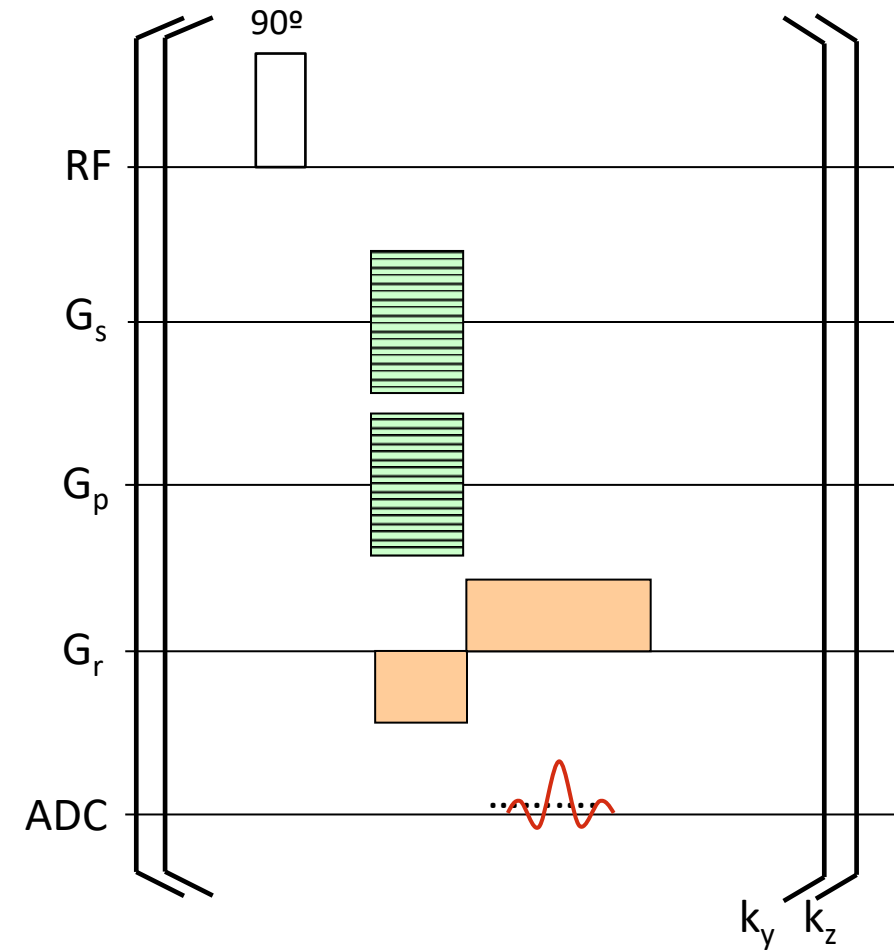
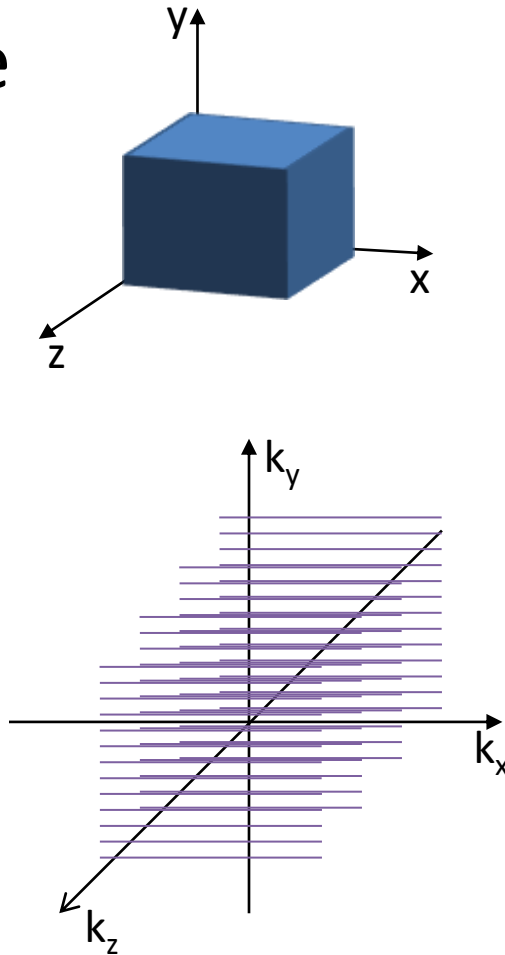
Gradient Echo with Optimized Acquisition Order



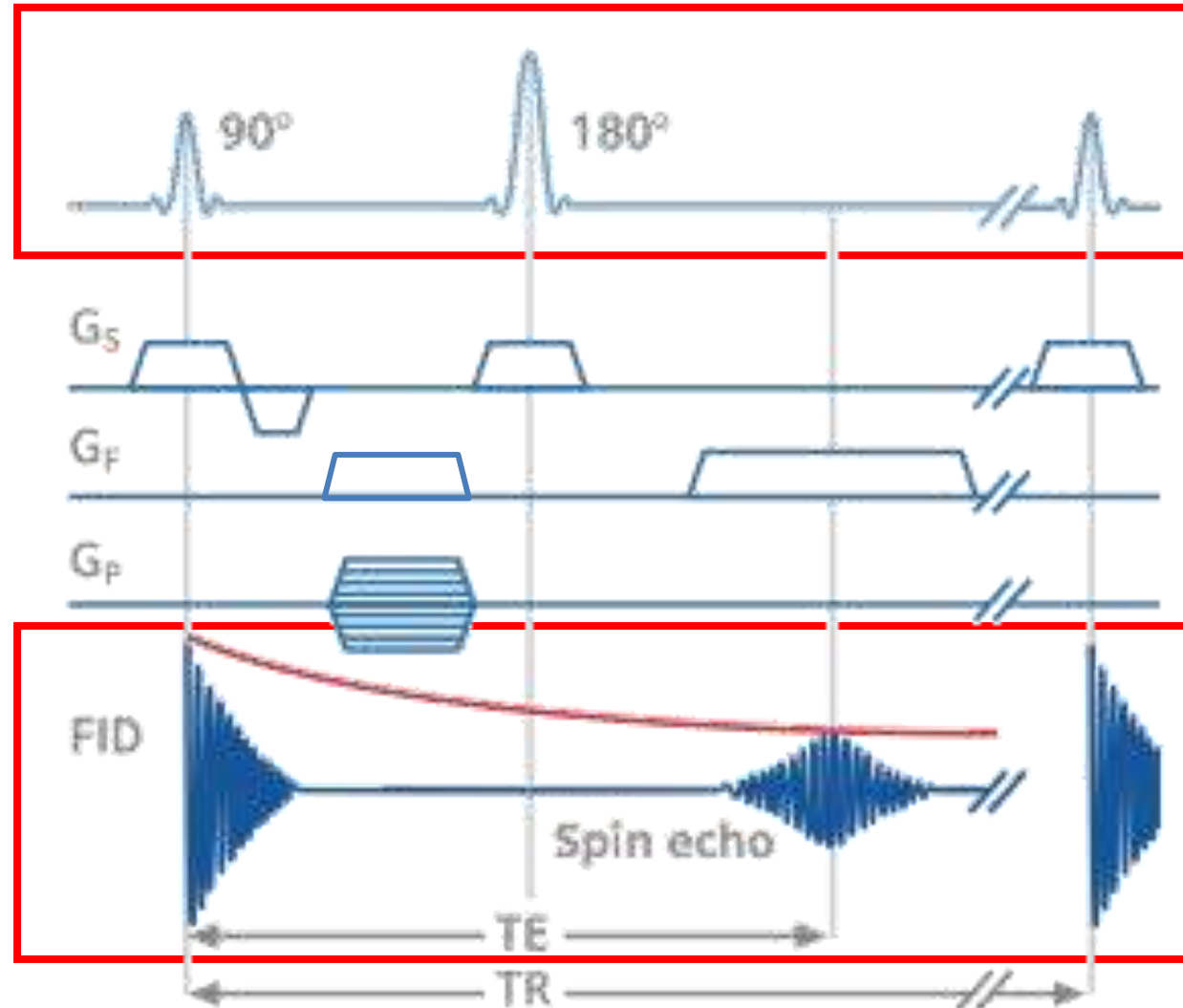
3D Gradient Echo Sequences

Alternative strategy for 3D imaging is to acquire a 3D k-space:

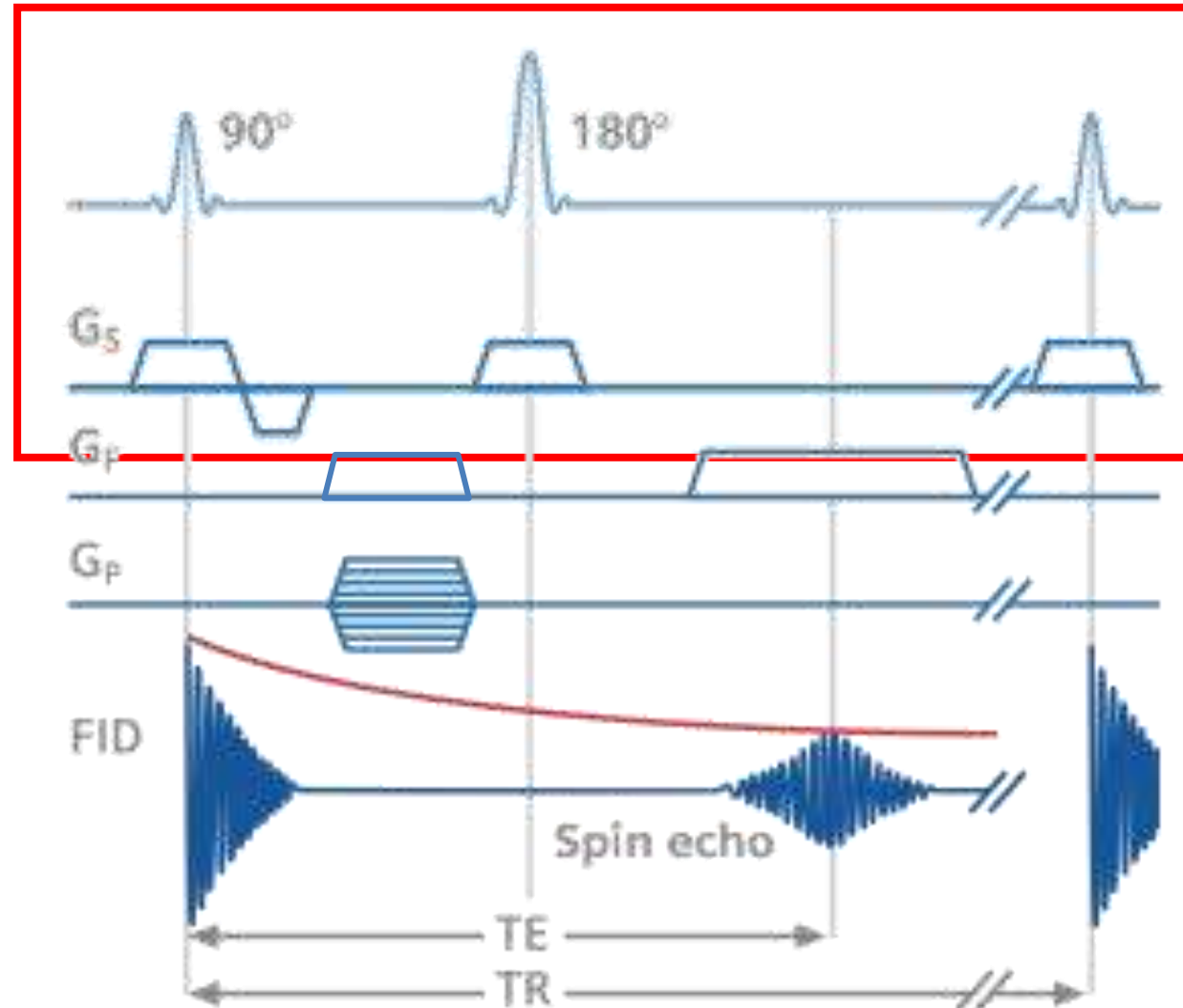
- Excite the whole volume
- Perform phase encoding in two dimensions



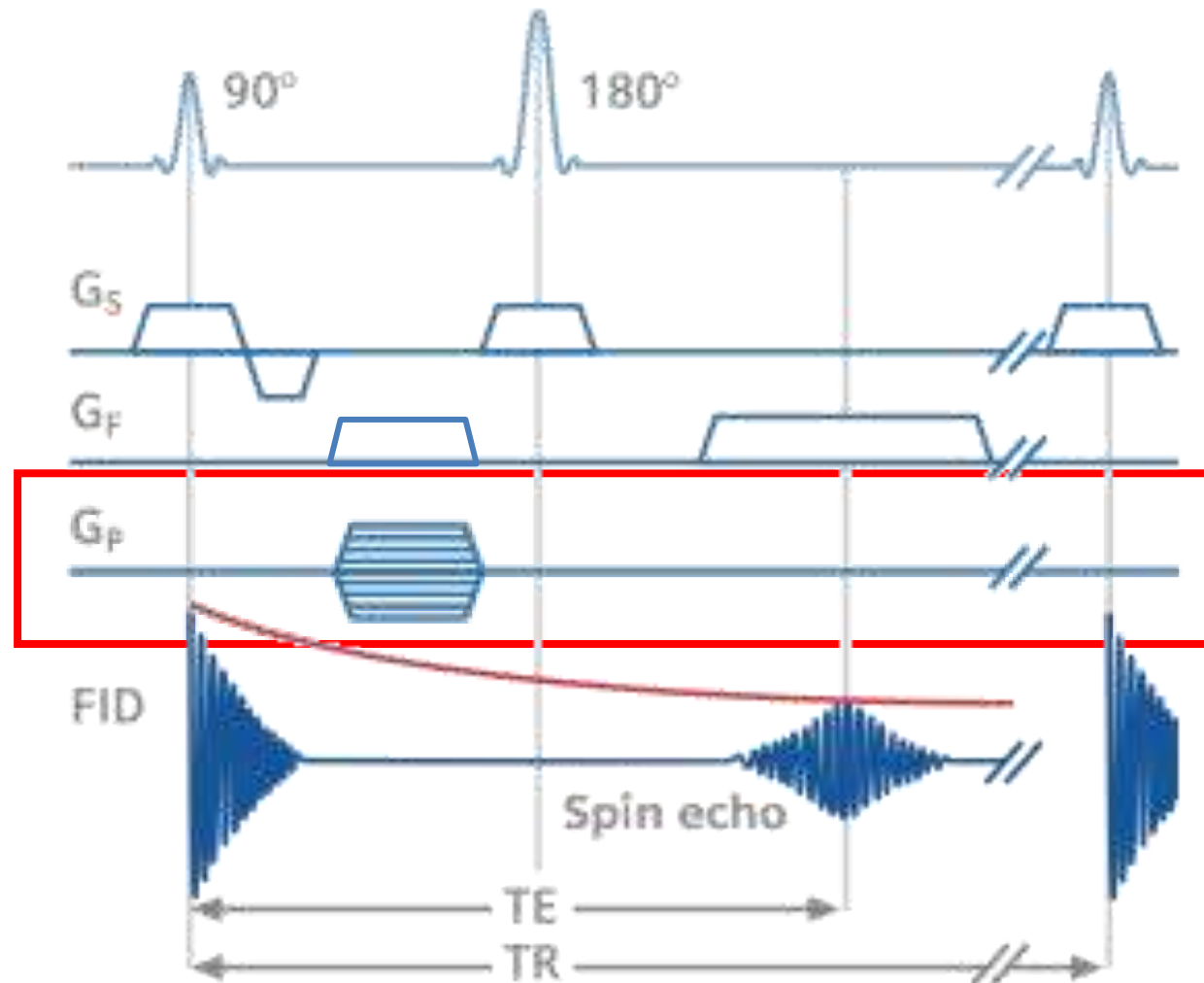
Spin Echo Sequence: Echo Generation



Spin Echo Sequence: Slice Selection



Spin Echo Sequence: Phase Encoding



Spin Echo Sequence: Frequency Encoding

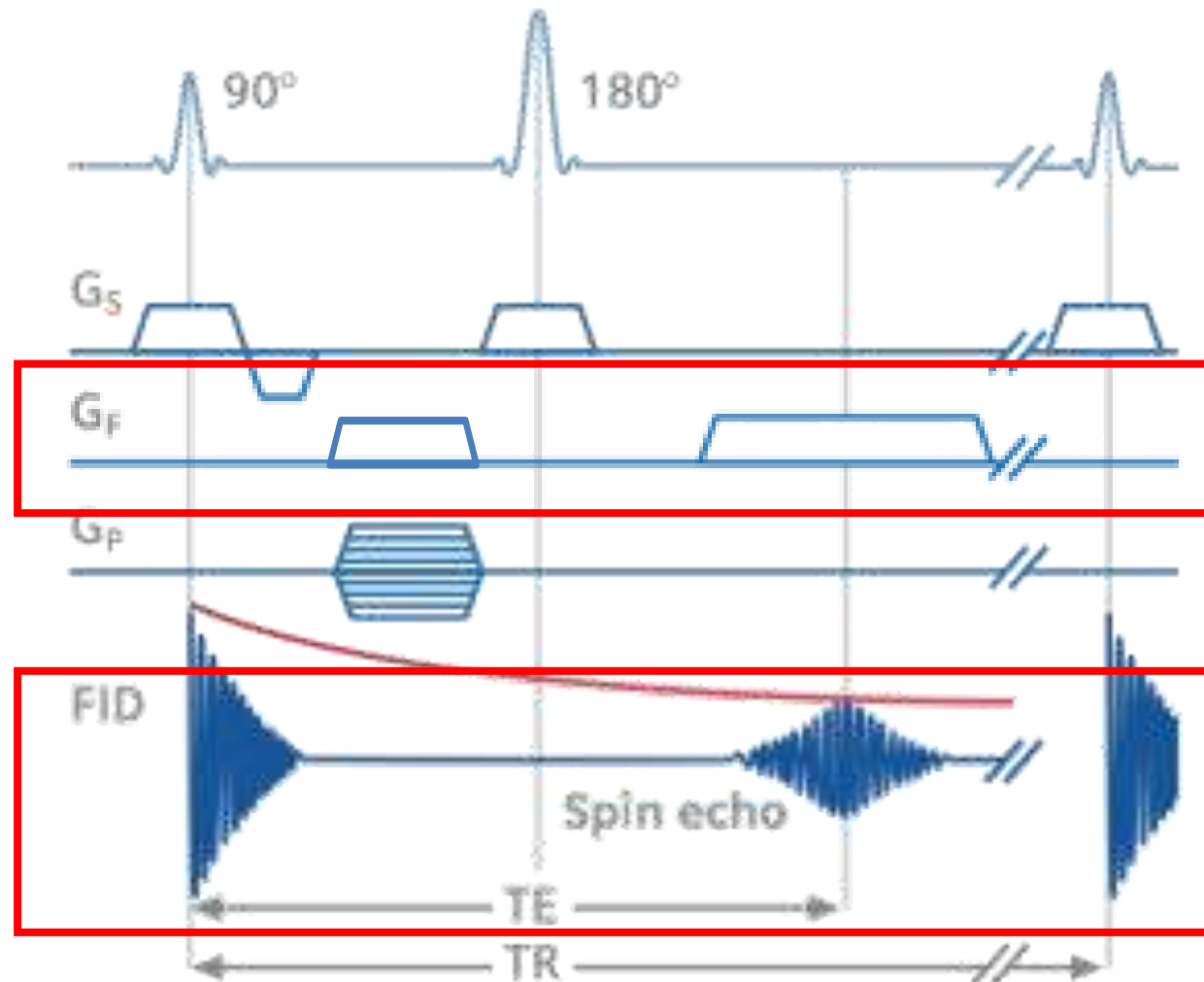
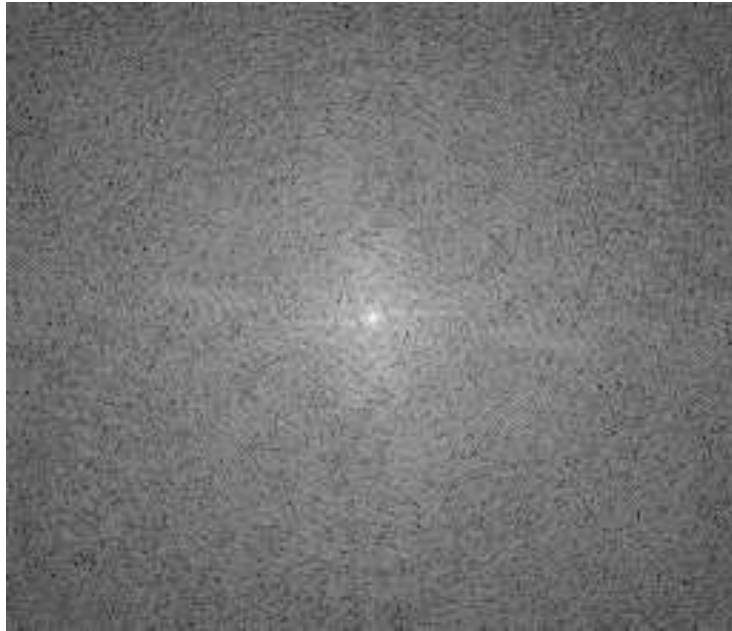


Image Reconstruction

Application of inverse 2D Fourier Transform:



k -space

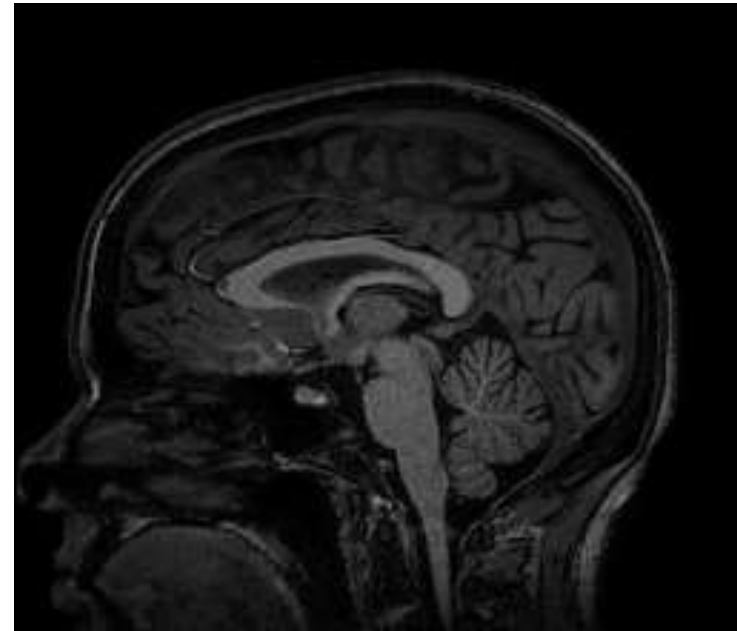
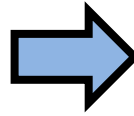
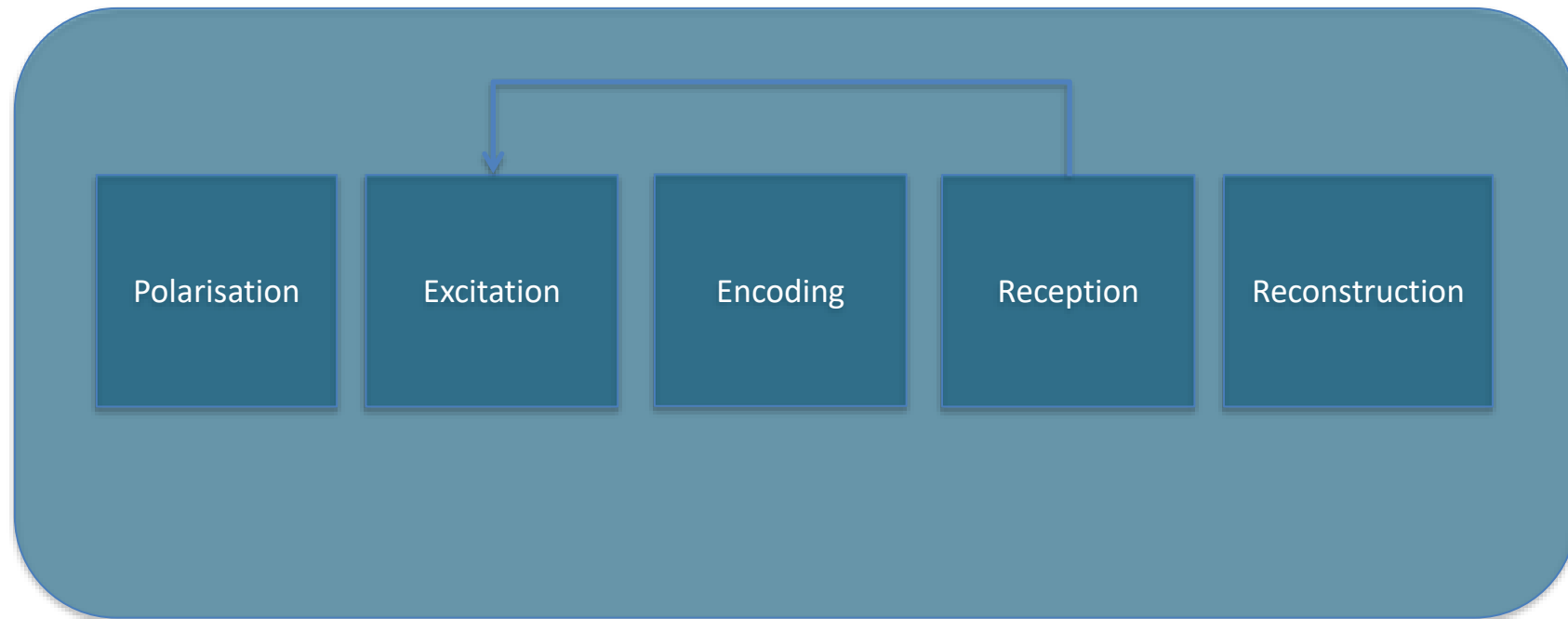


Image space

In theory, it should be enough to acquire half of k -space.
In practice, full sampling improves SNR.

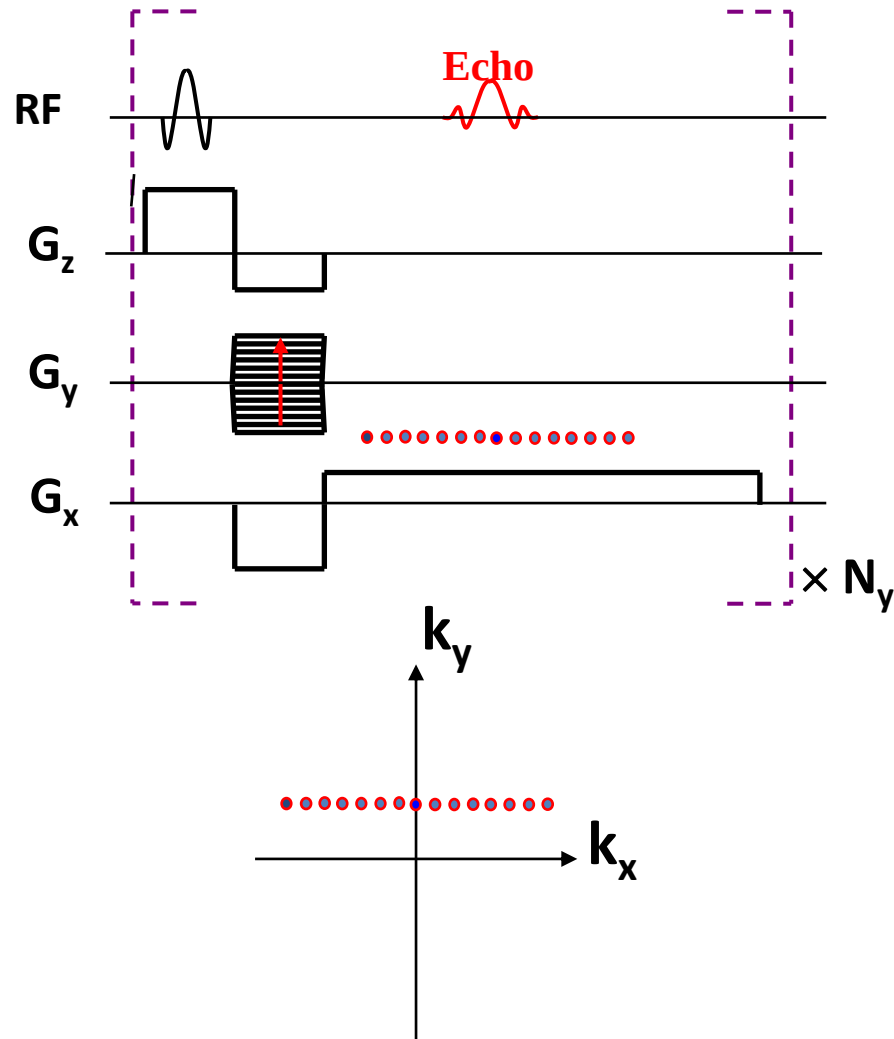
The Whole Picture



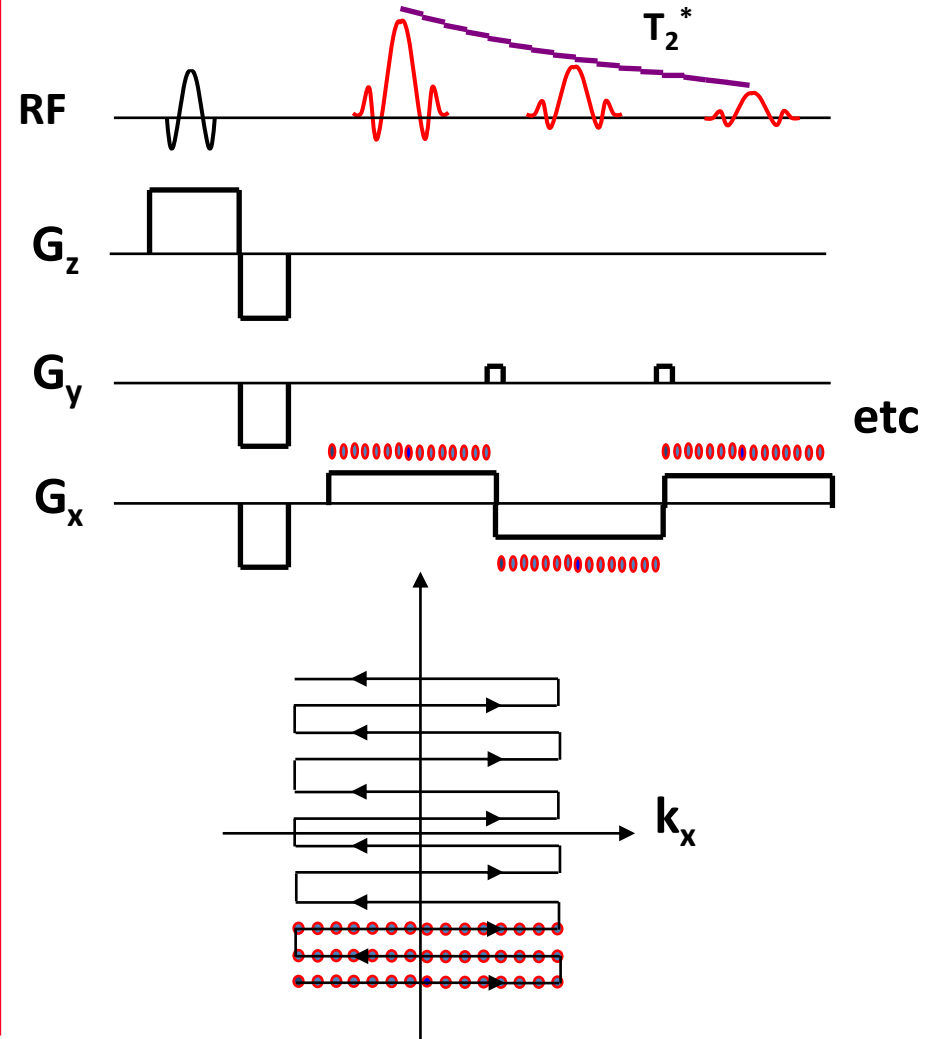
Steps of an MRI experiment

EPI: single shot imaging

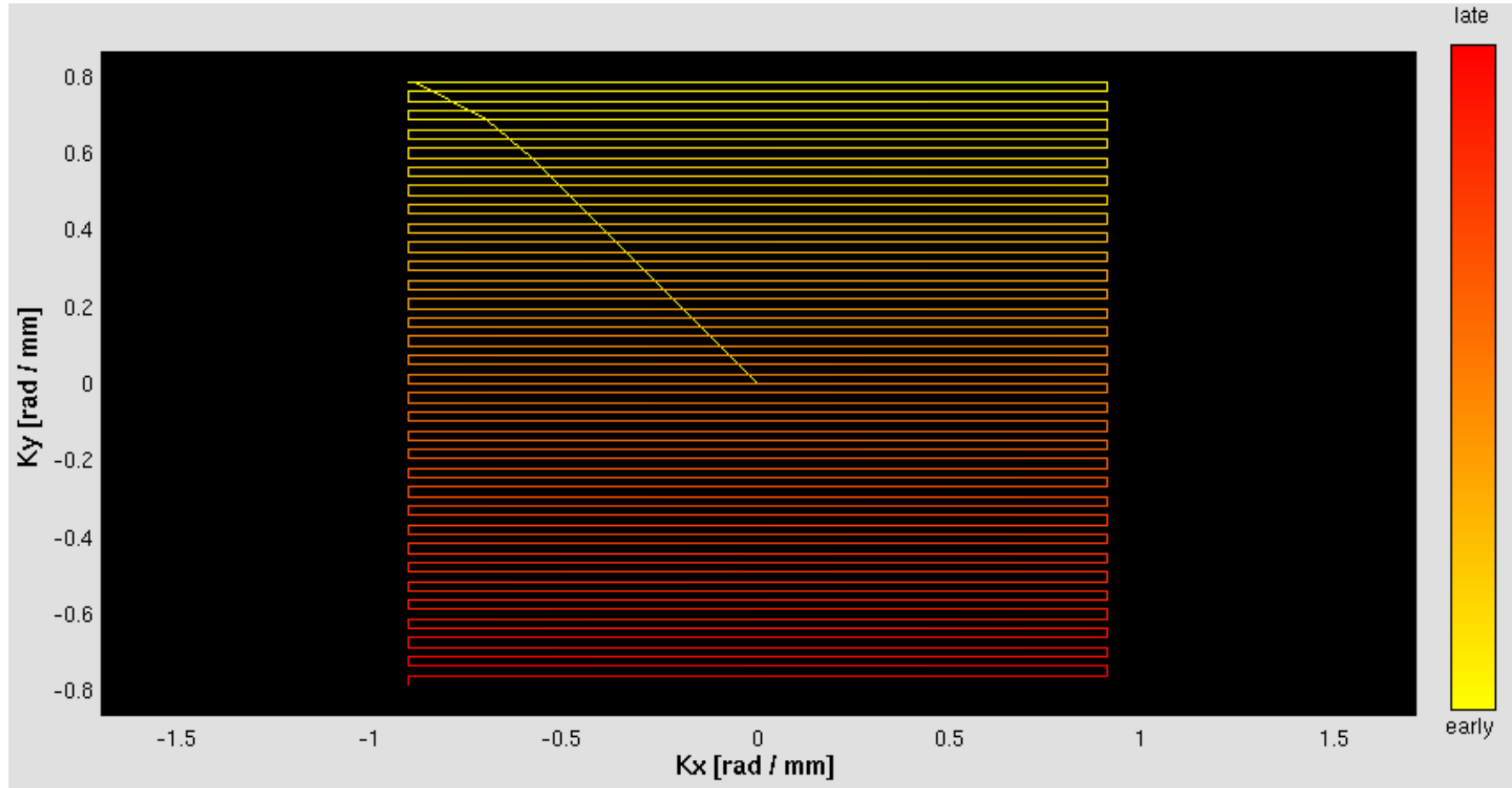
conventional MRI



Echo Planar Imaging

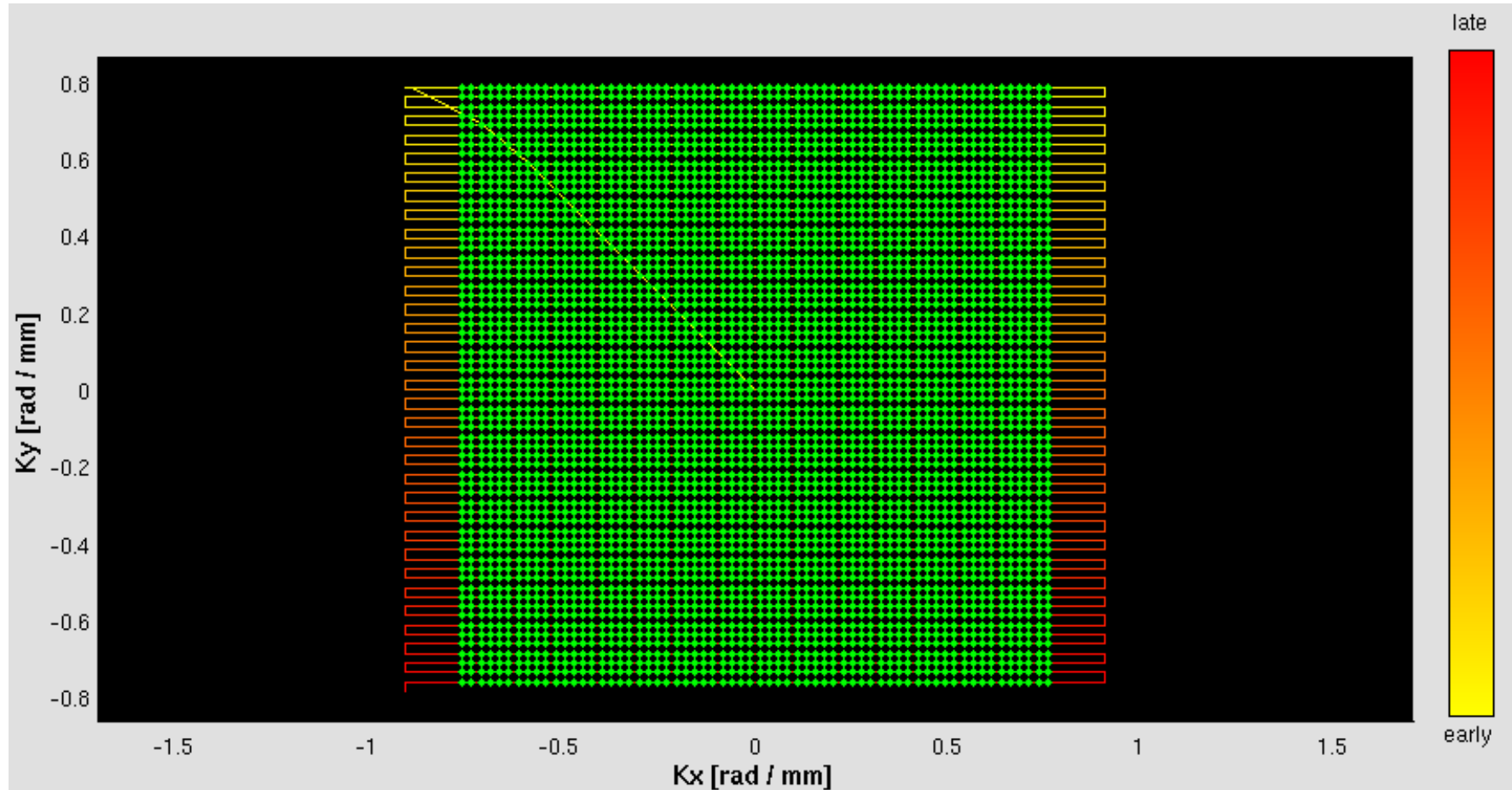


EPI (Echo-Planar-Imaging)



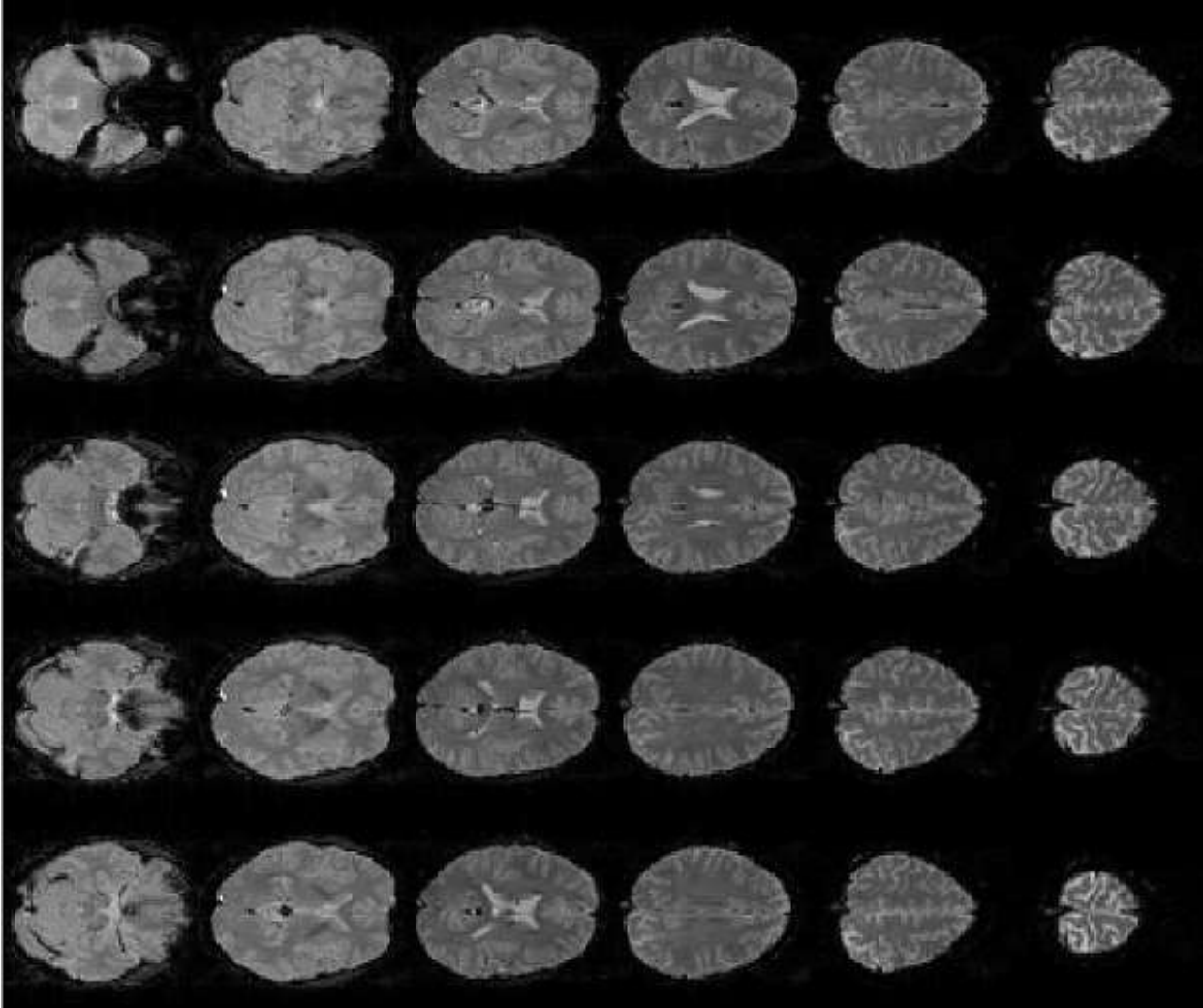
Single-shot EPI K-space trajectory

EPI (Echo-Planar-Imaging)



Single-shot EPI K-space trajectory + ADCs

EPI: workhorse for fMRI & dMRI



EPI with 30 slices,
each 3mm,
resolution of 64x64 pixels

= 122880 Voxel

T_2^* weighted

scan time $\approx$ 2 sec. !

Note on Oblique Slices

Logical and Physical Gradient System

The true physical orientation of the slice does not change the physics, and therefore we are dealing mainly with logical coordinates:

- Slice-Select Gradient: G_S
- Frequency-Encoding (Readout) Gradient: G_R
- Phase-Encoding Gradient: G_P

Transform from logical coordinate system to physical coordinate system with rotation matrices:

$$\begin{pmatrix} G_x \\ G_y \\ G_z \end{pmatrix} = \mathbf{R}_\phi(\alpha) \begin{pmatrix} G_R \\ G_P \\ G_S \end{pmatrix}$$

Often, $G_x=G_R$, $G_y=G_P$, $G_z=G_S$ is used in textbooks.

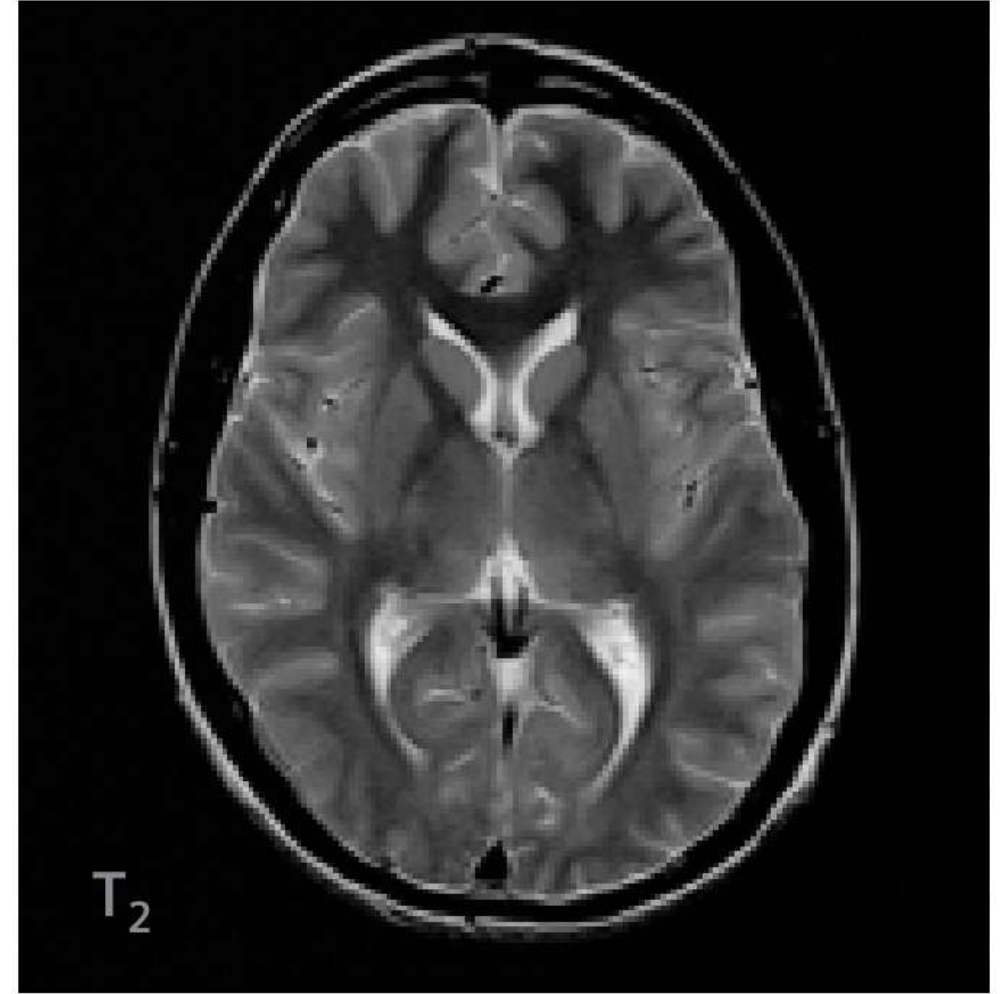
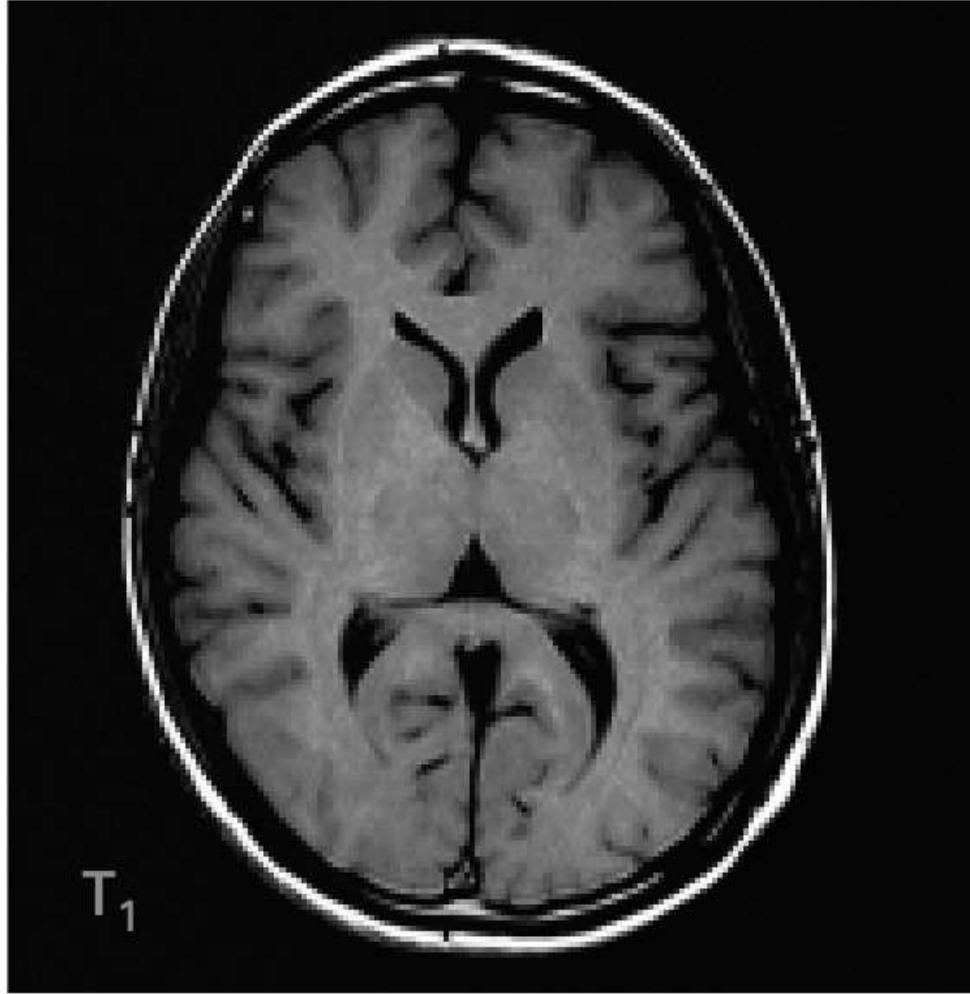
Summary

- MR Imaging uses **gradients** for a linear spatial variation of the main field
- Slice-wise imaging:
 - **Slice Selection**: Gradient applied *during excitation*
 - **Frequency Encoding**: Gradient pulse *during data acquisition*
 - **Phase Encoding**: Repeated measurements; each time, a linearly varying phase-shift is encoded *between excitation and acquisition*
 - **Echo Planar Imaging** acquires a full slice with a single excitation, at a reduced spatial resolution
- In 3D imaging, slice selection is replaced by 2D phase encoding

3.4 Different Contrasts in MRI

Based on slides by Prof. Tony Stöcker

Reminder: Multi-Contrast (T1/T2-weighting)



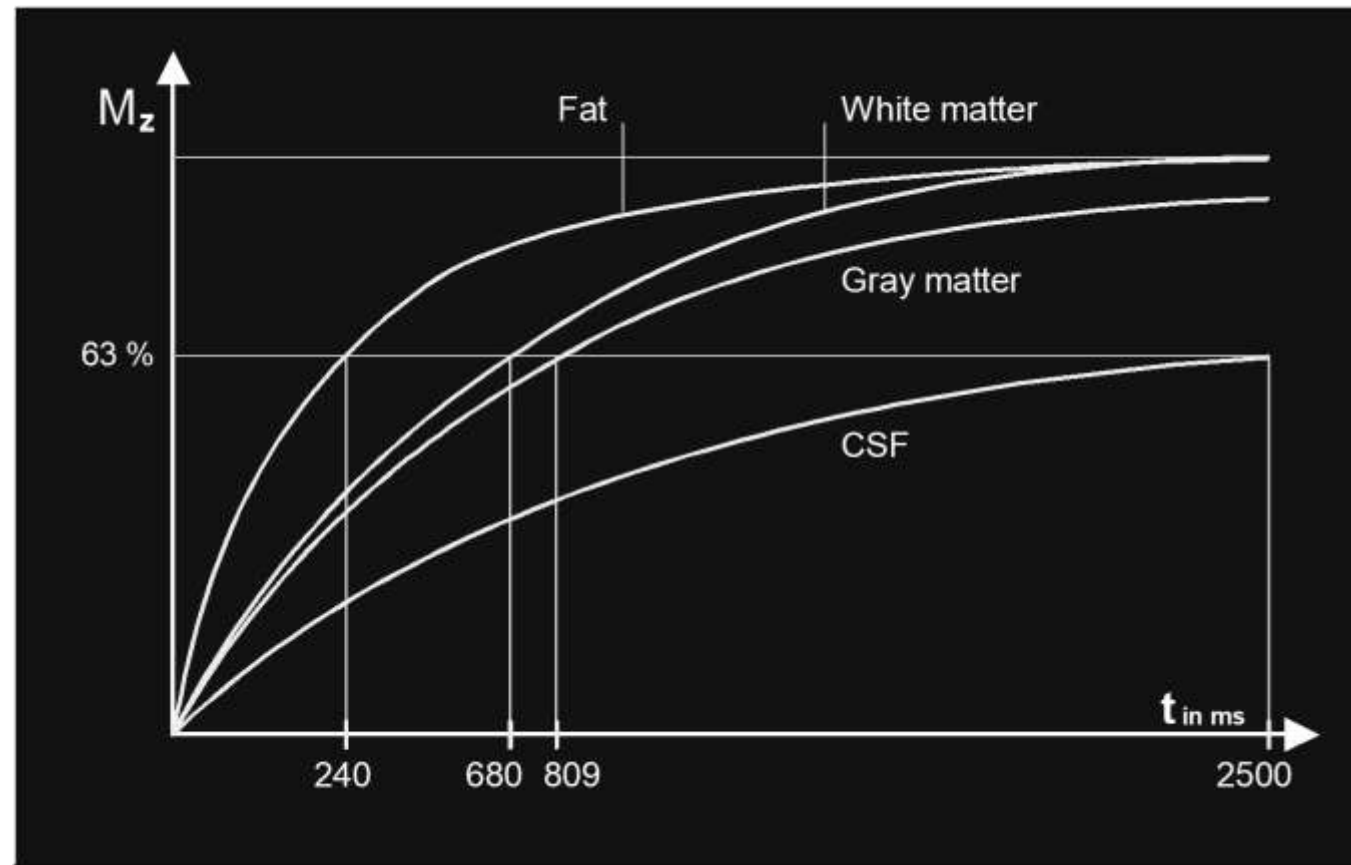
Signal Equation For Spin-Echo Sequence

$$S \propto \rho \exp\left(-\frac{TE}{T_2}\right) \left(1 - \exp\left(-\frac{TR}{T_1}\right)\right)$$

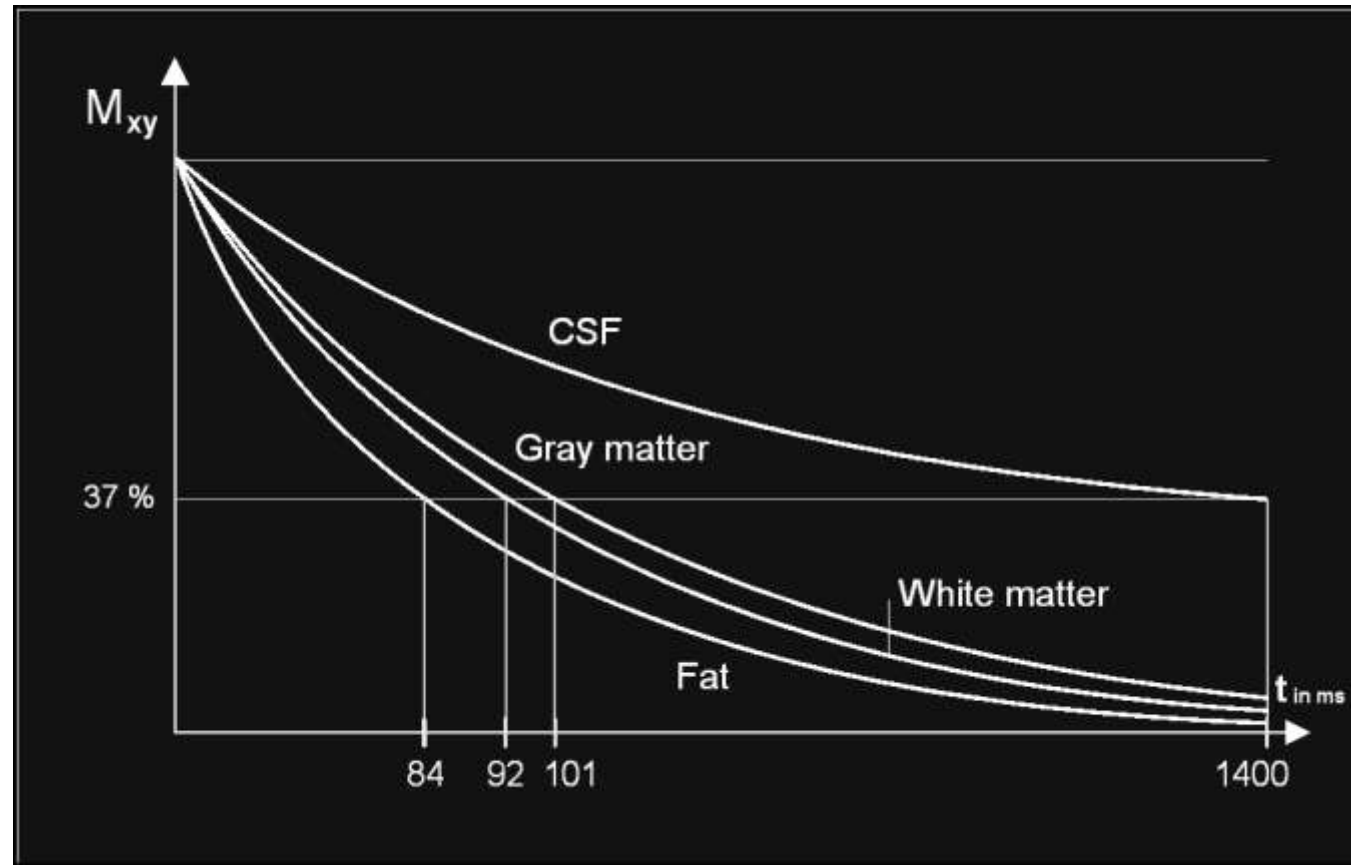
- $TE \rightarrow 0$ ($TR \approx T_1$)
 $S \propto \rho \cdot 1 \cdot \left(1 - \exp\left(-\frac{TR}{T_1}\right)\right) \Rightarrow S = S(\rho, T_1)$
- $TR \rightarrow \infty$ ($TE \approx T_2$)
 $S \propto \rho \cdot \exp\left(-\frac{TE}{T_2}\right) \cdot (1 - 0) \Rightarrow S = S(\rho, T_2)$
- $TR \rightarrow \infty, TE \rightarrow 0$
 $S \propto \rho \cdot 1 \cdot (1 - 0) \Rightarrow S = S(\rho)$

Same for gradient echo with 90° flip angle, if you replace T2 with T2*!

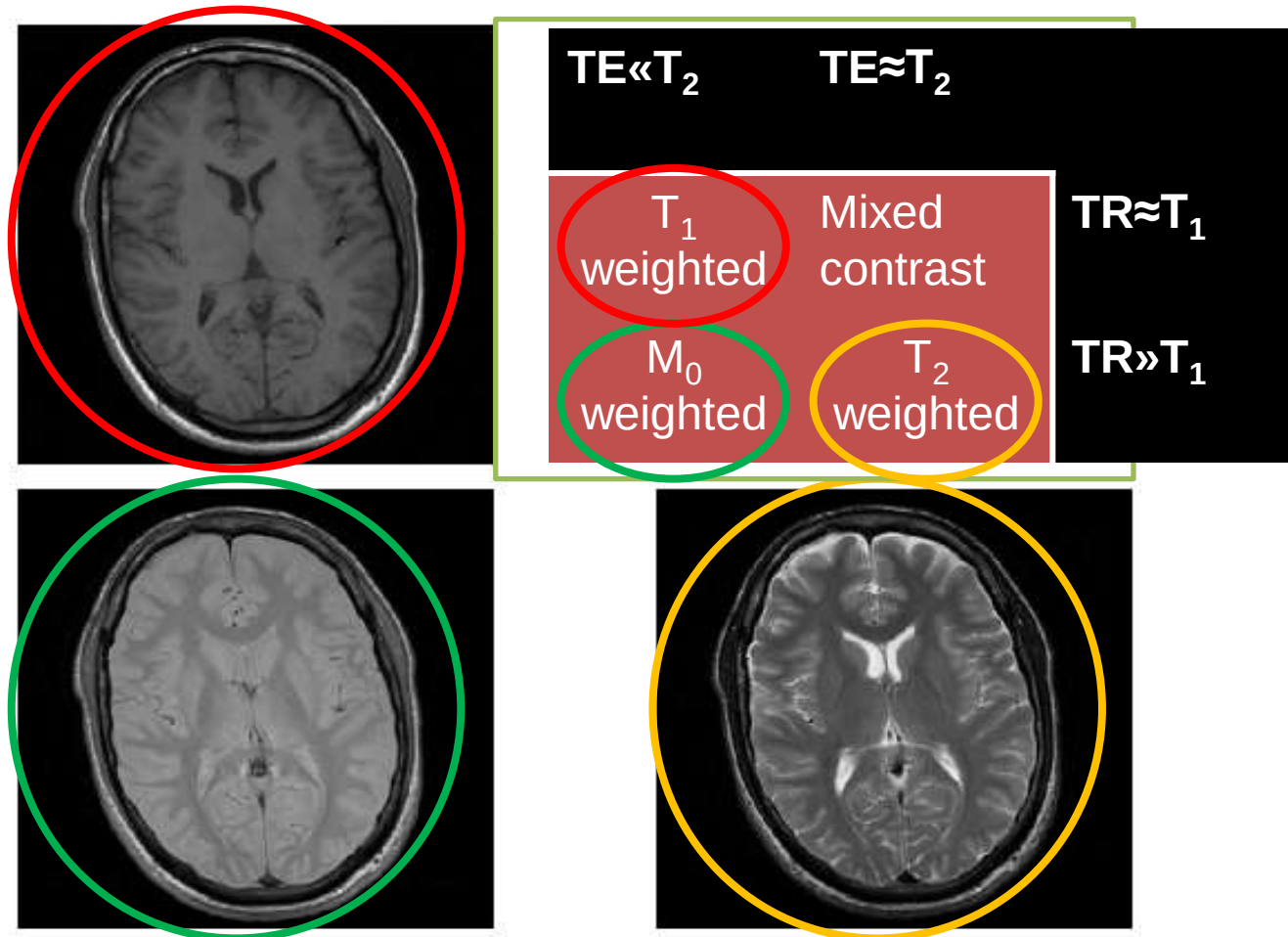
T_1 relaxation in the brain at 1T



T_2 relaxation in the brain at 1T

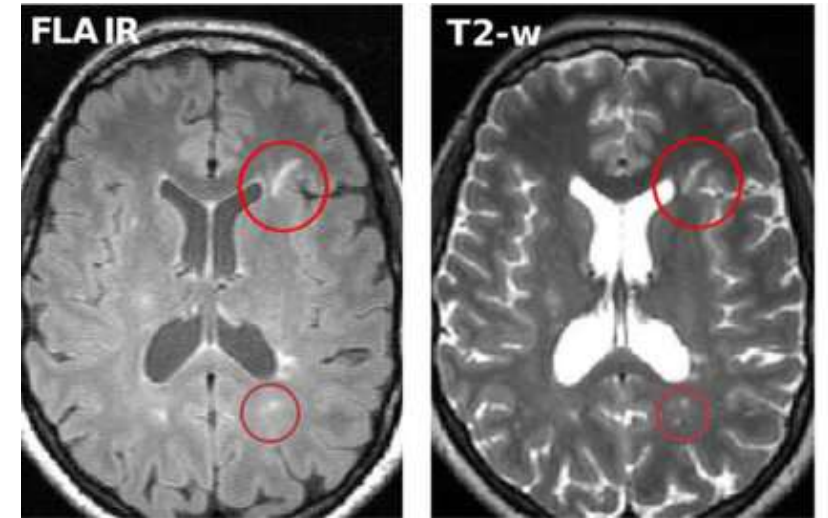


In vivo examples: different contrasts

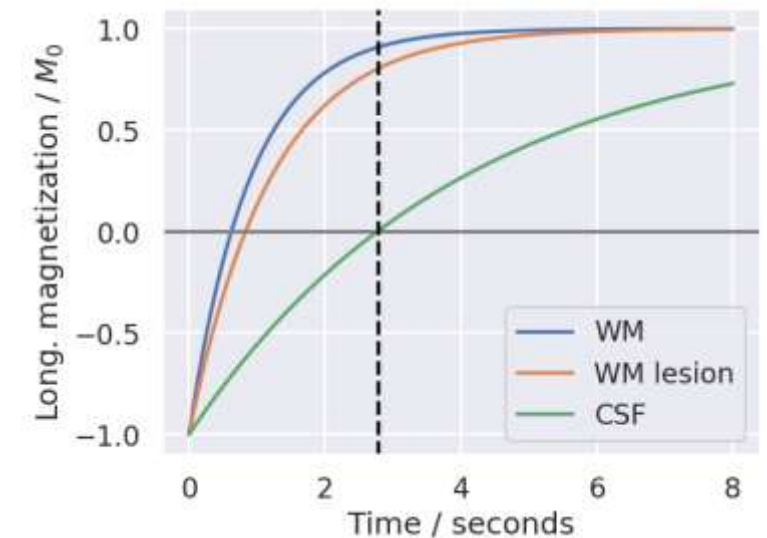


Fluid Attenuation Inversion Recovery (FLAIR)

- **Problem:** In T2-w images, white matter lesions are bright, but difficult to distinguish from cerebrospinal fluid
- **FLAIR** eliminates signal contributions from CSF by preceding a T2-weighted MR sequence by a 180° inversion pulse
 - Longitudinal magnetization at time t :
$$M_z(t) = M_0(1 - 2e^{-\frac{t}{T_1}})$$
 - Thus, a 90° pulse at $TI = \ln(2) \times T_{1,CSF}$ will not excite spins in the CSF



[Ma et al. JBHI 2022]



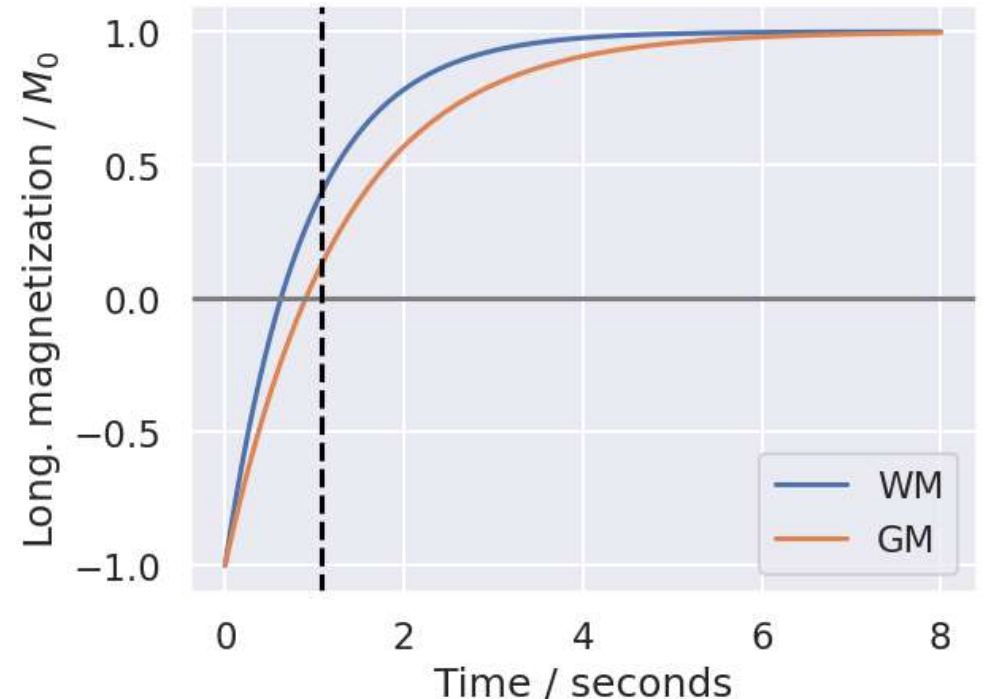
Magnetization Preparation for Optimal Contrast

- Inversion pulses can also be timed to optimize contrast between specific tissues, e.g., with $T_{1,WM}$ and $T_{1,GM}$
- Consider the contrast as the difference in resulting M_z :

$$C(t) = M_{z,WM}(t) - M_{z,GM}(t)$$

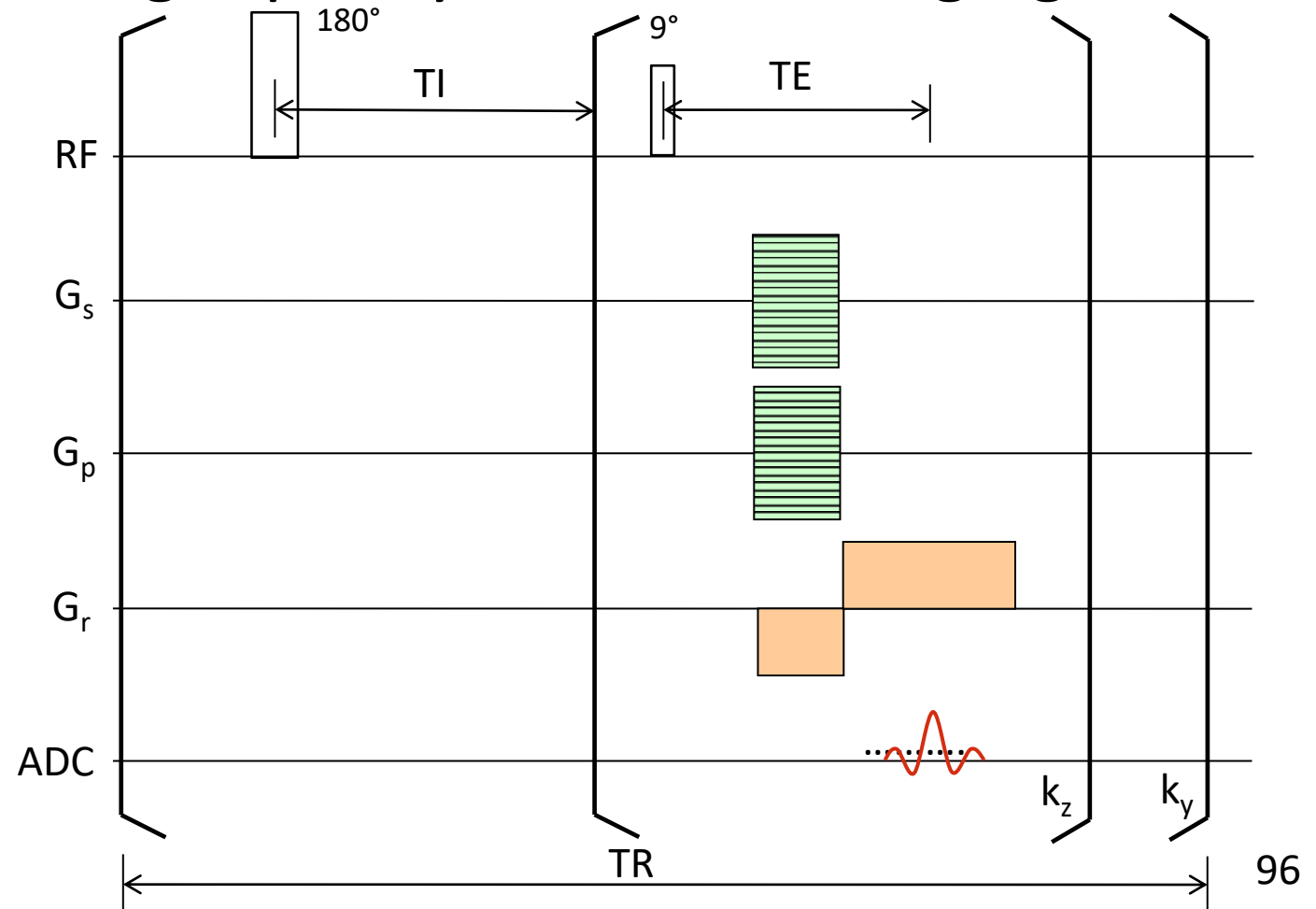
- Setting $\frac{d}{dt} C(t) := 0$ yields

$$TI = - \frac{\ln \frac{T_{1,WM}}{T_{1,GM}}}{\frac{1}{T_{1,WM}} - \frac{1}{T_{1,GM}}}$$



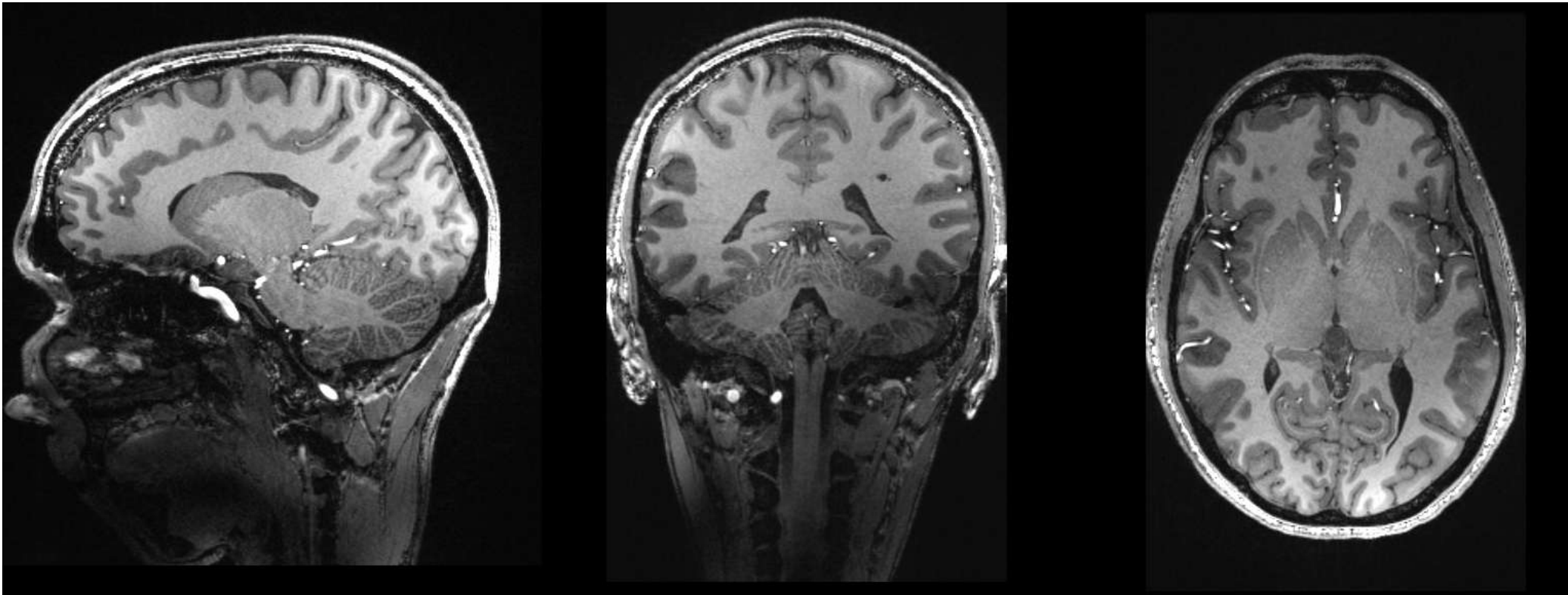
MP-RAGE

- **Magnetization Prepared Rapid Gradient Echo (MP-RAGE)** sequence is widely used for high quality structural imaging
- It combines
 - Inversion pulse for magnetization preparation (see previous slide)
 - 3D Rapid Gradient Echo readout
 - Gradient recalled echoes are “spoiled” to remove signal residuals



MP-RAGE: Example

- *Example:* MP-RAGE aquired at 7 Tesla (Tony Stöcker / DZNE)
 - 256x256x160 mm Field-of-View, 0.7x0.7x0.7 mm³ resolution
 - TE / TR / TI = 3 / 2250 / 900 ms
 - 160 phase-encode steps per TR



Summary

- **Different contrasts in MRI** are achieved by varying
 - Echo time: Short TE reduces effect of T_2 / T_2^* relaxation
 - Repetition time: Long TR reduces effect of T_1 relaxation
 - Inversion time: Initial inversion pulse with suitable TI can
 - eliminate specific contributions (e.g., zero out CSF in FLAIR)
 - optimize contrast between specific tissues (e.g., GM/WM in MP-RAGE)
- Details of the **signal readout** also affect the contrast
 - T_2 weighting with spin echo vs. T_2^* weighting with gradient echo

3.5 Image Artifacts in MRI

Based on slides by Prof. Tony Stöcker

Real world imaging... artefacts

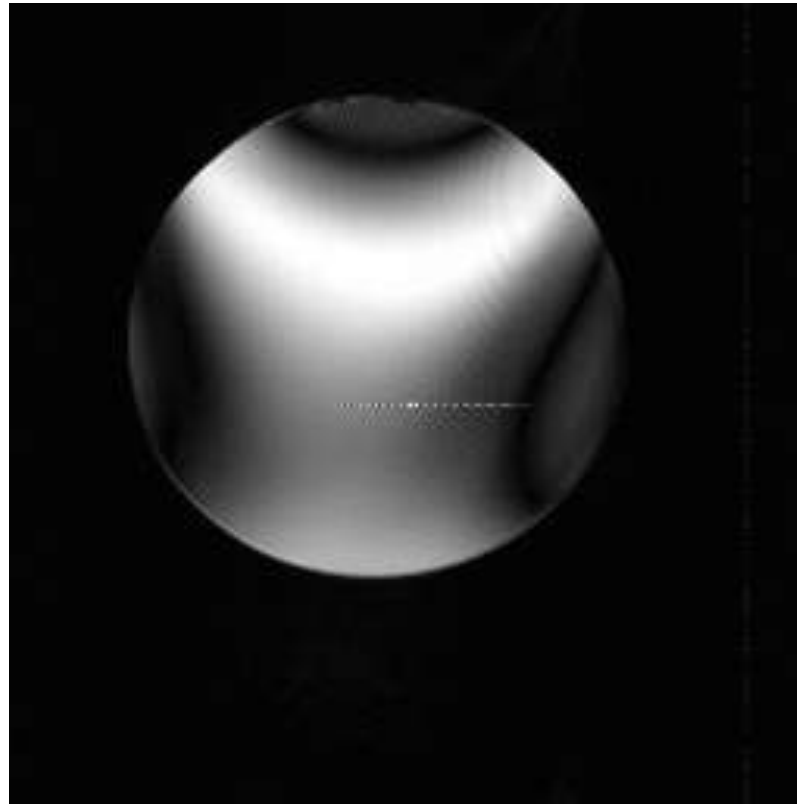
Artefact sources:

- Hardware (magnet, coils, gradients) imperfections
- Eddy currents caused by gradients
- Inhomogeneous B_0 caused by patient
- Human body is not rigid (motion, flow)
- Limits of the sequence itself (PSF, etc.)

→ influence on MR image

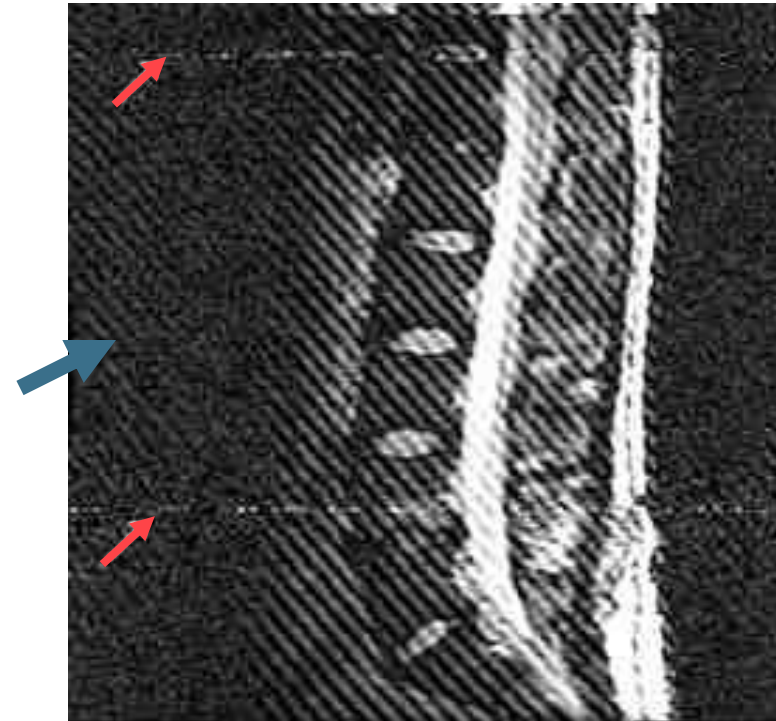
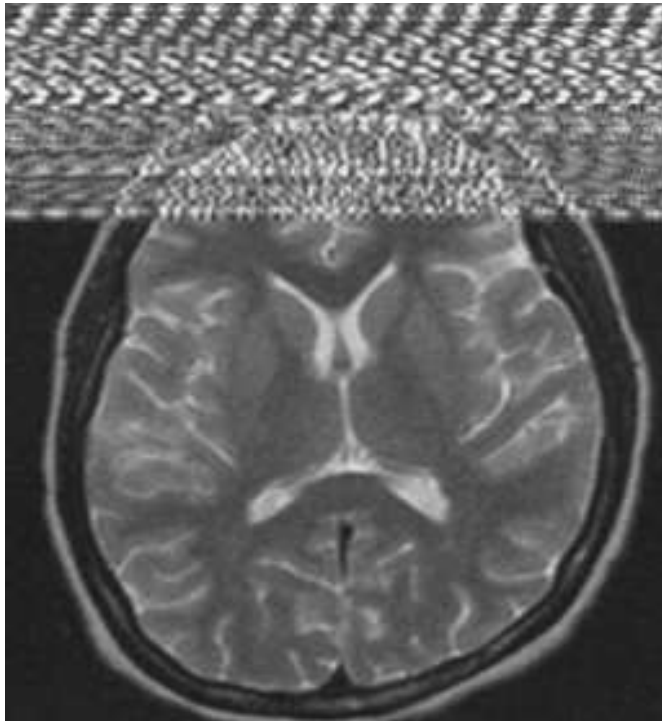
Inhomogenous Field

E.g. mis-set shim currents



RF Noise

- Spurious RF signals inside the magnet room.
- E.g. from break-down of the shielding, or electric devices inside the room.
- Spikes (single peak at one time-point of the acquisition)
- Zipper (continuous RF interference)



Gibbs Ringing



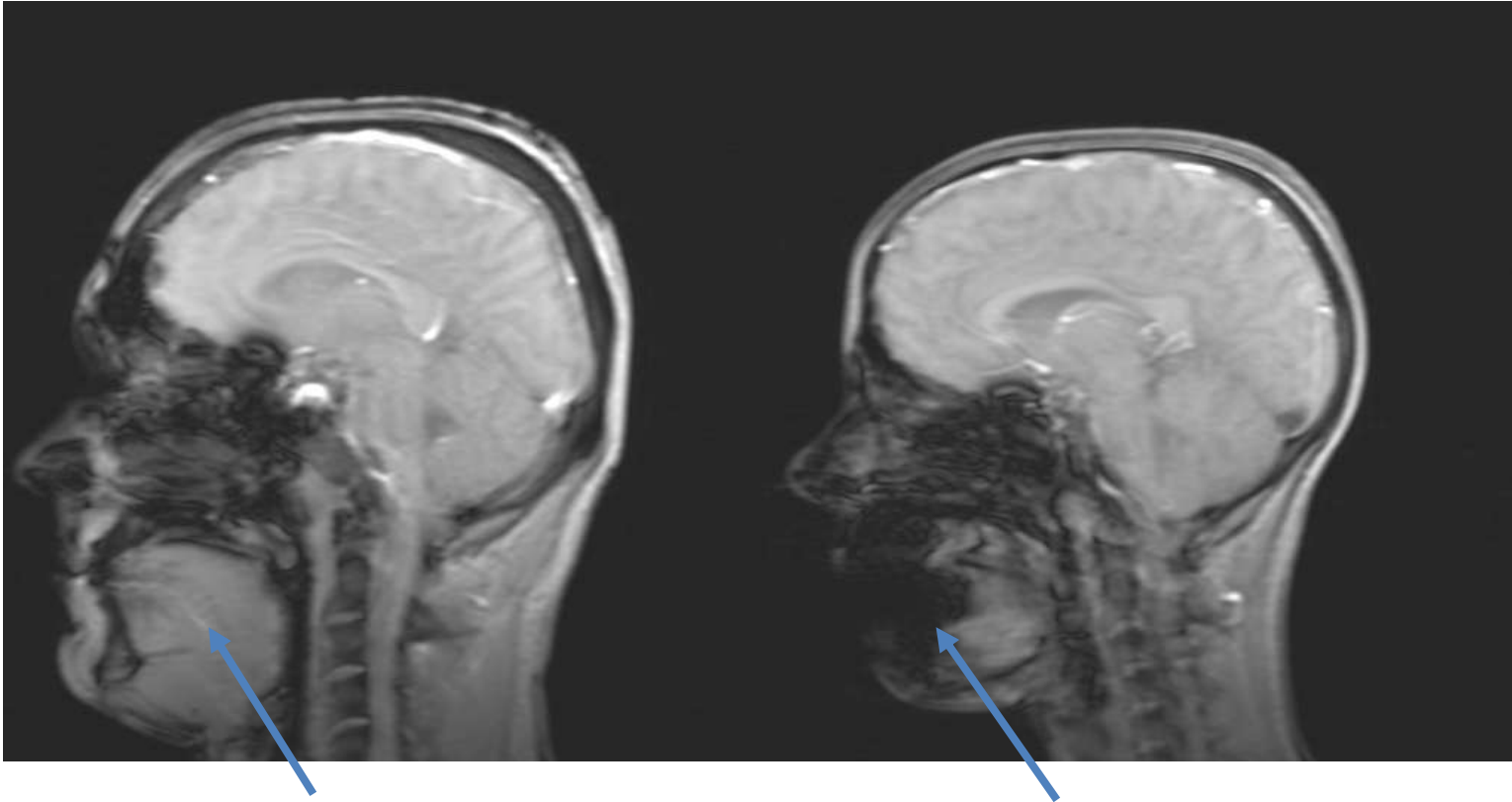
Truncation artifact due to finite sampling of k-space signal.
(Box-car transforms to sinc function!)

Can be suppressed by acquiring more lines and/or digital filtering.

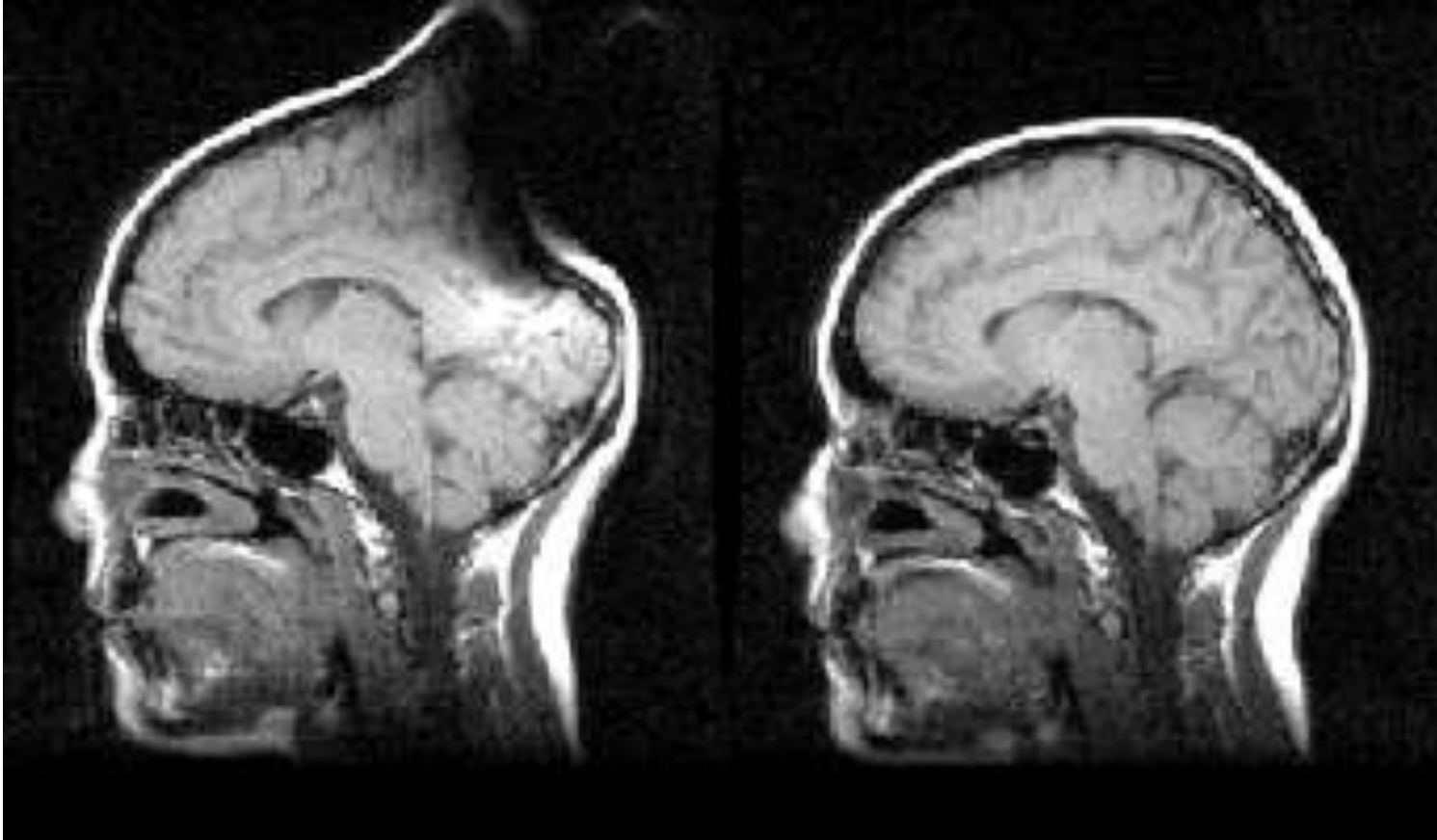
Metal artefacts

- Metals strongly distort the static magnetic field
- Therefore T_2^* is extremely shortened
- Possible consequences for imaging:
 - Image distortions
 - Signal dropout
- Remedies:
 - Remove metal
 - Shimming (but: might not be sufficient)

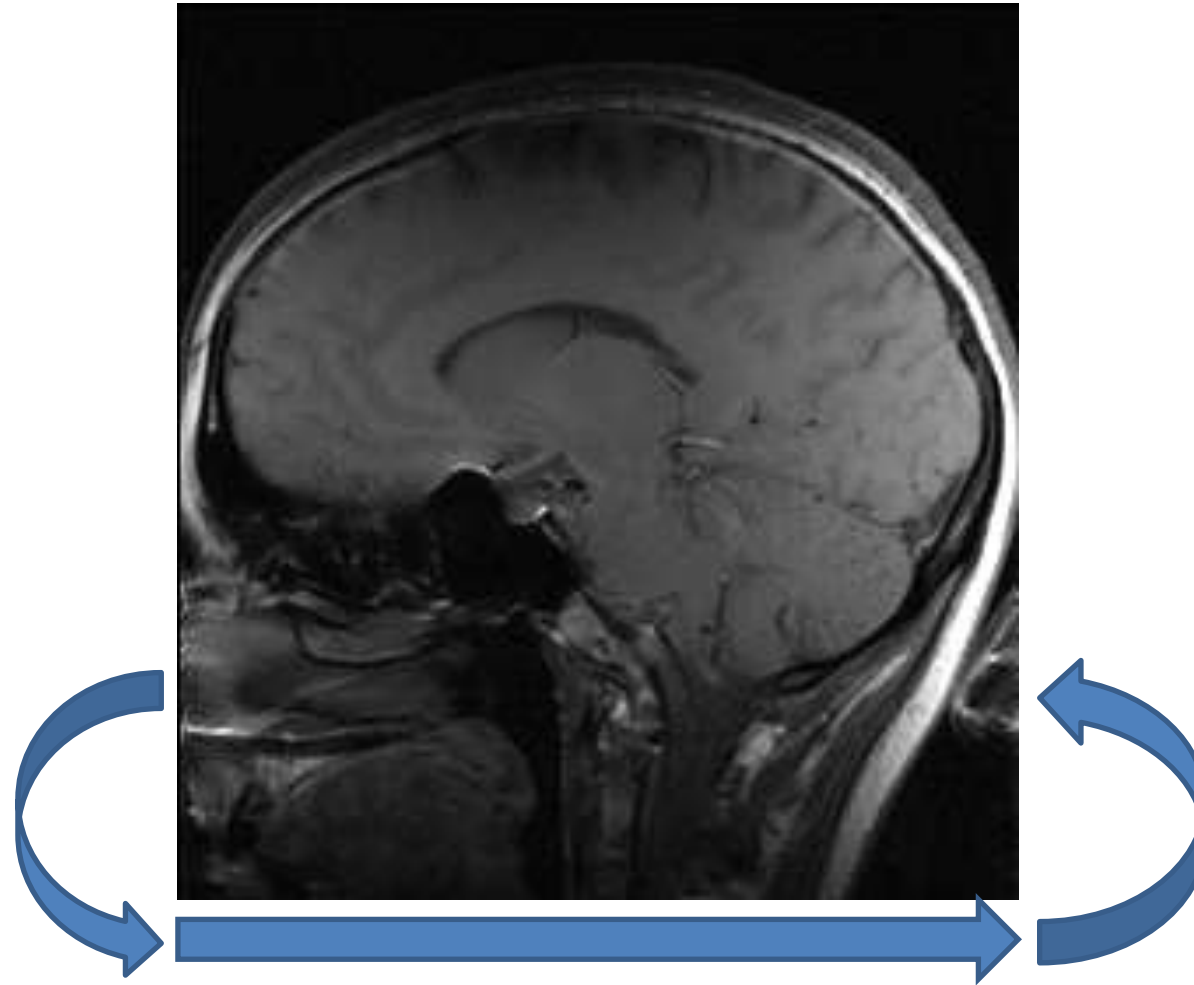
Metal artefacts: Retainer wire



Metal artefacts: Hair band



Fold-in artefacts



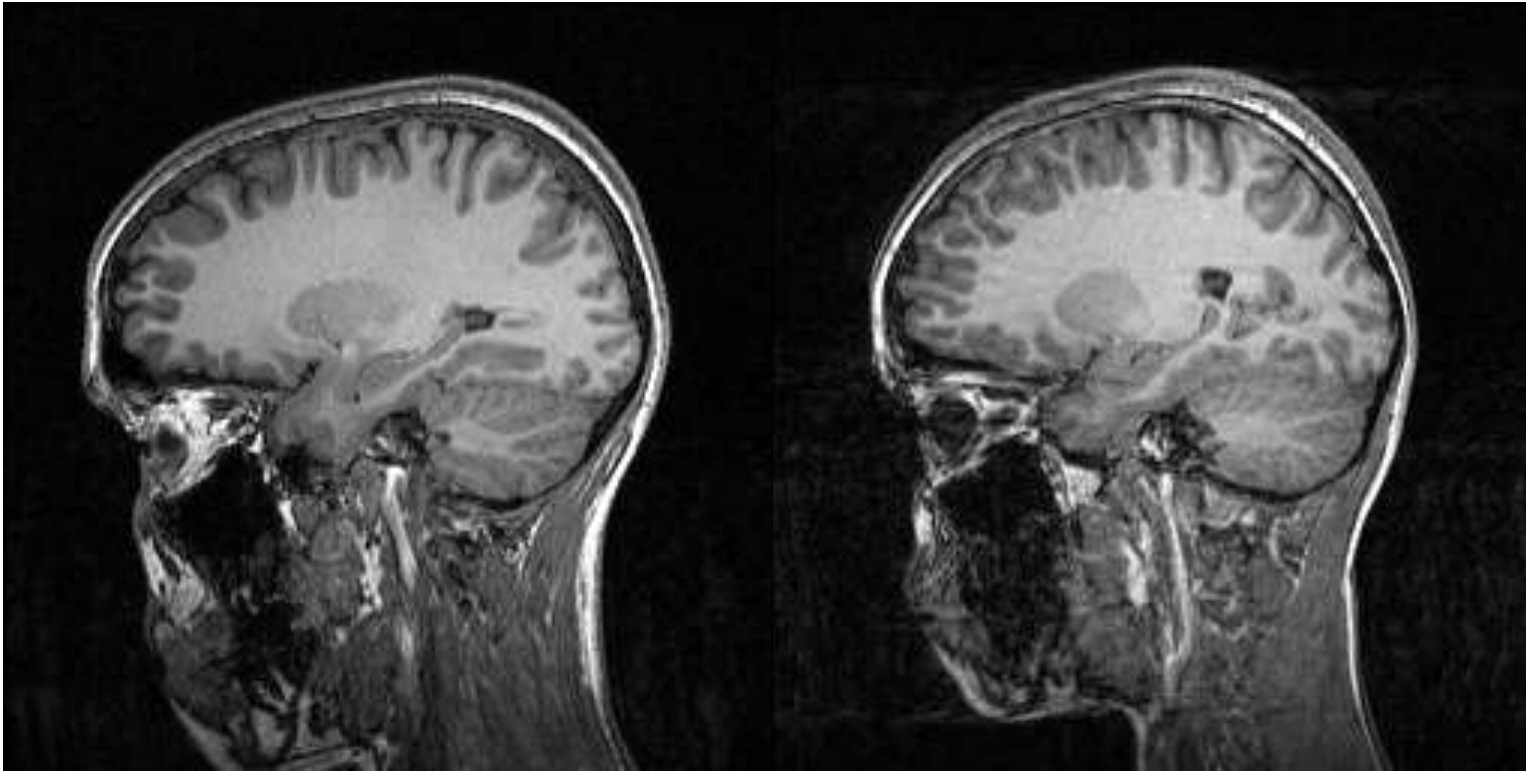
Motion artefacts

- General assumption: Patient does not move during measurement
- If motion occurs, the k -space to be measured changes
- Result: lines of different k -spaces are measured
- Possible consequences for imaging:
 - Blurring
 - Repetition of structures (ringing)
- Remedies:
 - Speed up image acquisition
 - Restrict patient movement

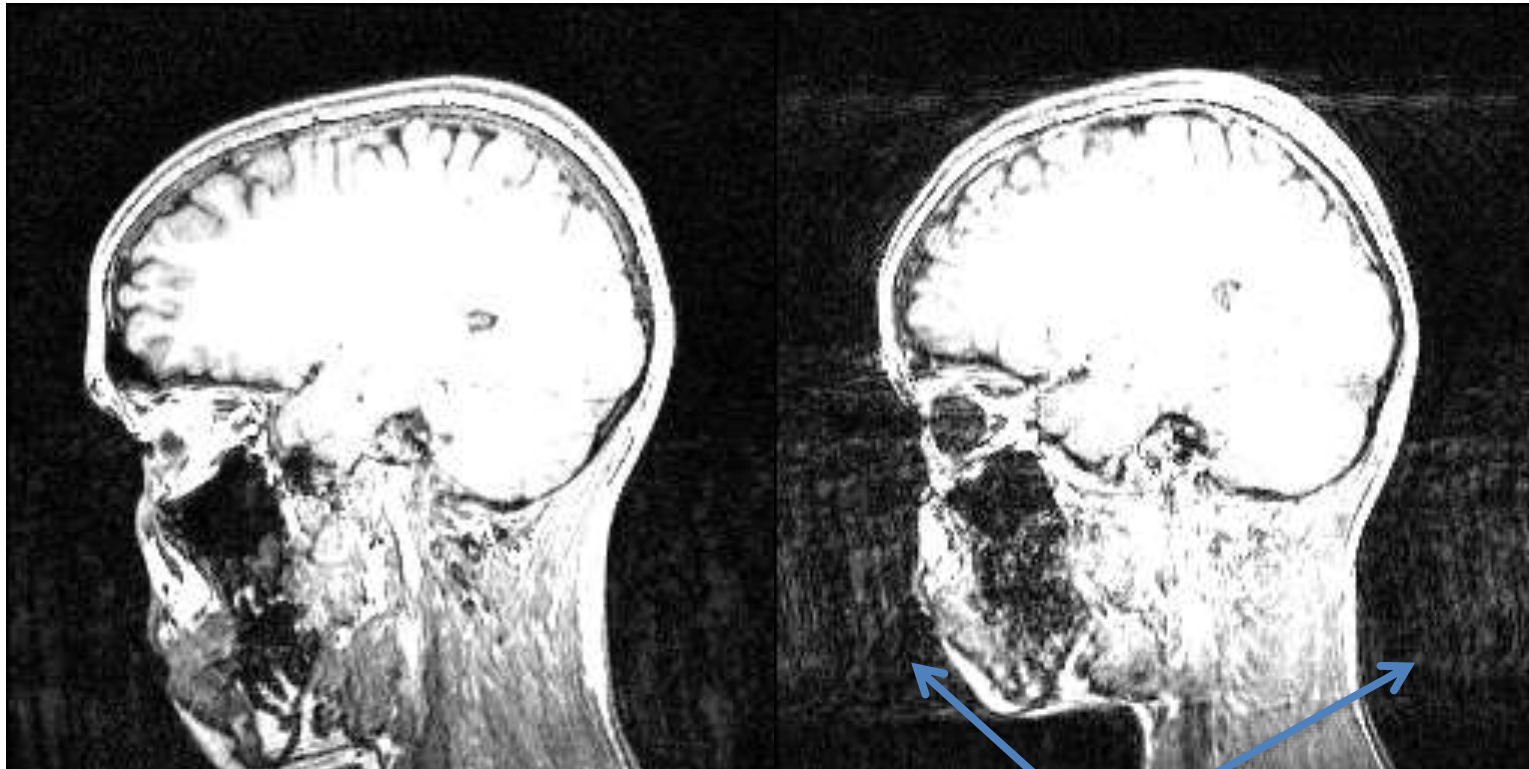
Motion artefacts

No motion

Motion artefacts



Motion artefacts 2

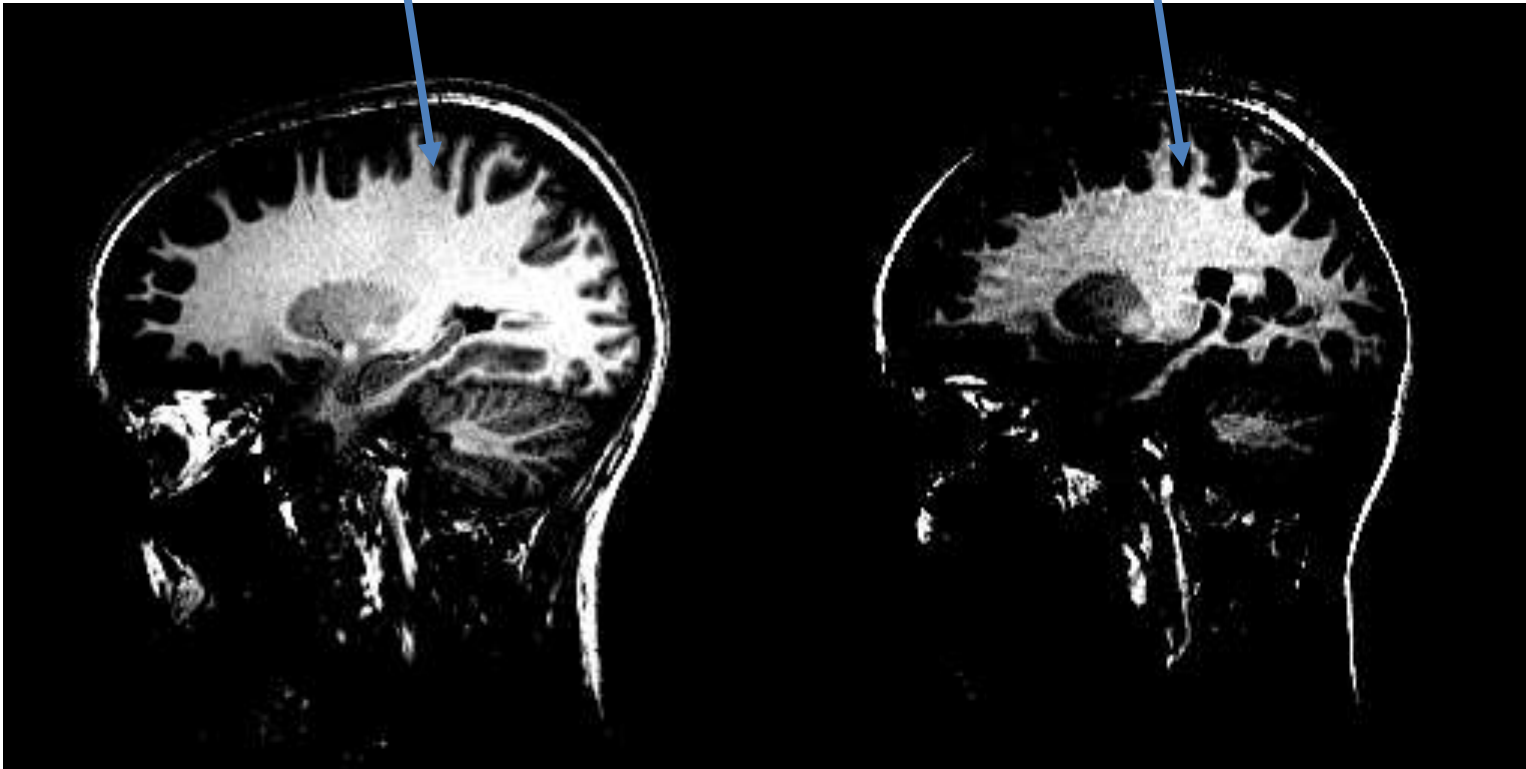


Blurring

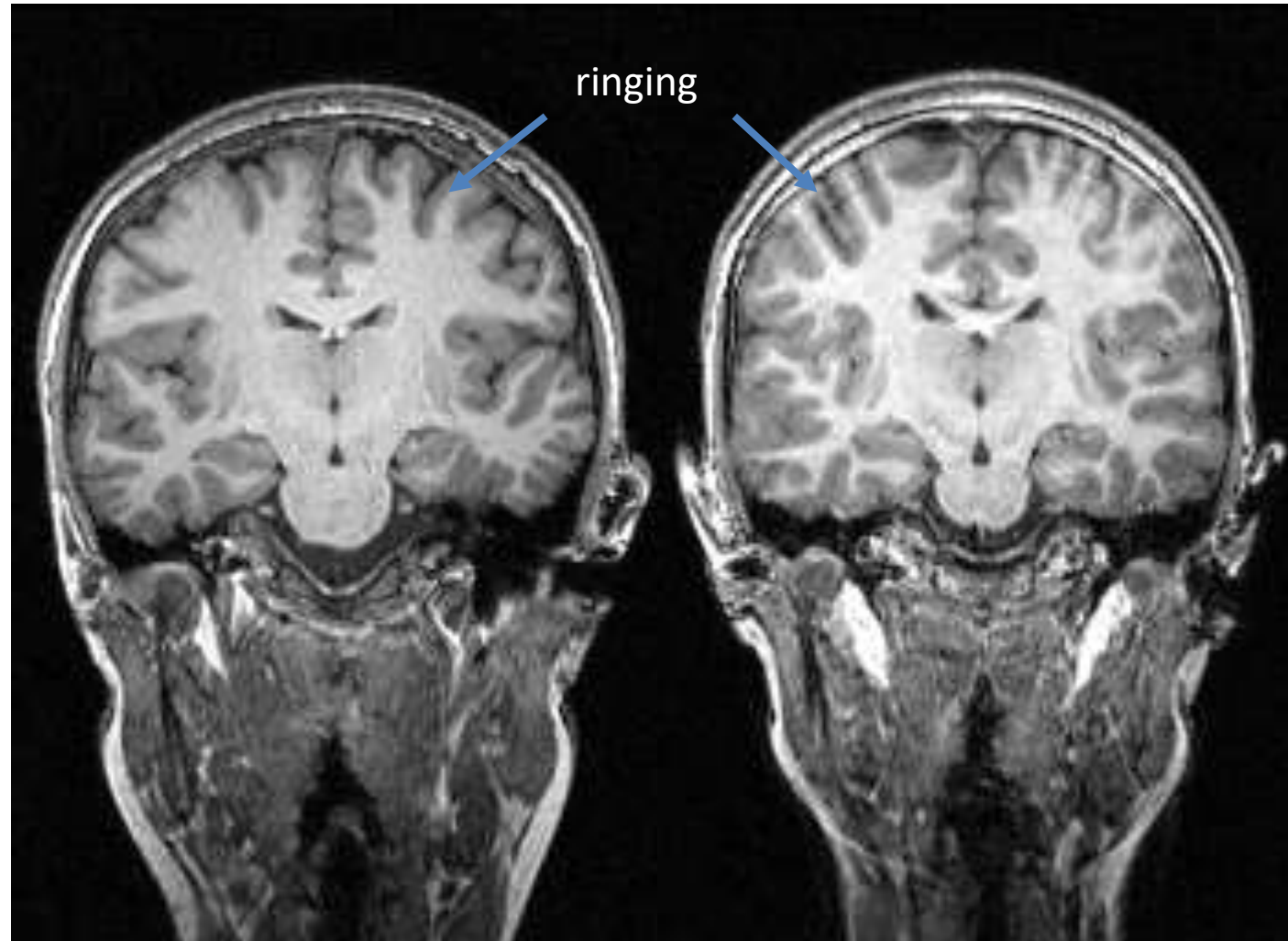
Motion artefacts 3

Smooth white matter

Artificially structured white matter

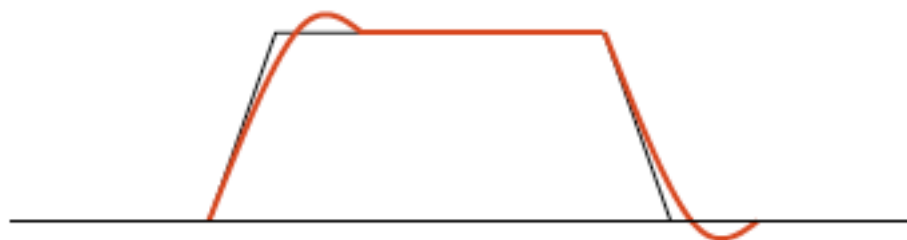


Motion 4

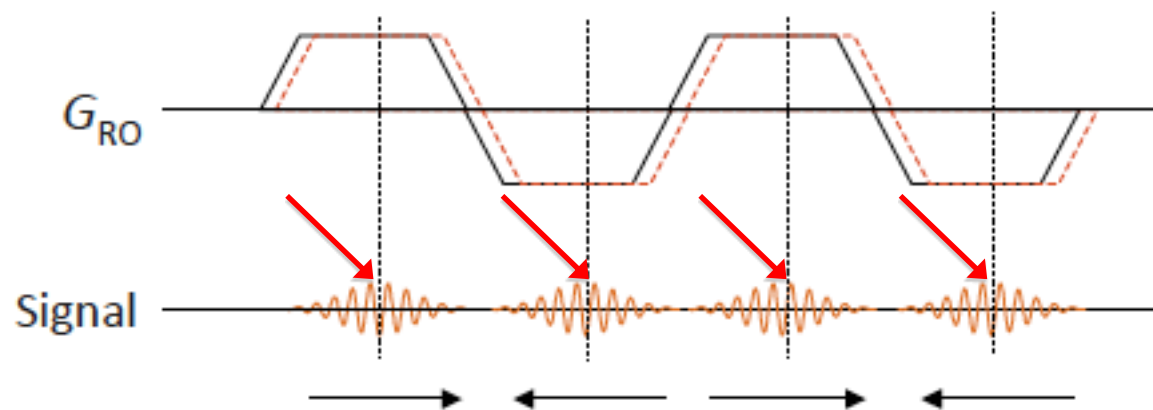


EPI image artefacts: “N/2 Ghost”

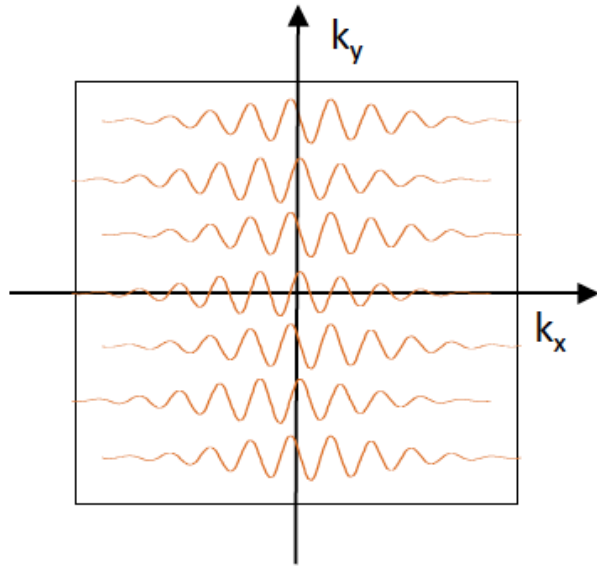
Because of Eddy currents, the gradients shape is modified:



And so is the K-Space trajectory, and the signal echo doesn't appear at the K-Space centre anymore:



EPI image artefacts: “N/2 Ghost”



The artefact is 2 lines-redundant in the K-Space:

$$(\text{Period} = 2 \times \Delta k_y = 2/\text{FoV}_y)$$



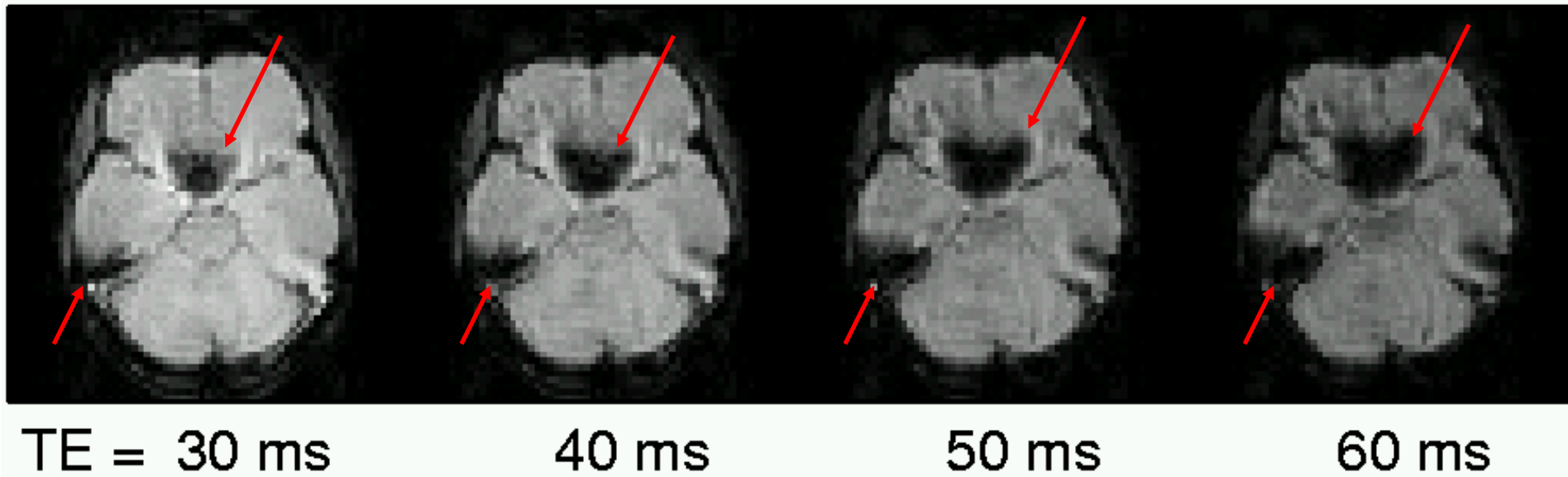
Image space periodicity
=
inverse of the frequency
space (K-Space)
periodicity

Artefact is
 $1/(2/\text{FoV}_y) = \text{FoV}_y/2$
periodic on the image

EPI: susceptibility artefacts

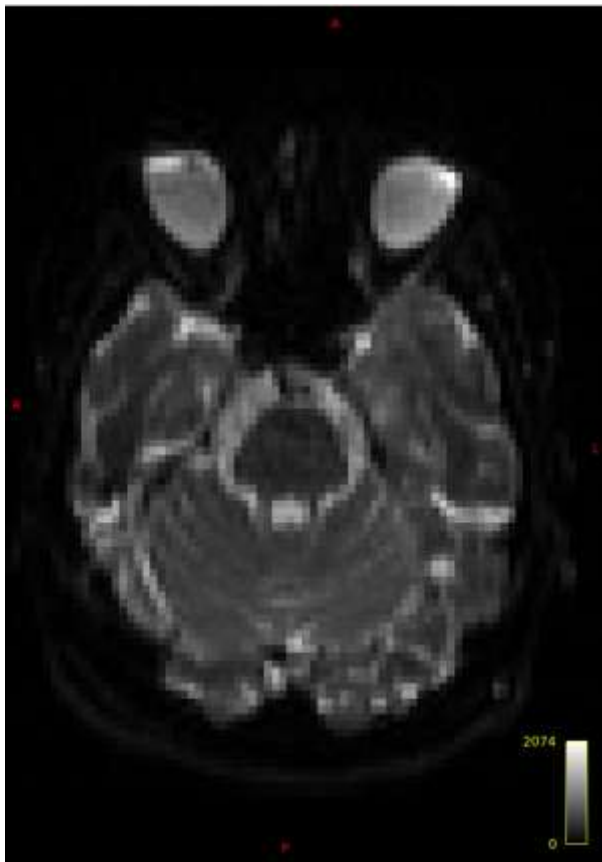
- Induced by local changes of the main magnetic field due to different magnetic properties of tissue.
- Strong effect at air-tissue boundaries (sinuses, ear canal)
- Especially pronounced in T_2^* -weighted sequences with long echo time (=> BOLD EPI)

EPI: susceptibility-induced signal dropout and image distortions

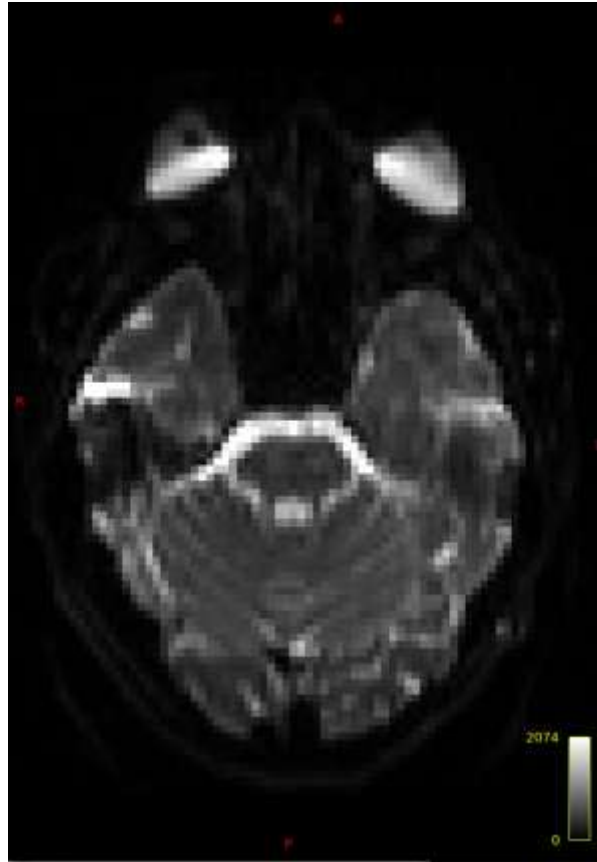


Example: Correcting Geometric EPI Distortions

- EPI distortions depend on the polarity of the phase encoding
- We can reduce them by combining information from both polarities



A>>P



P>>A



Unwarped A>>P

Summary

- **Artifacts** that we will have to account for during MR image processing include:
 - **Measurement noise:** Additive complex Gaussian leads to Rician distribution on magnitude images
 - **Bias fields:** Low-frequency variations in image intensity
 - **Motion artifacts:**
 - Between acquisitions: Repositioning of head
 - During acquisitions: Distortions, signal loss, blurring, noise
 - **Susceptibility artifacts:** Signal loss, distortions

3.6 Compressed Sensing MRI

Motivation and Main Idea

- Acquiring MRI scans faster allows us to
 - Reduce patient discomfort and motion artifacts
 - Acquire more images (temporal resolution, different contrasts)
 - Increase throughput / decrease cost
- *Idea:* Can we break Nyquist limit by exploiting signal structure?
 - *Example:* 256×256 slice image corresponds to 65536-dimensional vector space, plausible images form a lower-dimensional manifold
 - Can we reconstruct an image on that manifold with fewer measurements?

Image Compression / Discrete Cosine Transform

- Lossy compression stores natural images at a fraction of their original size by omitting perceptually irrelevant information
- *Example: JPEG*
 - Based on Discrete Cosine Transform
 - Concentrates most of the signal in a few coefficients
 - High frequencies can be quantized strongly
 - Many entries can be rounded to zero (*sparsity*)

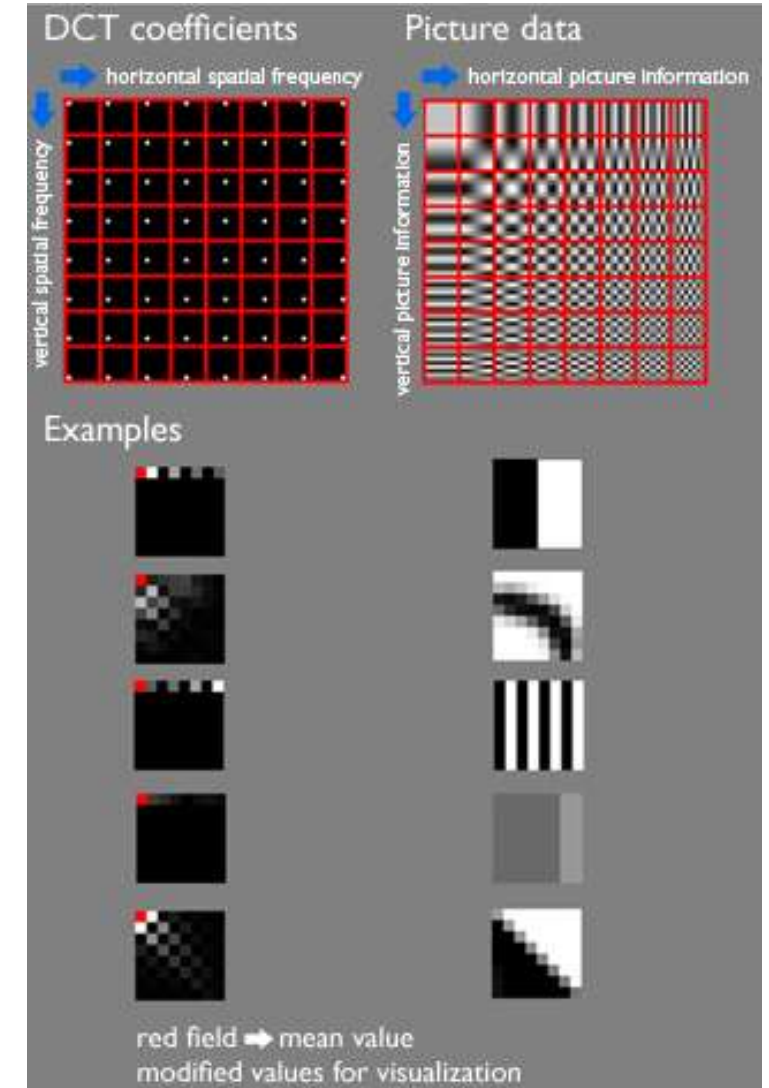
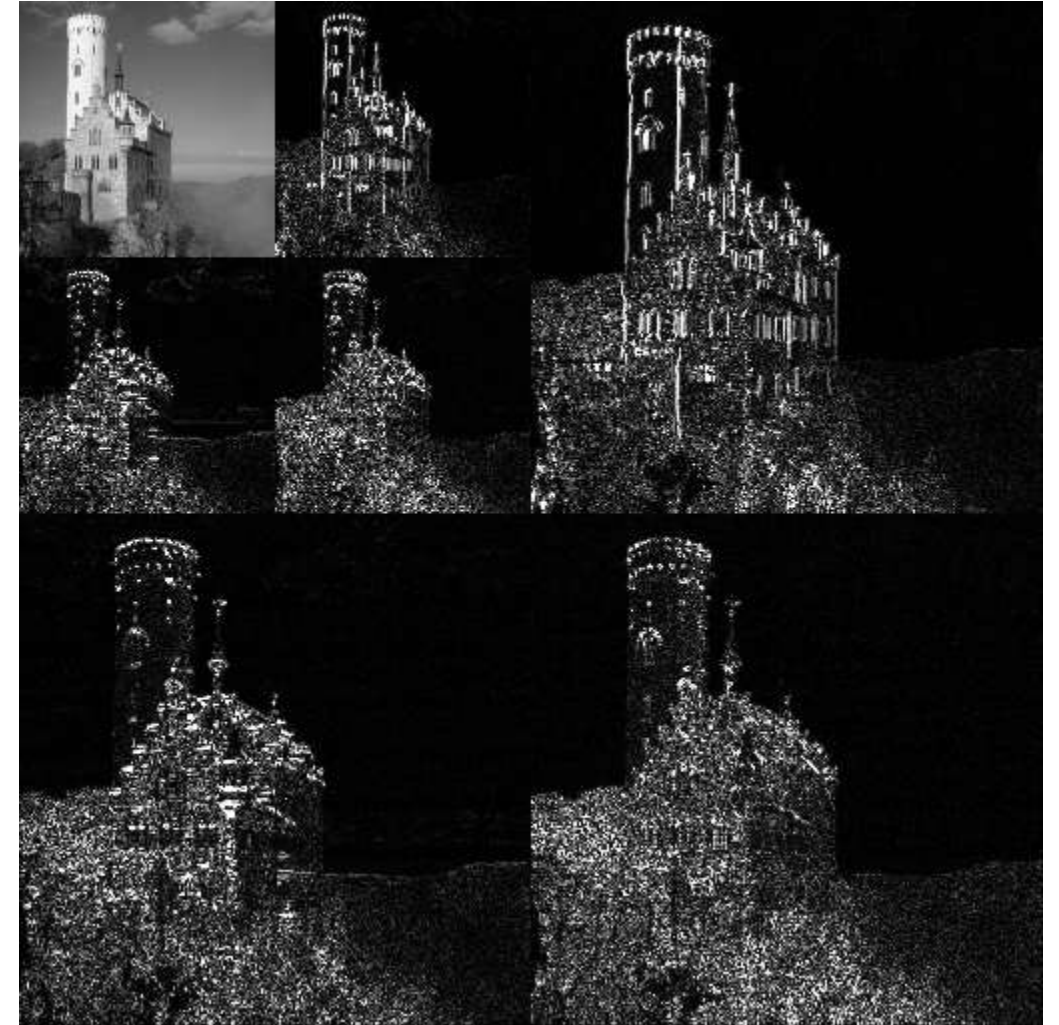


Image Compression / Discrete Wavelet Transform

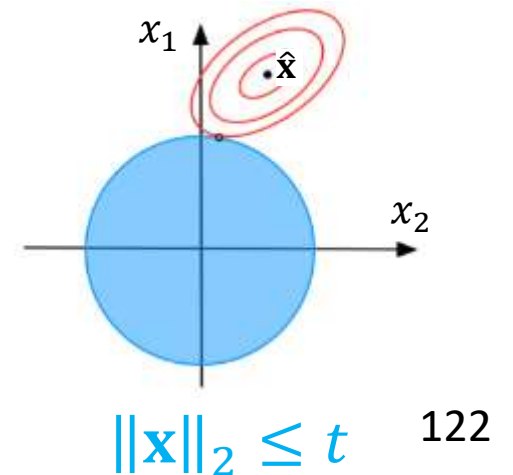
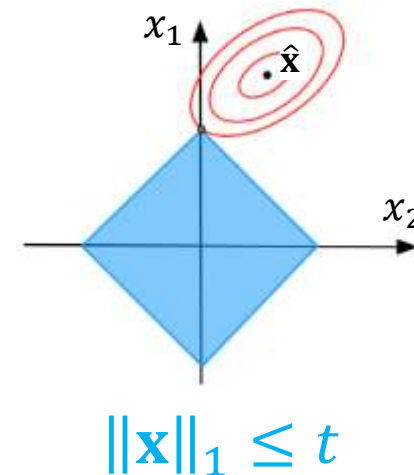
- JPEG2000 avoids blocking artifacts and achieves even higher compression rates via the Discrete Wavelet Transform
 - Recursive decomposition into low and high frequency components
 - Results in a sparse multi-resolution representation
- *Next step:* Exploit sparsity to avoid having to acquire redundant measurements



Sparse Signal Recovery

- We will assume a linear relationship between
 - signal $\mathbf{x} \in \mathcal{R}^n$
 - measurement $\mathbf{b} \in \mathcal{R}^m$
 - described by $\mathbf{A} \in \mathcal{R}^{m \times n}$ so that $\mathbf{Ax} = \mathbf{b}$
- If $m < n$ (undersampling), linear system is underdetermined
- We can obtain a sparse solution by minimizing its l_1 norm:

$$\min_{\mathbf{x}} \|\mathbf{x}\|_1 \text{ s.t. } \|\mathbf{Ax} - \mathbf{b}\|_2 \leq \epsilon$$

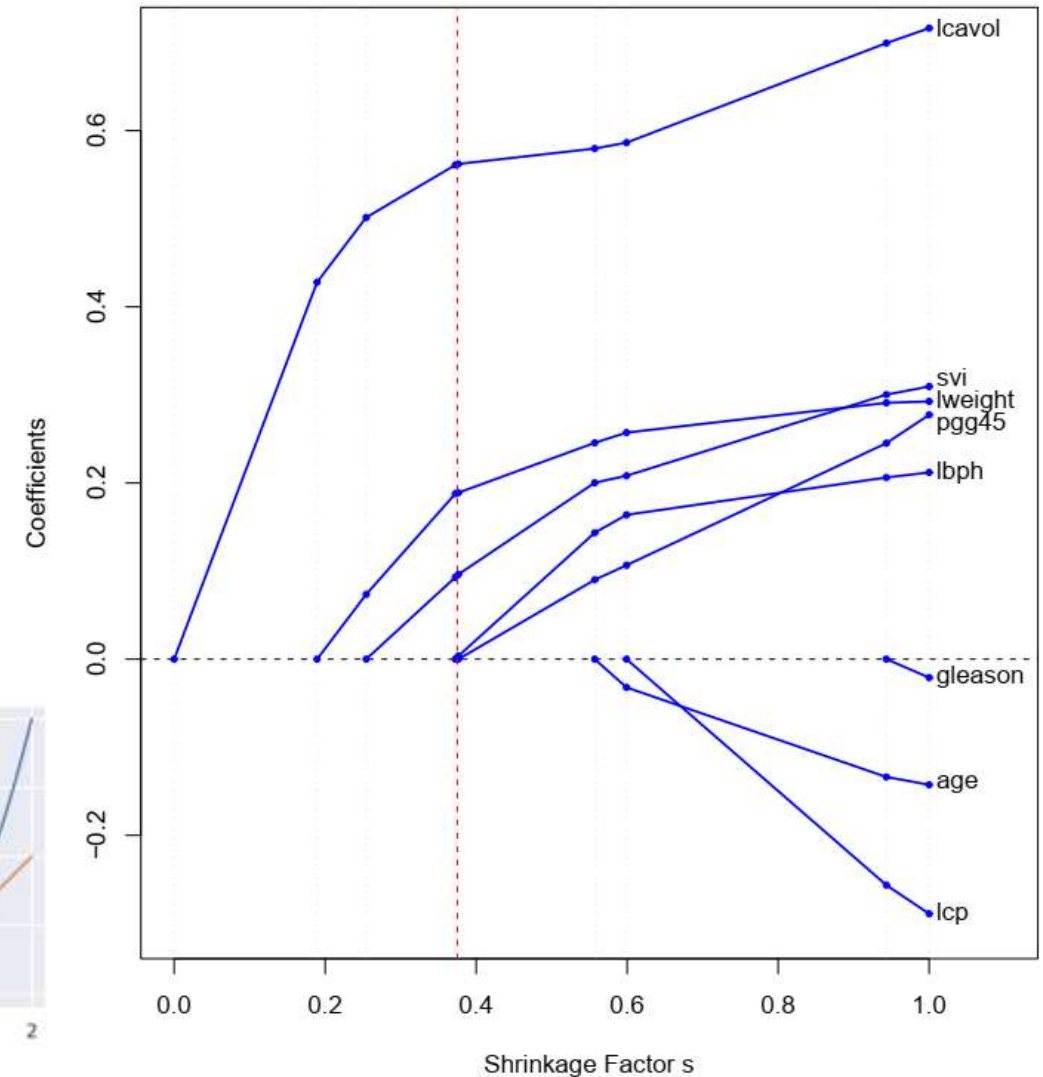
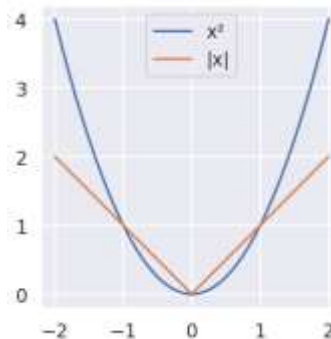


Related Concept: Lasso Regression

- **Lasso regression** combines an MSE loss with an l_1 regularizer:

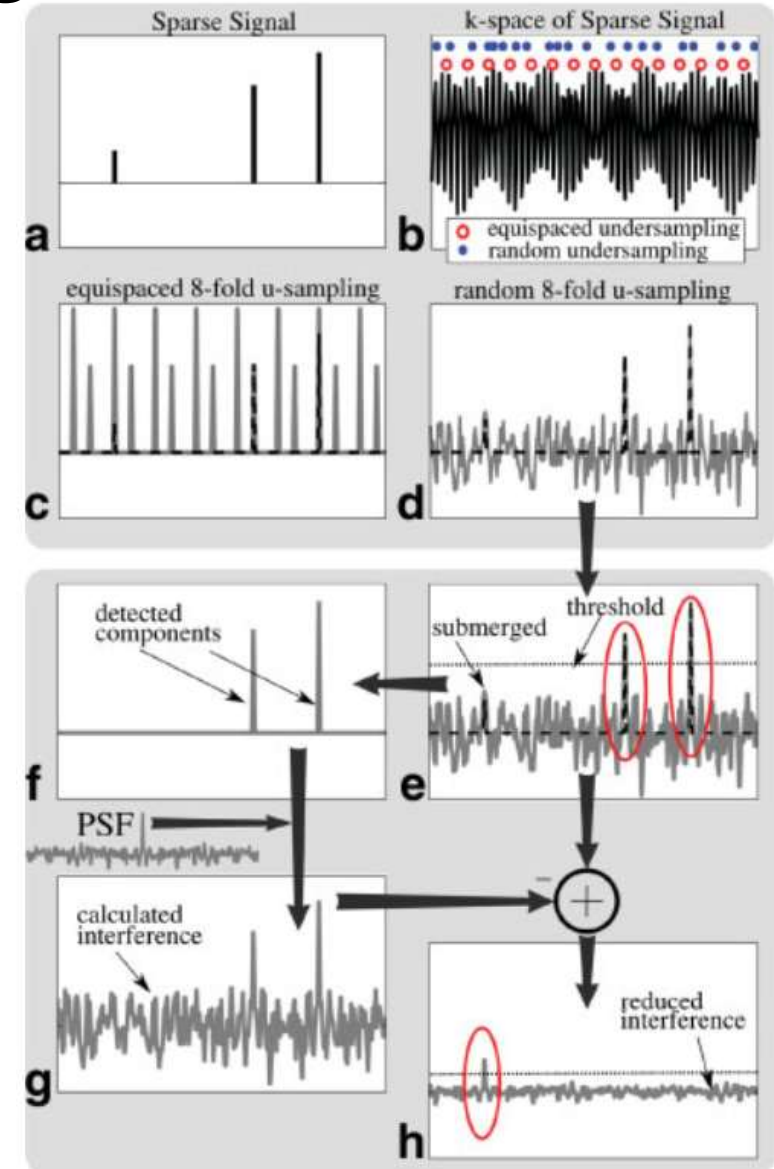
$$\sum_{i=1}^N \left(y_i - \beta_0 - \sum_{j=1}^p x_{ij} \beta_j \right)^2 + \lambda \sum_{j=1}^p |\beta_j|$$

- Lasso = “Least absolute shrinkage and **selection** operator”
- *Observation:* Coefficients β_j
 - shrink with increasing λ (decreasing shrinkage factor)
 - stop contributing at some point



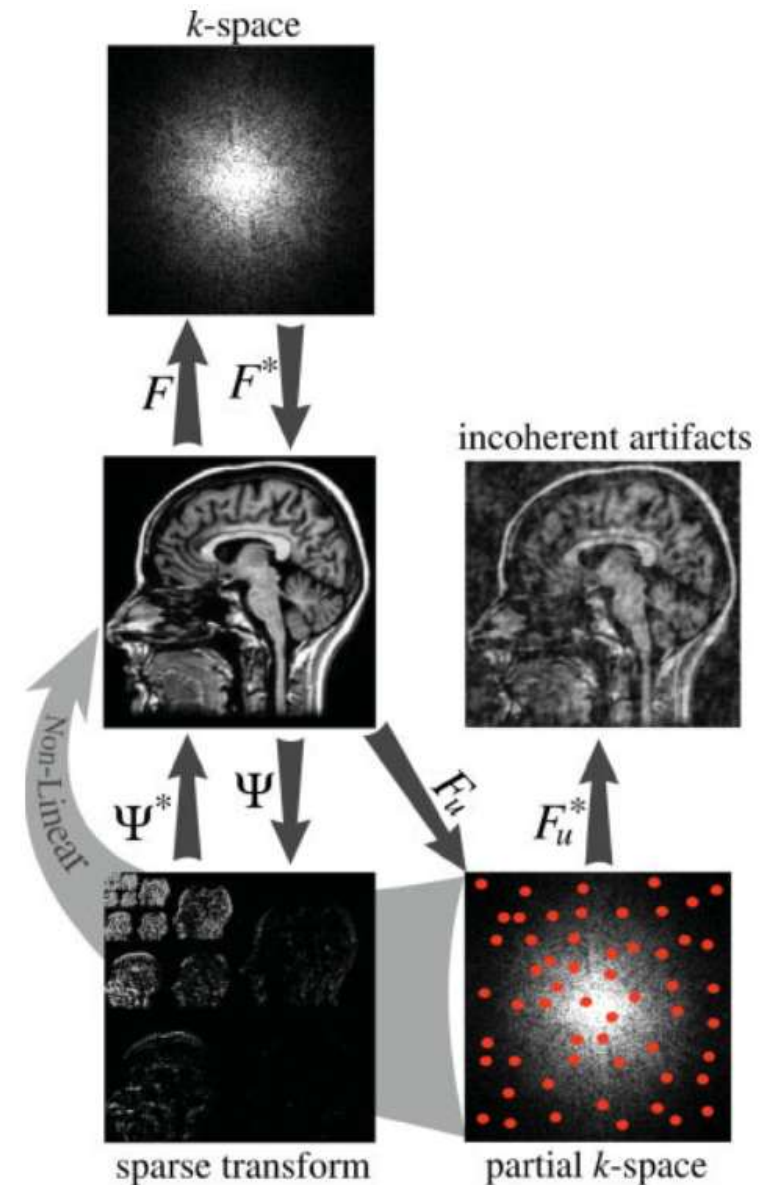
Incoherent Subsampling

- Regular subsampling leads to aliasing (c)
 - Impossible to recover the original signal
- Random subsampling leads to noise-like artifacts (d), sparse signals stand out
 - Can be reconstructed by l_1 minimization if $\mathbf{A}$ satisfies specific conditions
- *Main assumption*: sampling domain and sparse domain are incoherent
 - Intuitively, sparse vectors in one domain correspond to dense signals in the other
 - Canonical and Fourier bases are maximally incoherent



Compressed Sensing MRI

- Compressed sensing in MRI involves three domains, connected via linear transformations
 - Sampling domain: k space
 - Signal domain: Image space
 - Sparse domain: DFT, Wavelet, etc.
- l_1 minimization in the sparse domain permits high quality image reconstructions from suitably undersampled k space



How Sparse Are Our Images?

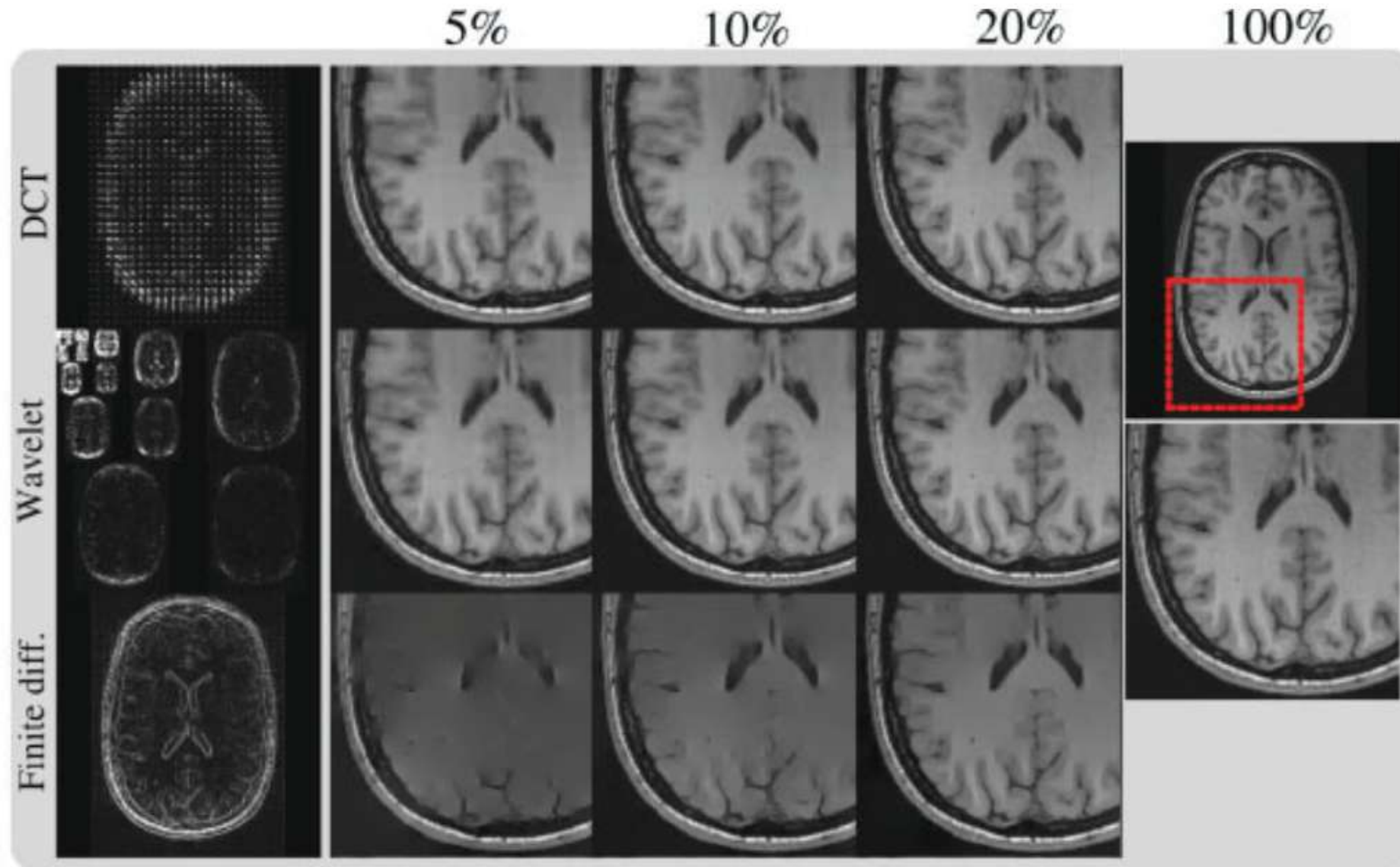


Figure: [Lustig et al. 2007]

Undersampling Schemes

- Since we acquire one line in k space at a time, undersampling is often done randomly in phase encoding direction
- Since most of the signal is at the center of k space, it makes sense to sample more densely there
- In multi-slice acquisition, we can randomize per-slice and alternate between two slicing direction during reconstruction
- *Example: Acceleration by 2.4*

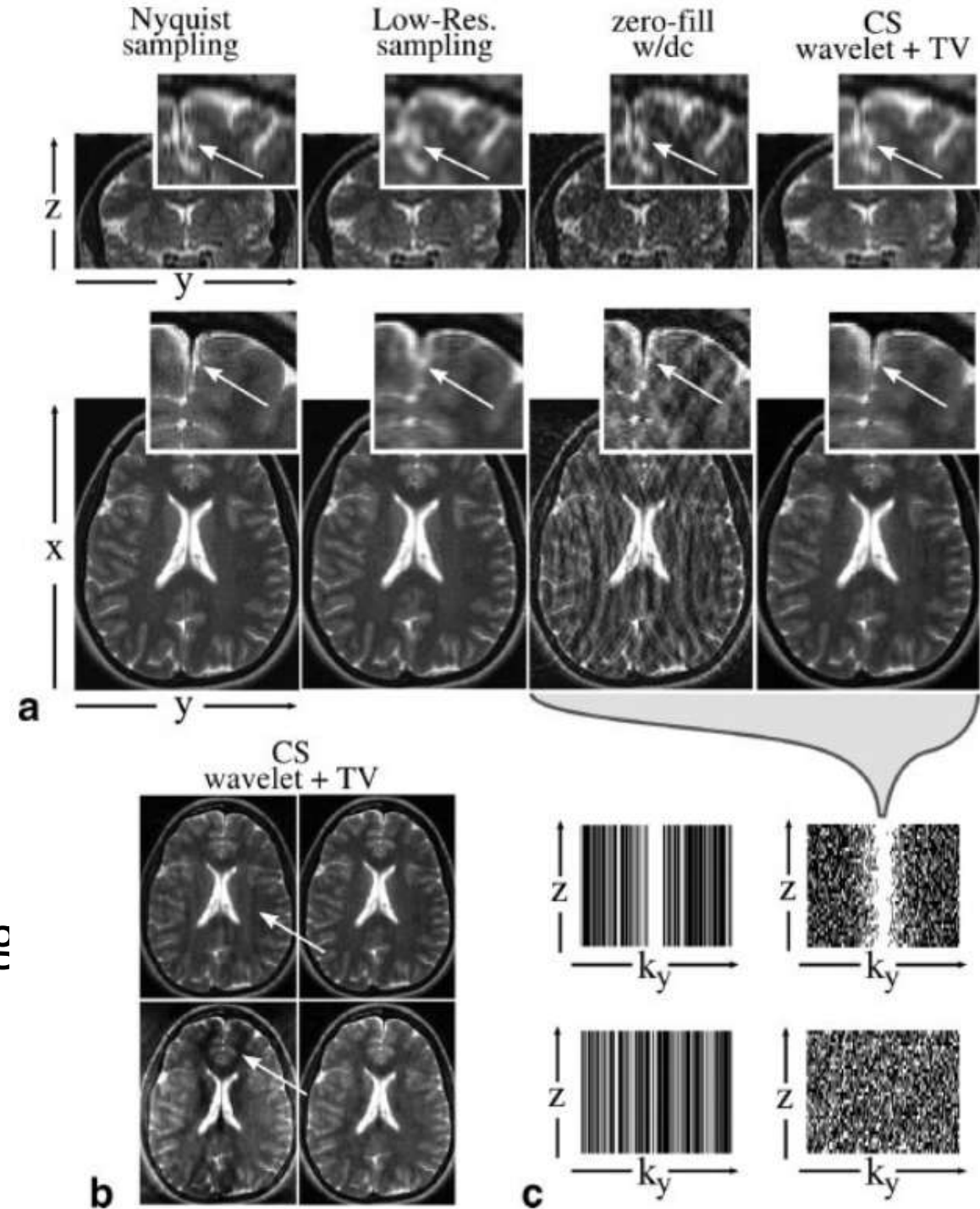


Figure: [Lustig et al. 2007]

Summary: Compressed Sensing in a Nutshell

- **Compressed sensing** can accelerate image acquisition
 - If we can represent images as sparse vectors
 - If measurement and sparsity domains are incoherent
- Specific **transformations** represent images as sparse vectors
 - Discrete Cosine Transform
 - Discrete Wavelet Transform
- MRI is well-suited for compressed sensing since it acquires images in k space, which is incoherent with image space
- Sparse solutions are often computed with **l_1 minimization**

3.7 Neural Network Based Image Reconstruction

Motivation

- **Artificial neural networks (NNs)** can approximate any continuous function
- Benefits of training NNs to reconstruct MR images $\mathbf{y}$ from k space measurements $\mathbf{x}$:
 - Learning a more specific prior of plausible images (or even head MR images) amplifies advantages of compressed sensing
 - Improved ability to separate noise from signal
 - Can directly account for systematic measurement errors
- Important role in current **ultra-low field MRI** (e.g., operating at 0.064 T)

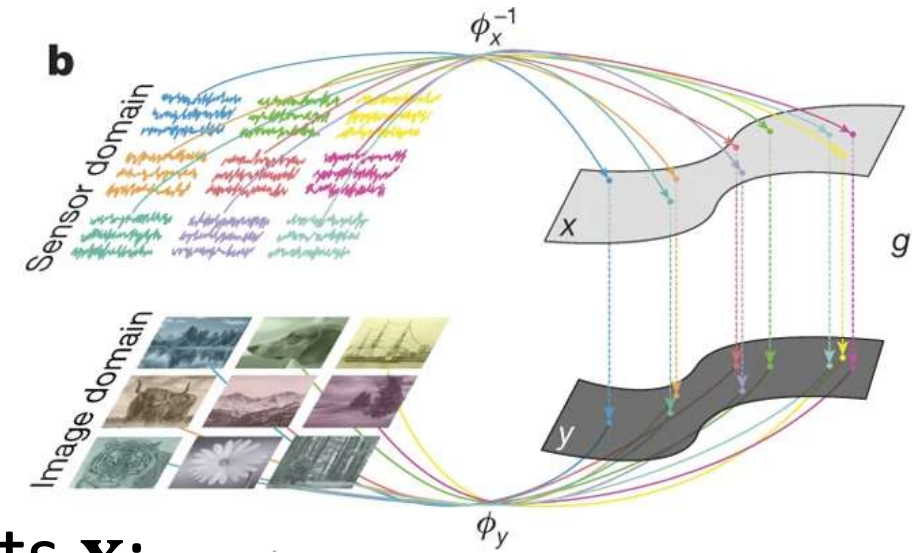
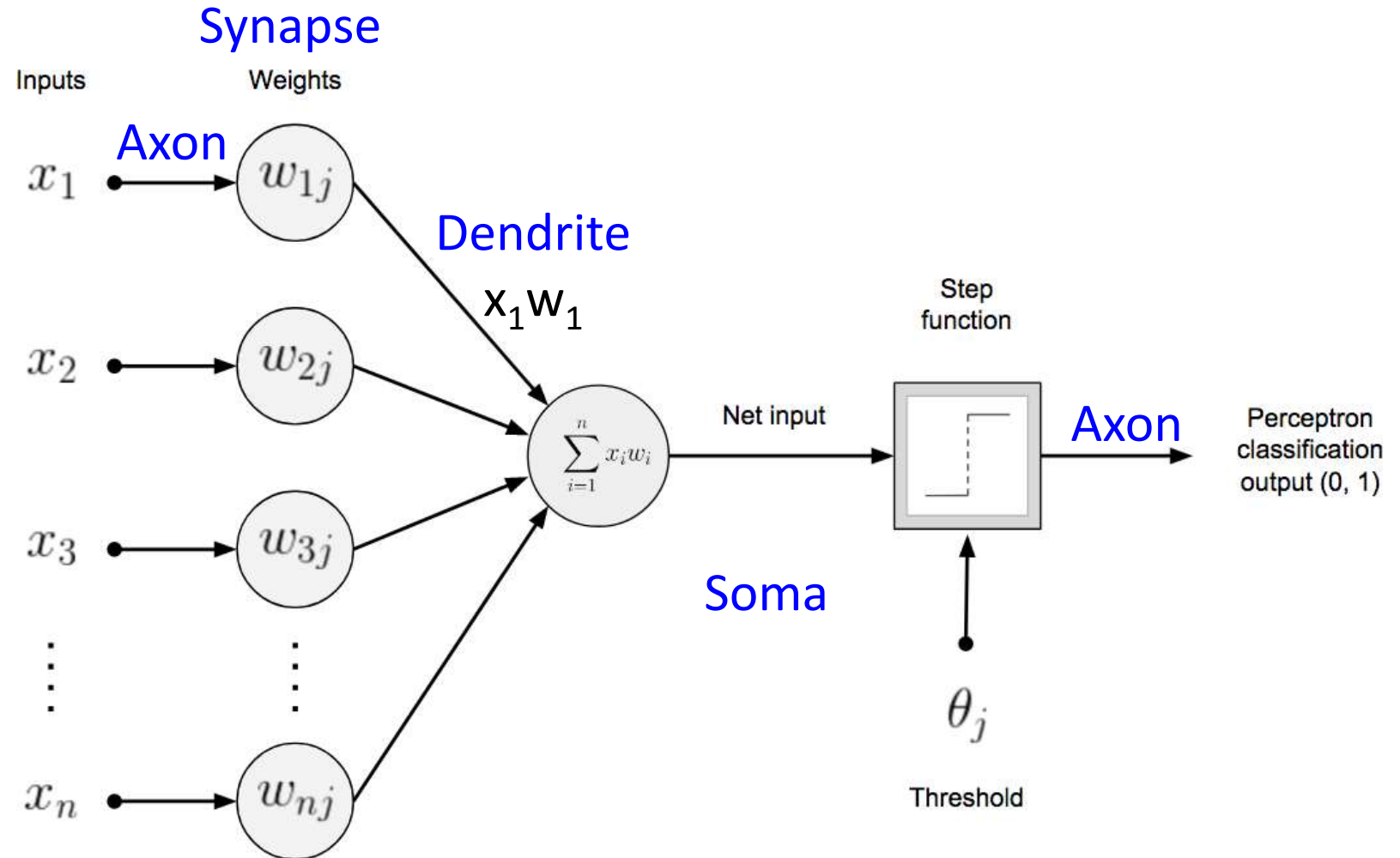
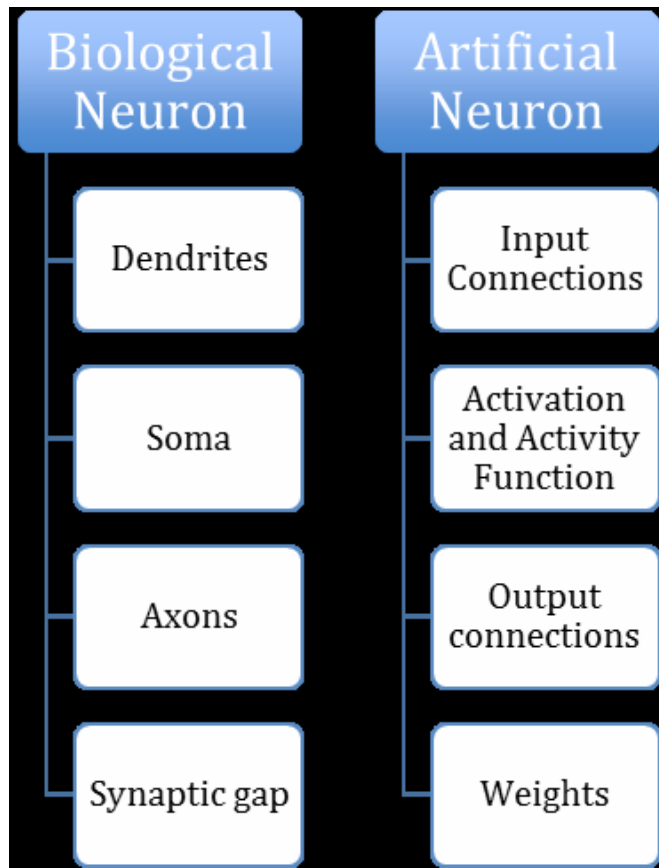


Image © Hyperfine

Brief Introduction to Neural Networks

- Will introduce some foundations of neural networks
 - Also required for state-of-the-art methods in later chapters
- We will focus on
 - Aspects that are immediately relevant for two specific approaches, AUTOMAP [Zhu et al. 2018] and image-to-image [Wang et al. 2016]
 - use of trained networks (“inference”), no details on training process
- Additional concepts will be introduced in later chapters
- Deeper understanding to design and train your own networks is conveyed in lectures such as “Einführung in Deep Learning für Visual Computing“, „Advanced Computer Vision“

From Biological to Artificial Neuron



Artificial Neuron: Formal Definition

Artificial neurons are inspired by biological ones, but are ultimately nothing more than the following mathematical equation:

$$y_k = f \left(\sum_{j=1}^m w_{kj} x_j + b_k \right)$$

- y_k is the output of the k th neuron
- x_j are the inputs
- w_{kj} are the weights
- b_k is a bias term
- f is a nonlinear transfer (activation) function

Disclaimer:
This is a useful building block to learn
functions from data, NOT an accurate
model of biology

Artificial Neuron: Alternative Equations

A common alternative notation sets $x_0 := 1$, $w_{k0} := b_k$, so that

$$y_k = f \left(\sum_{j=0}^m w_{kj} x_j \right)$$

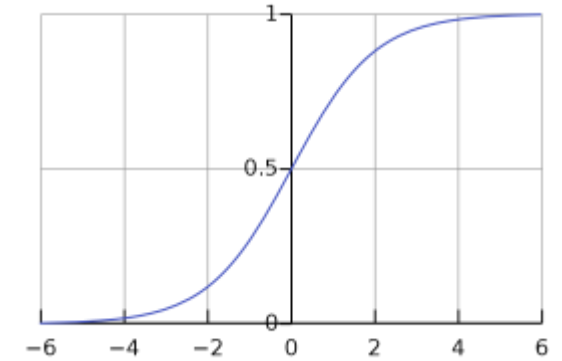
In matrix notation:

$$\mathbf{y} = f(\mathbf{W}\mathbf{x})$$

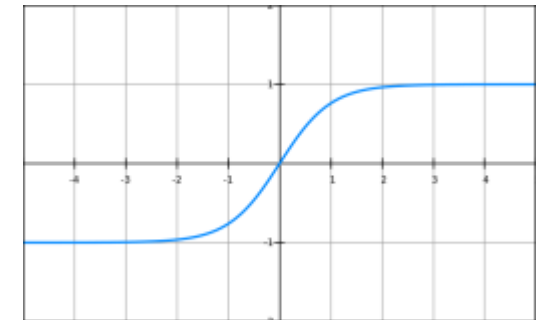
- f is applied element-wise
- weight matrix $\mathbf{W}$ collects w_{kj}

Sigmoid Activation Functions

- **Logistic sigmoid** $f(\alpha) = 1/(1 + e^{-\alpha})$
 - Output in (0,1), can be interpreted as a probability
 - Thresholding-like behaviour, but differentiable
 - Simple derivative: $f'(\alpha) = f(\alpha)(1 - f(\alpha))$
 - **But:** Derivative (nearly) zero over a wide range
 - Sometimes used in output layers, rarely in hidden layers

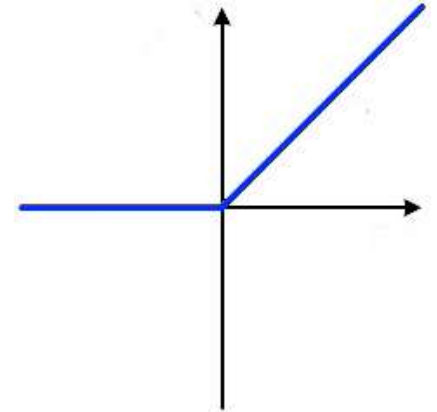


- **Hyperbolic tangent** $f(\alpha) = \frac{e^{\alpha} - e^{-\alpha}}{e^{\alpha} + e^{-\alpha}}$
 - Thresholding-like and differentiable
 - Output centered around zero
 - helpful when using basic optimization algorithms
 - Simple derivative: $f'(\alpha) = 1 - (f(\alpha))^2$
 - **But:** Derivative still (nearly) zero over a wide range

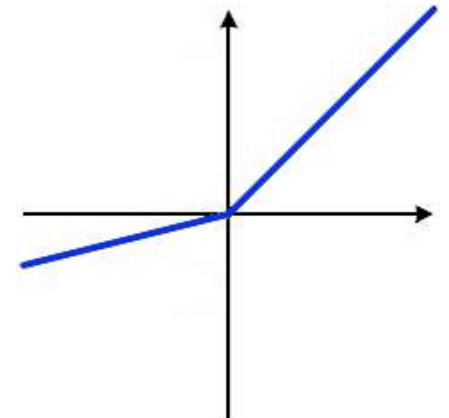


Rectified Linear Activation Functions

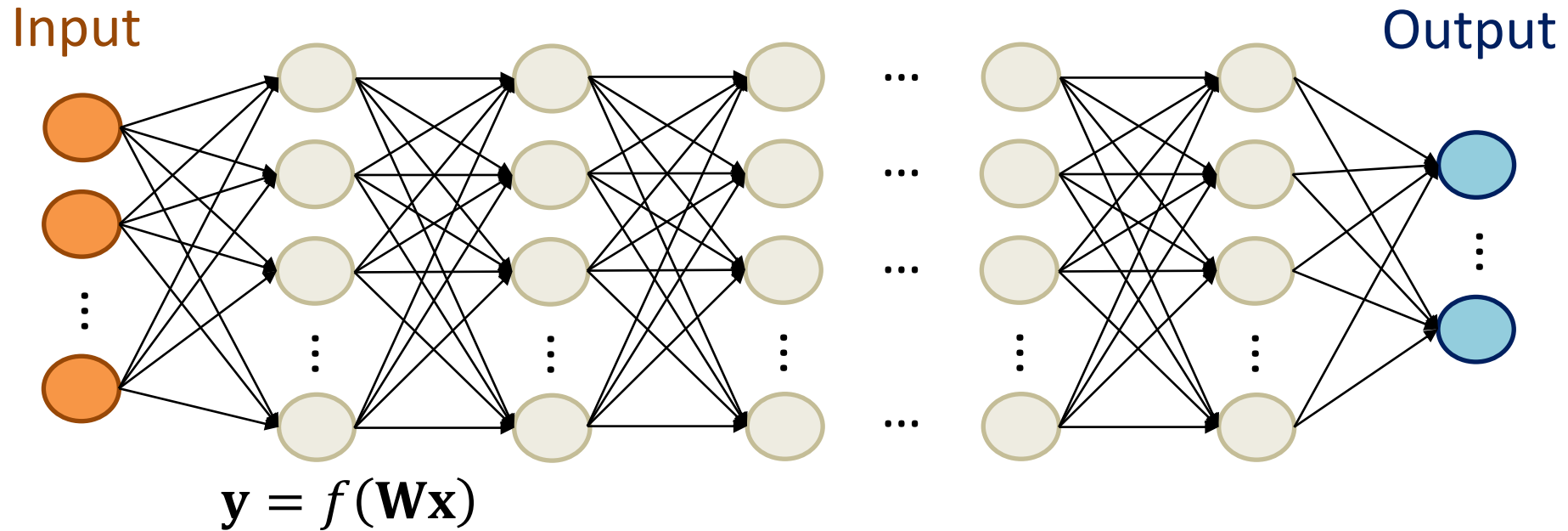
- **Rectified Linear Unit (ReLU)** $f(\alpha) = \max(\alpha, 0)$
 - Simple and computationally efficient
 - No saturation, gradient-based learning remains effective
 - **But:** No gradient for negative inputs, ReLU might “die” (i.e., become inactive for all training samples)



- **Leaky ReLU** $f(\alpha) = \max(0.01\alpha, \alpha)$
 - Similar to ReLU, but prevents neurons from “dying”
 - Parametric Rectifier: Factor 0.01 treated as learnable parameter



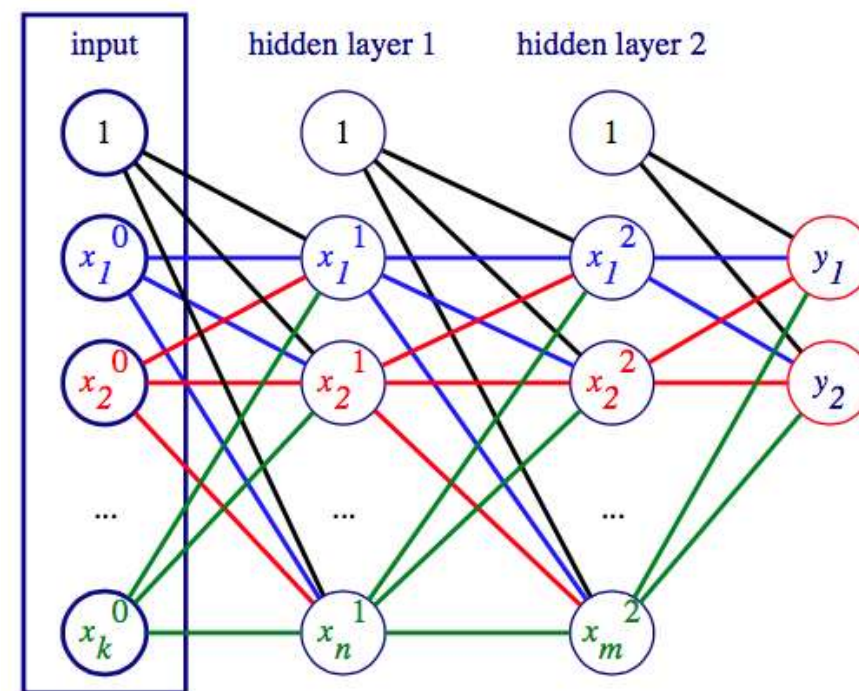
Deep Neural Networks



- In **feedforward neural networks**, neurons are arranged in layers so that the output of earlier layers is only used in later layers
- **Deep neural networks** have multiple (often a large number of) hidden layers

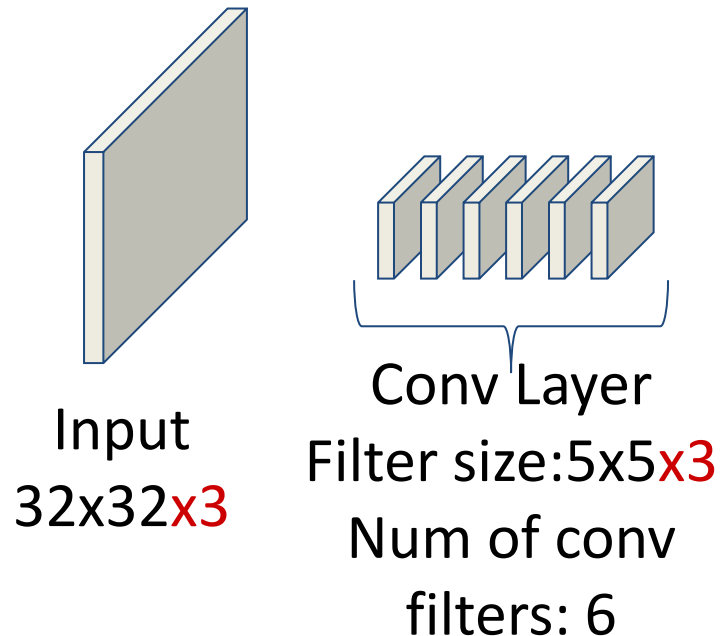
Fully Connected Layer

- A layer is called *fully-connected* if all neurons in adjacent layers are pairwise connected
- Does not scale well to high-dimensional inputs, e.g., images.
 - Huge number of parameters would lead to memory issues and overfitting
 - *Alternative:* Convolutional Layers



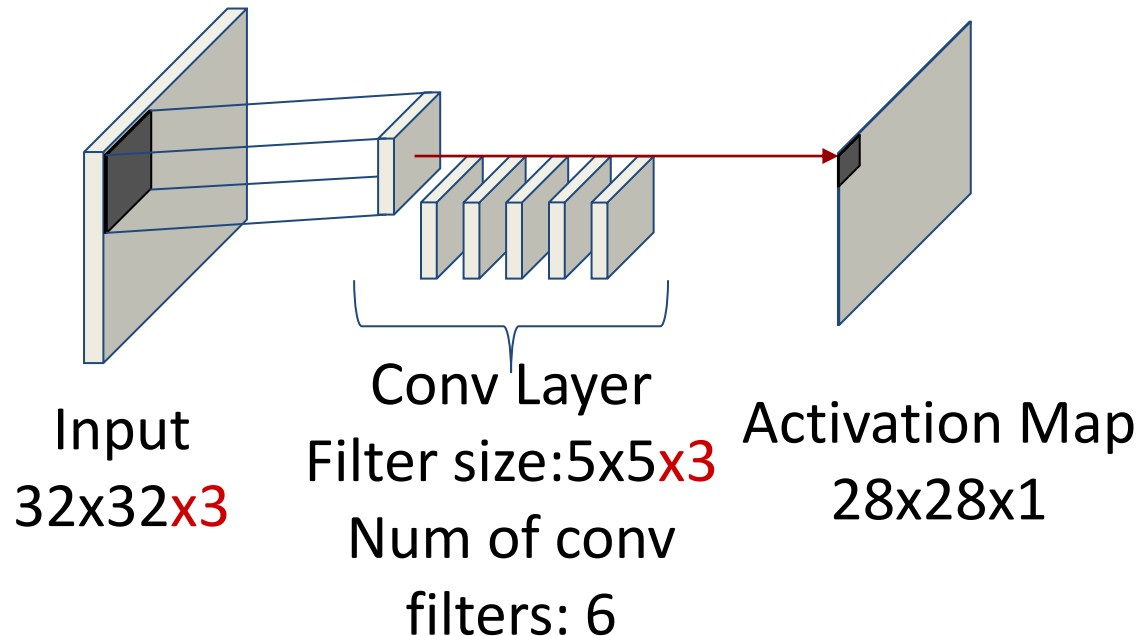
Convolution Layer

- In a convolution layer, the coefficients of a number of convolution kernels (plus biases) are the trainable parameters
- Filter size and number of filters are defined as part of the architecture, the filter implicitly spans the depth of the input volume (example: 3)



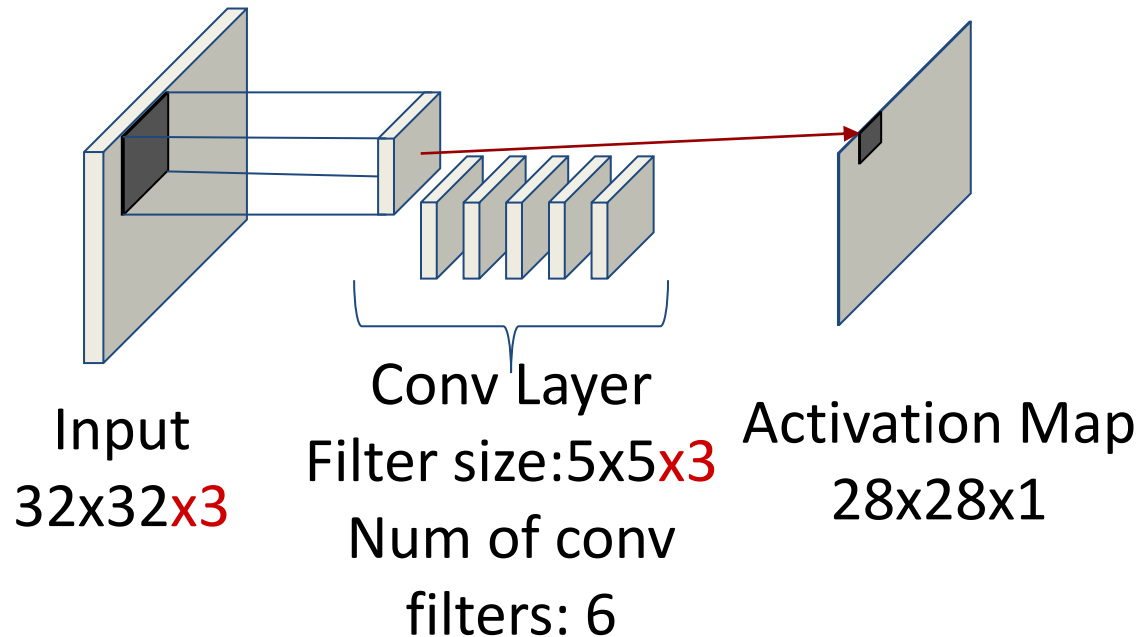
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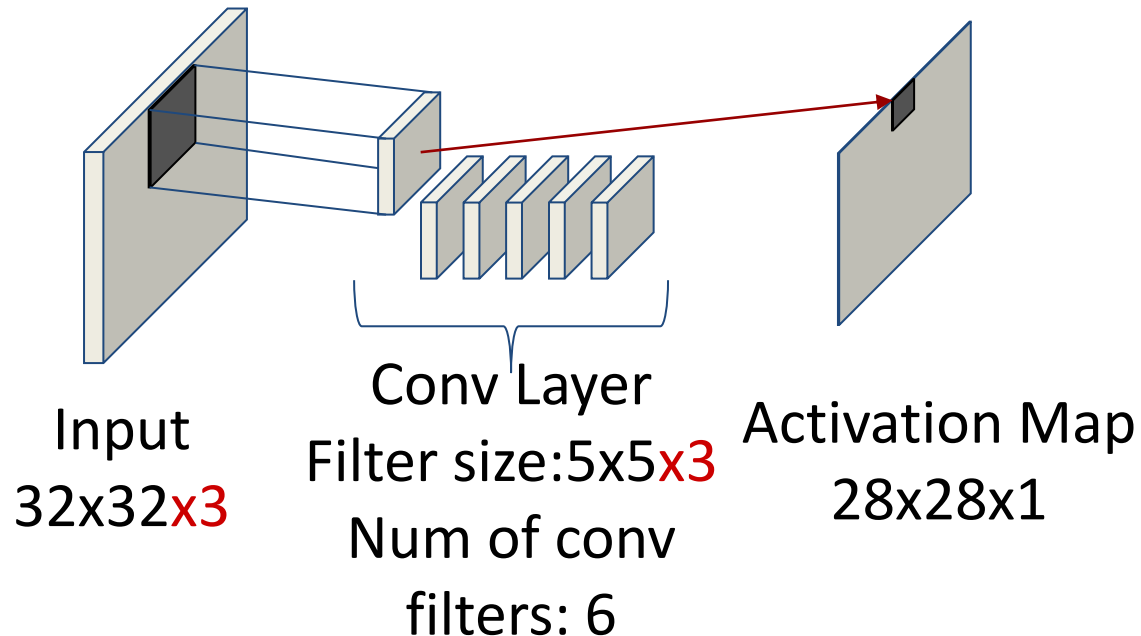
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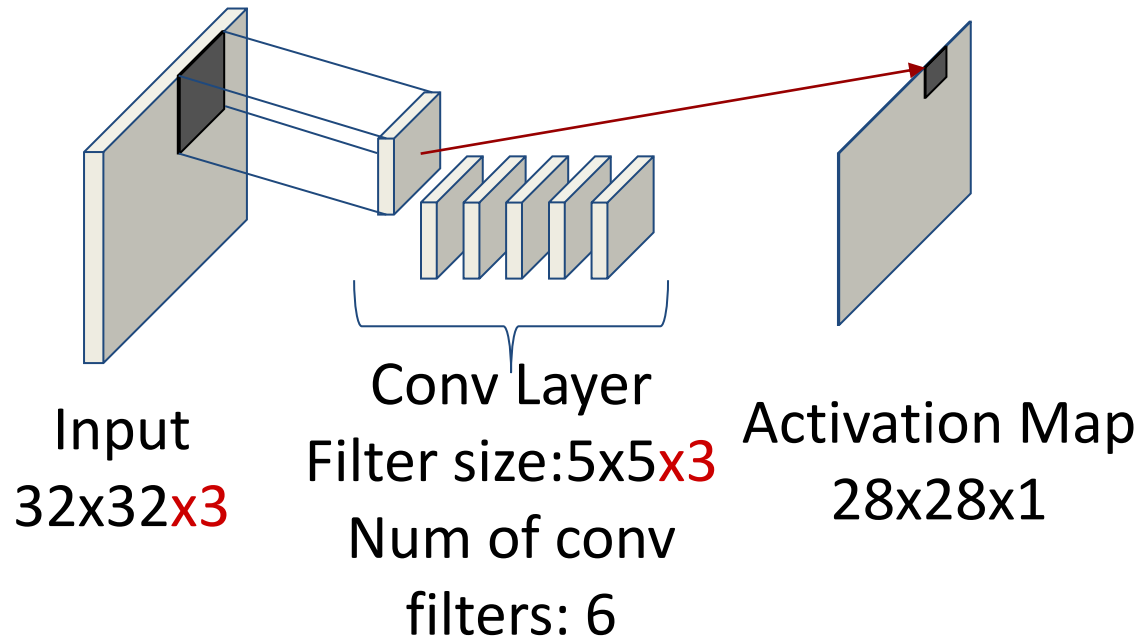
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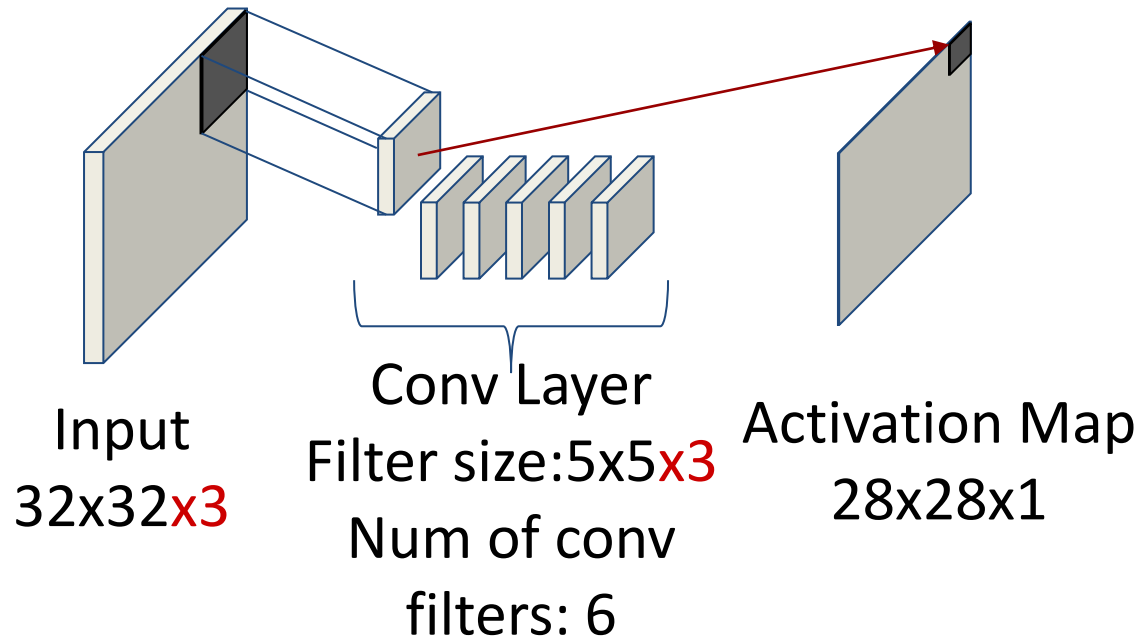
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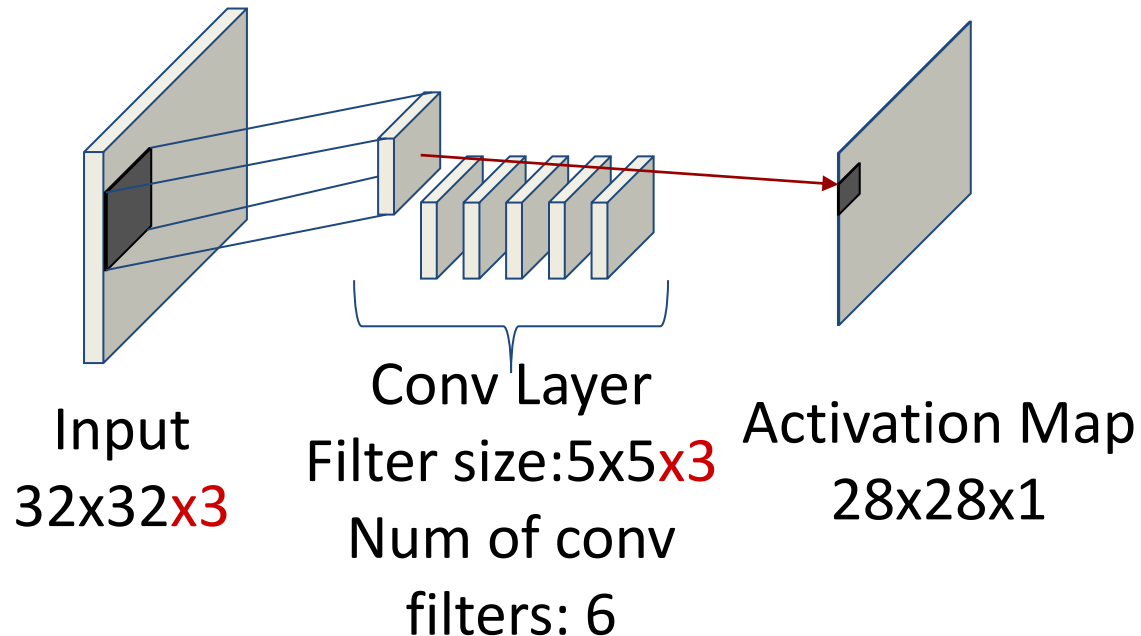
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Convolution Layer

- In a convolution layer, the coefficients of a number of convolution kernels (plus biases) are the trainable parameters
- Filter size and number of filters are defined as part of the architecture, the filter implicitly spans the depth of the input volume (example: 3)



Terminology: Convolution vs. Cross-Correlation

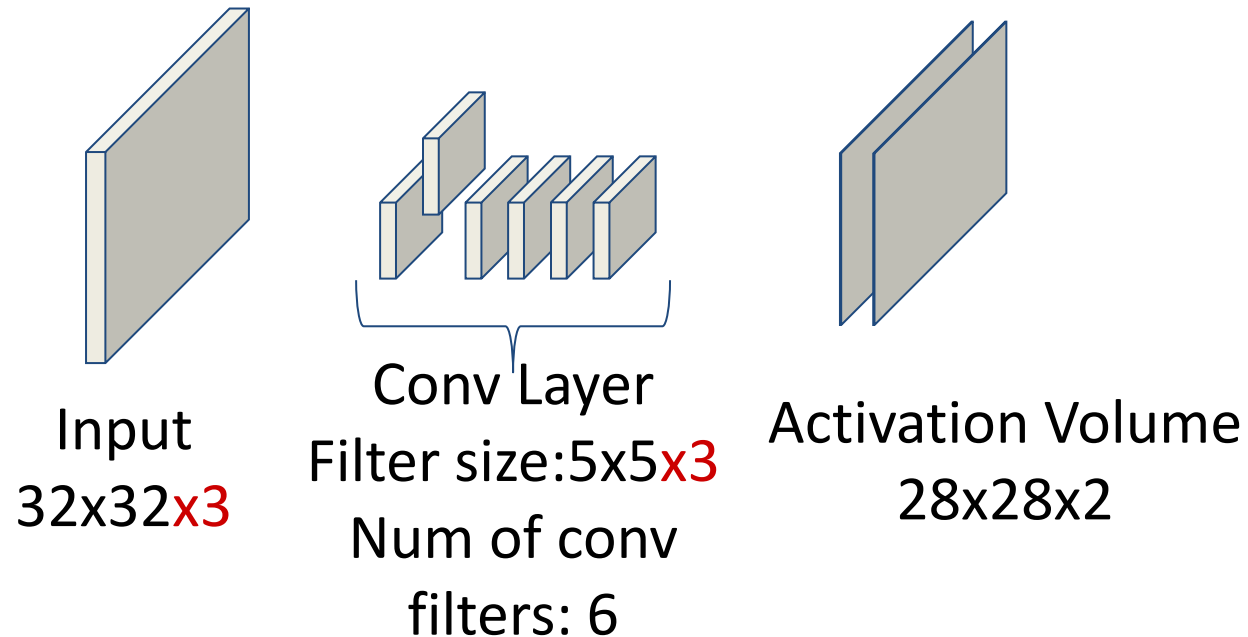
- **Cross-correlation** of g and h produces a function f
 - Almost the same as convolution, except that h does not get “flipped”

$$f(i, j) = (g \otimes h)(i, j) = \sum_{u=-k}^k \sum_{v=-k}^k g(u, v) \cdot h(i + u, j + v)$$

- Unlike cross-correlation, convolution is *commutative* and *associative*
- In CNN literature, flipping is often omitted, but the operation is still (incorrectly) called “convolution”
 - Can view this issue as learning a flipped convolution kernel h ; in this sense, imprecise use of terminology does not matter much
 - In this chapter, we will stick with CNN terminology

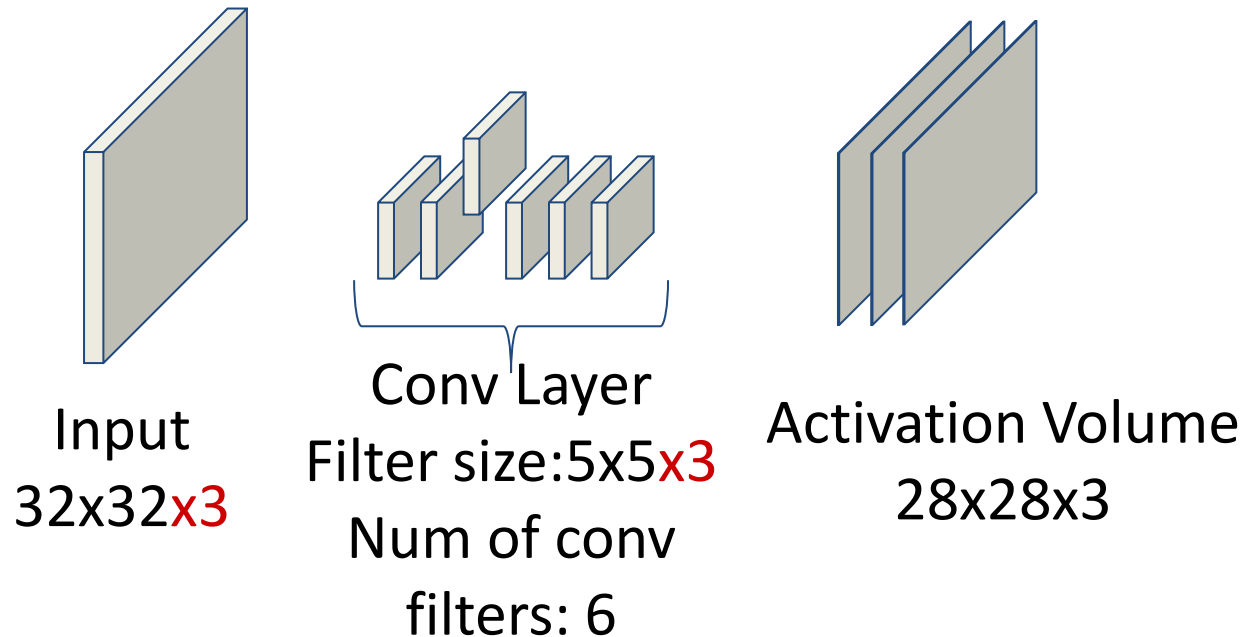
Convolution Layer

- Each kernel yields an activation map, forming an activation volume



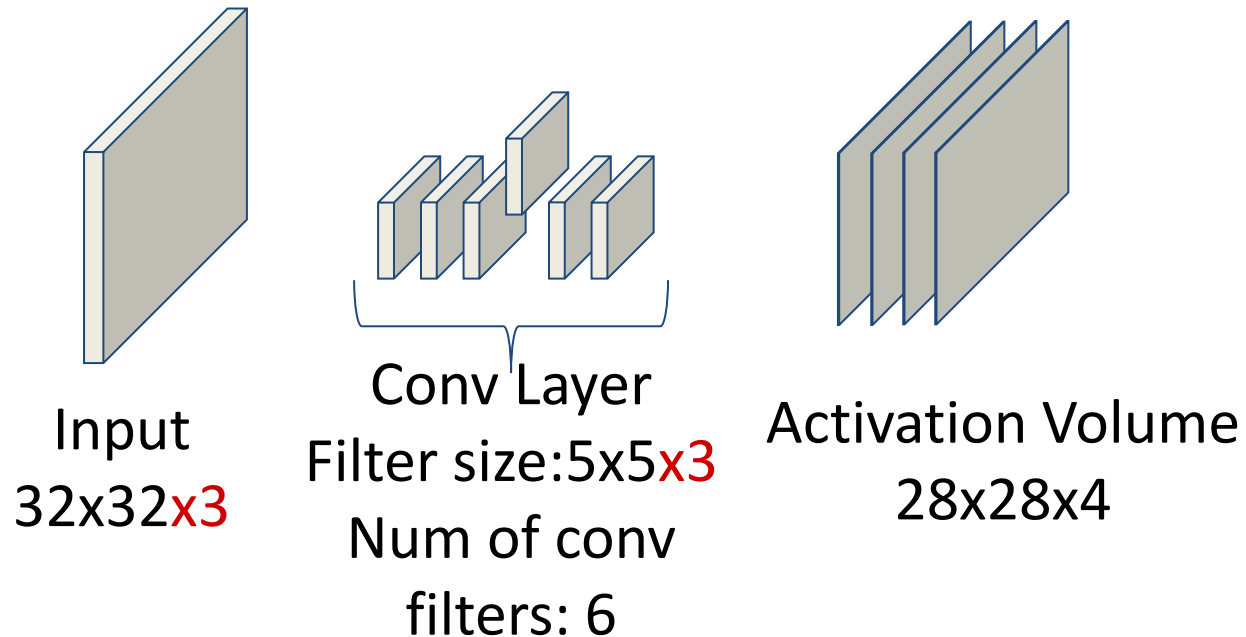
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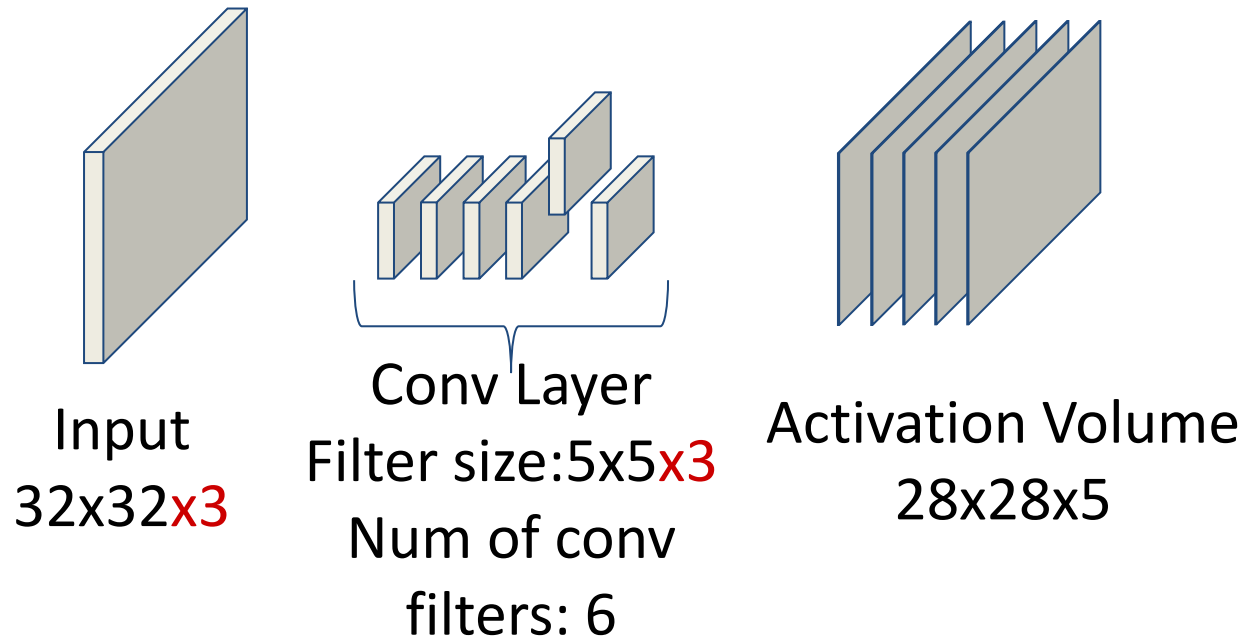
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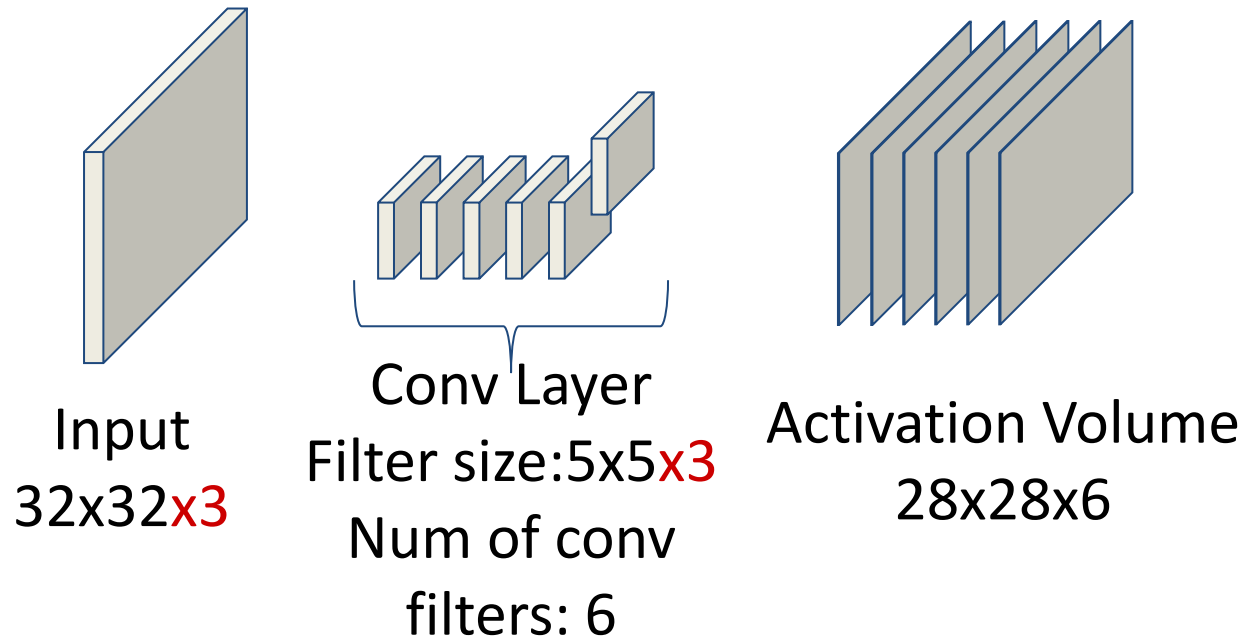
Convolution Layer

- Each kernel yields an activation map, forming an activation volume



Convolution Layer

- An activation function is applied point-wise to the resulting tensor
- It is then used as the input of the next layer
- Consequently, effort increases quadratically with the number of filters



Convolution without Padding

1	0	2	1	1	2	2
1	0	1	0	1	0	1
2	1	2	1	1	1	1
-1	0	1	2	1	2	2
1	2	0	-1	0	1	-1
6	1	7	-1	0	1	0
1	1	1	2	1	0	2

7x7

*

1	-1	0
-1	0	0
1	0	1

3x3

Quiz: What will be the size of the output? **5x5**

Convolution with Padding

0	0	0	0	0	0	0	0	0
0	1	0	2	1	1	2	2	0
0	1	0	1	0	1	0	1	0
0	2	1	2	1	1	1	1	0
0	-1	0	1	2	1	2	2	0
0	1	2	0	-1	0	1	-1	0
0	6	1	7	-1	0	1	0	0
0	1	1	1	2	1	0	2	0
0	0	0	0	0	0	0	0	0

*

1	-1	0
-1	0	0
1	0	1

3x3

Quiz: What will be the size of output
with padding=1? **7x7**

Transposed Convolution

- In **transposed convolution**, the input map elements multiply the “up convolution filter” and are added together in the output

5	-2
3	1

Input map

*T

1	0
-1	2

Up conv. filter

=

5	0	
-5	10	

Output map

- 2x2 up conv. filter

Transposed Convolution

- In **transposed convolution**, the input map elements multiply the “up convolution filter” and are added together in the output:

5	-2
3	1

Input map

*T

1	0
-1	2

Up conv. filter

=

5	-2	0
-5	12	-4

Output map

- 2x2 up conv. filter

Transposed Convolution

- In **transposed convolution**, the input map elements multiply the “up convolution filter” and are added together in the output:

5	-2
3	1

Input map

*T

1	0
-1	2

Up conv. filter

=

5	-2	0
-2	12	-4
-3	6	

Output map

- 2x2 up conv. filter

Transposed Convolution

- In **transposed convolution**, the input map elements multiply the “up convolution filter” and are added together in the output:

5	-2
3	1

Input map

*T

1	0
-1	2

Up conv. filter

=

5	-2	0
-2	13	-4
-3	5	2

Output map

- 2x2 up conv. filter

Transposed Convolution

- In **transposed convolution**, the input map elements multiply the “up convolution filter” and are added together in the output:

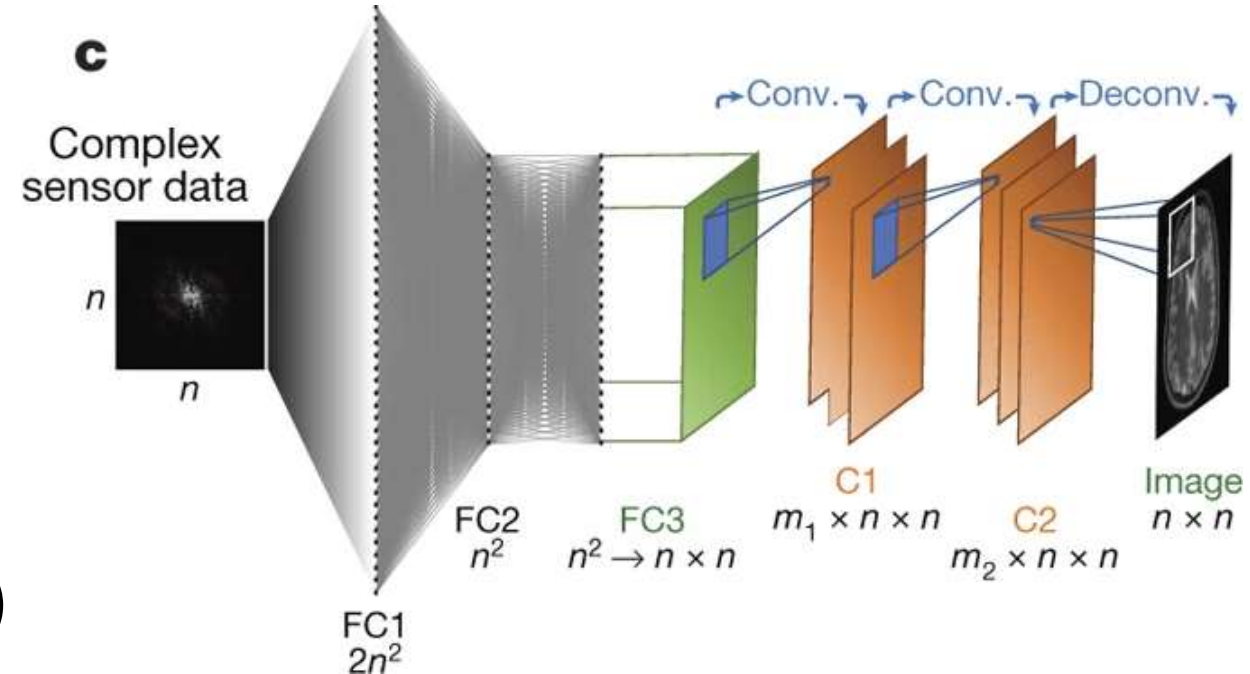
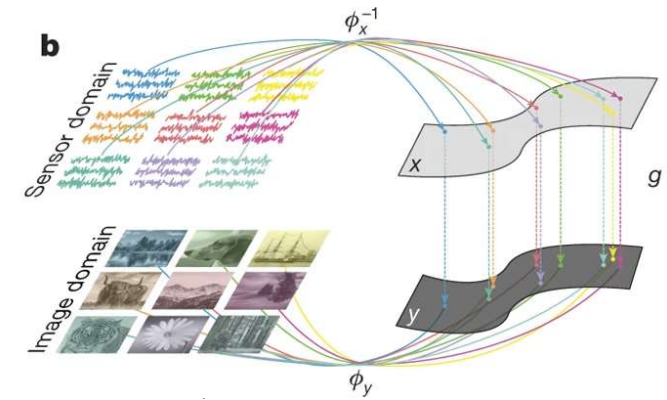
<table border="1"><tr><td>5</td><td>-2</td></tr><tr><td>3</td><td>1</td></tr></table>	5	-2	3	1	$\ast T$	<table border="1"><tr><td>1</td><td>0</td></tr><tr><td>-1</td><td>2</td></tr></table>	1	0	-1	2	$=$	<table border="1"><tr><td>5</td><td>-2</td><td>0</td></tr><tr><td>-2</td><td>13</td><td>-4</td></tr><tr><td>-3</td><td>5</td><td>2</td></tr></table>	5	-2	0	-2	13	-4	-3	5	2
5	-2																				
3	1																				
1	0																				
-1	2																				
5	-2	0																			
-2	13	-4																			
-3	5	2																			
Input map		Up conv. filter		Output map																	

- 2x2 up conv. filter

- Sometimes referred to as “deconvolution”
- Often used to generate images

AUTOMAP Architecture

- The AUTOMAP architecture [Zhu et al. 2018] aims to unify image reconstruction for different modalities (including MRI) using a NN that
 - Maps from the raw signal domain to the image manifold and from there to the image domain using fully connected layers with hyperbolic tangent
 - Decodes the initial representation to the pixel level using 5×5 convolutional layers with ReLU
 - Generates the final image using a 7×7 transposed convolution
 - Learns a sparse representation by imposing an l_1 regularizer on the final convolutional activations (C2) during training



Training AUTOMAP

- Training has been explored based on different datasets:
 - Luminance from 30,000 animal/plant/scene images from ImageNet
 - 50,000 slices from T1 weighted brain MRI from 131 subjects
 - Random Gaussian noise (for comparison)
- Corresponding measurements have been simulated according to various signal equations (one per network), including
 - Fourier Transform (with and without subsampling)
 - Fourier Transform with EPI-like misalignment
- Multiplicative noise is added to measurements during training to promote separation of signal and noise
- All images are downsampled to 128×128 pixels

Example Results: Simulated Noisy Measurements

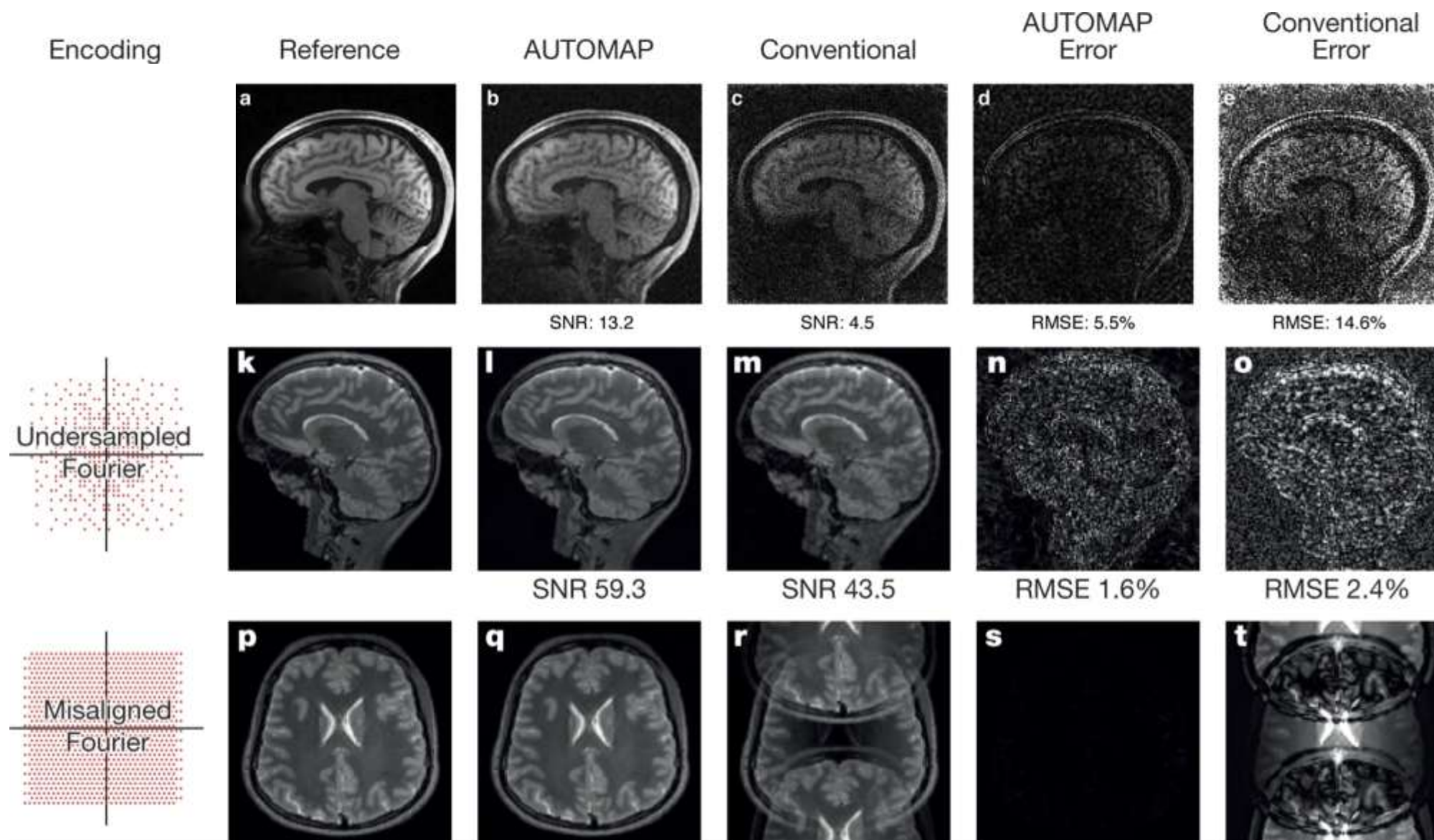
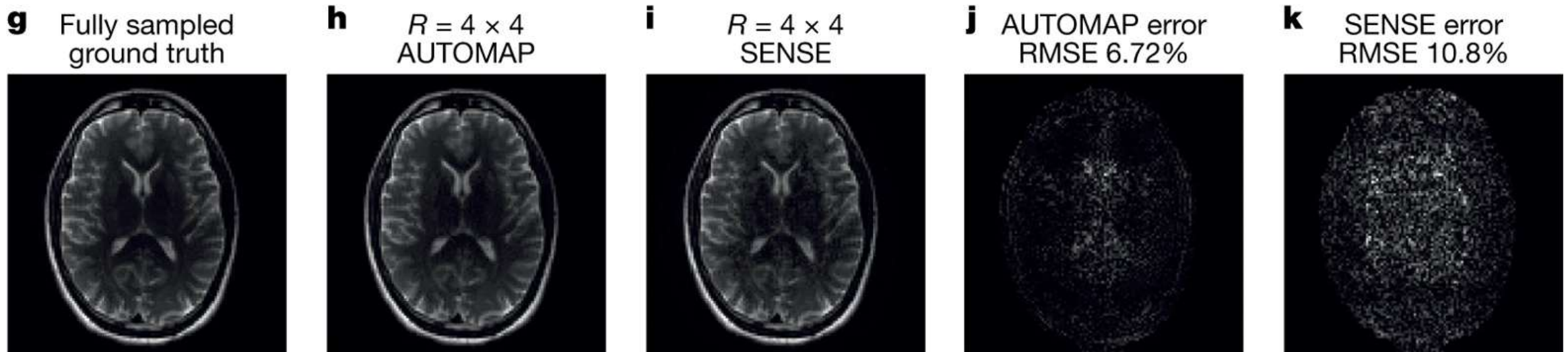


Figure: [Zhu et al. 2018]

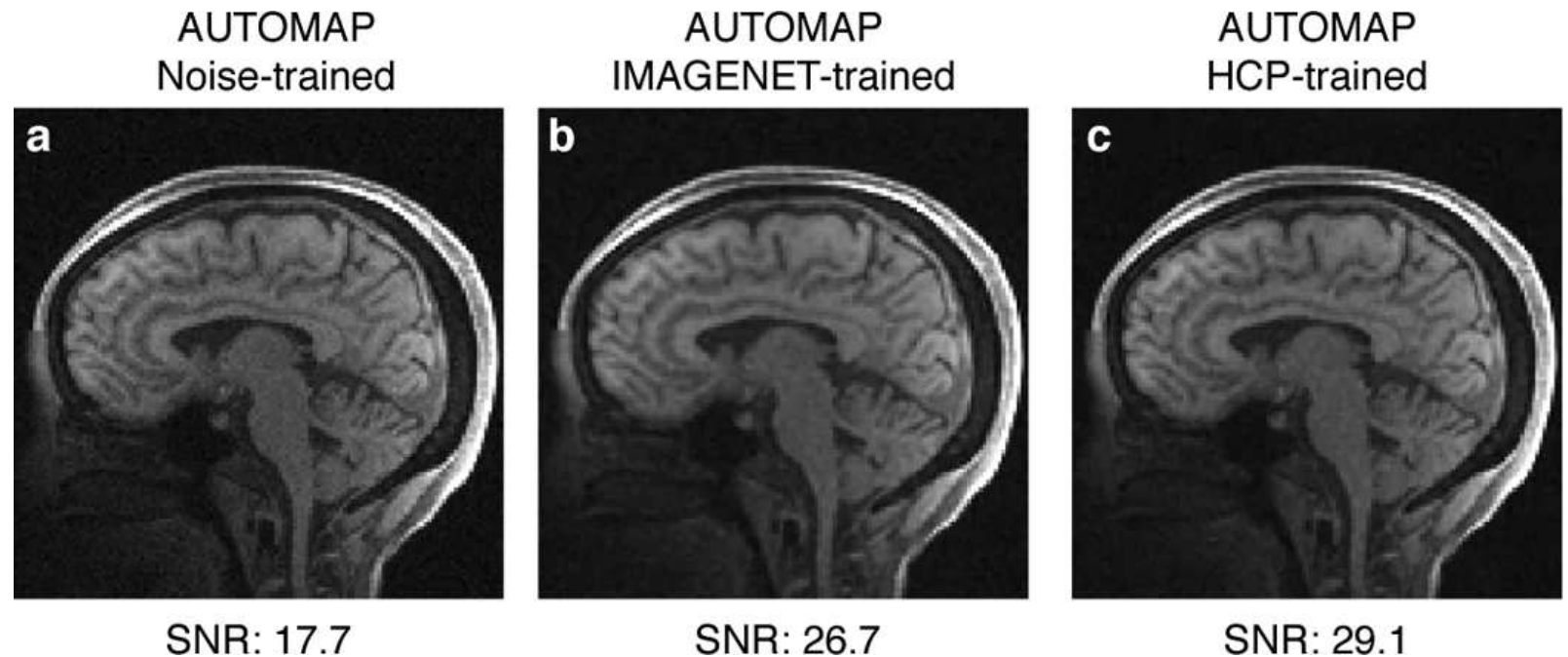
Example Results: Real Data

- *Example:* AUTOMAP reconstruction of subsampled MRI data more closely matches fully sampled reference image



Caveat: Bias from Training Data?

- *Concern:* If we train a NN on normal / healthy images, will it reconstruct (subtle) signs of pathology or will it rather eliminate them as if they were noise?
- *(Partial) reassurance:* The process still works when using non medical or even random images for training



Caveat: Robustness to Unforeseen Artifacts?

- *Concern:* NN based reconstruction handles artifacts that were seen during training very well, but it is difficult to predict how it will be affected by unseen artifacts
- *(Partial) reassurance:* Motion artifacts have a similar effect on AUTOMAP and conventional reconstruction

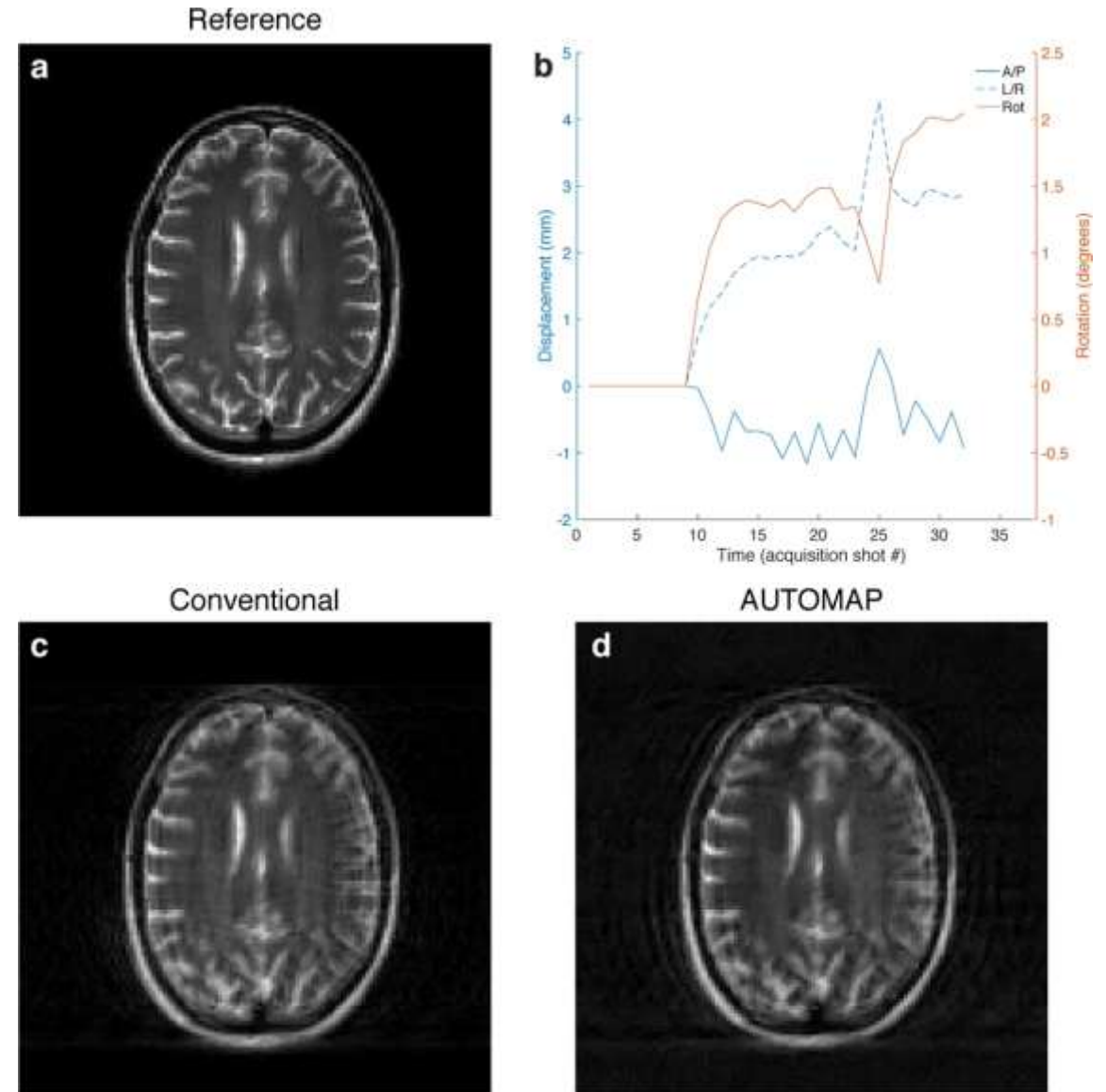


Image-to-Image Approaches

- **AUTOMAP** is very general, but has **limitations**:
 - Learning reconstruction with fully connected layers discards physical understanding and does not scale well to higher resolutions
 - Fast Fourier Transform works in $\mathcal{O}(n \log n)$, fully connected layers in $\mathcal{O}(n^2)$
 - No direct control of data fidelity
 - How compatible is the reconstructed image with the measurements?
- **Image-to-image** strategy [Wang et al. 2016]:
 1. Perform initial classical physics-based reconstruction, using compressed sensing if needed
 2. Train a lightweight CNN to improve image quality
 3. Constrain final physics-based reconstruction to be similar to the CNN prediction while remaining faithful to the measurements

Image-to-Image (Wang et al.): Details

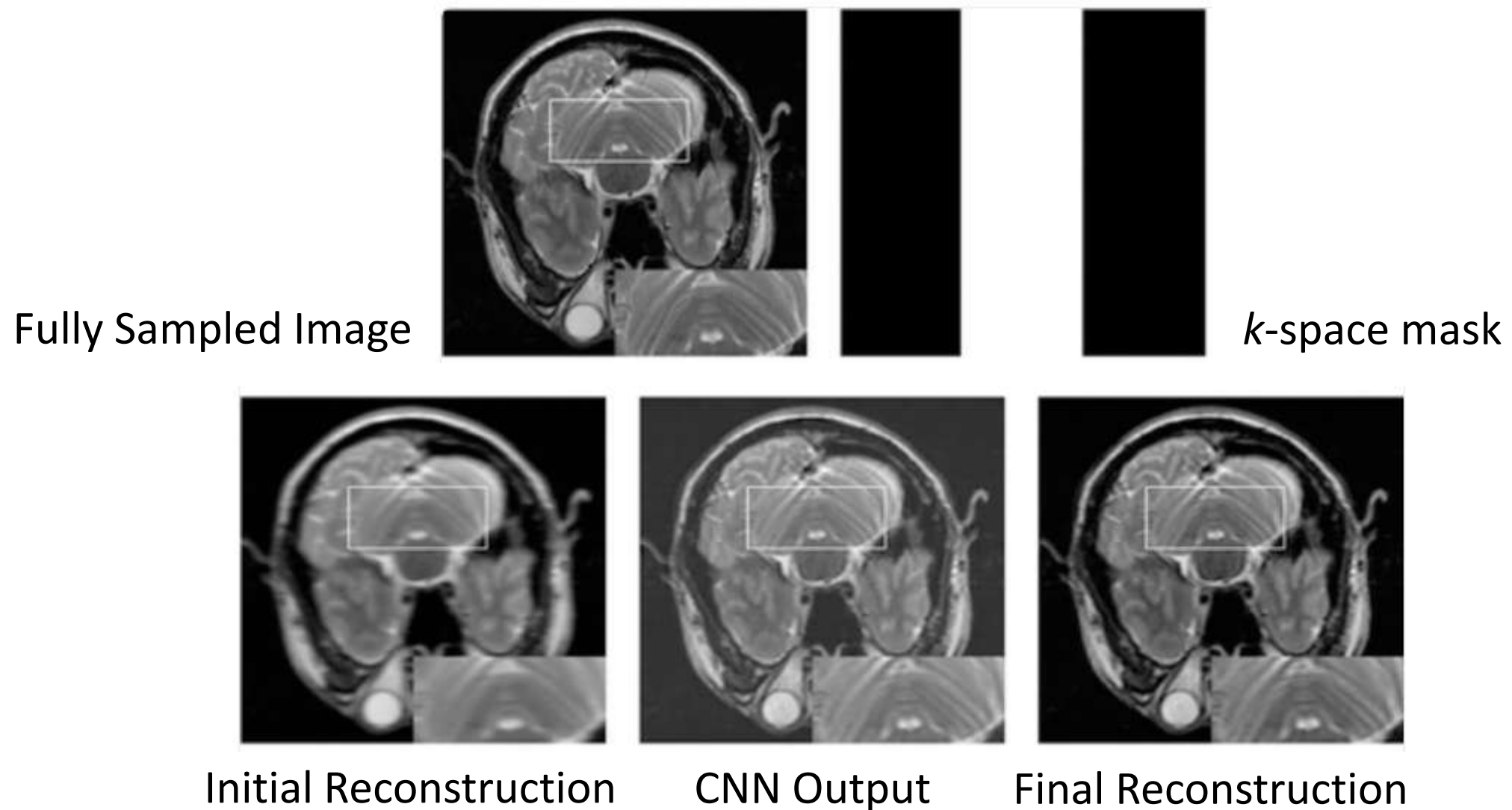
- CNN architecture in [Wang et al. 2016]:
 - Three convolutional layers ($9 \times 9, 5 \times 5, 5 \times 5$) with hidden channel dimensions 64 and 32 and ReLU activations
- Training and validation used
 - 500 MR slice images with different orientations and contrasts
 - Pairs of original images and reconstructions after subsampling
- Final constrained reconstruction u according to

$$\arg \min_u \left\| \underbrace{C(F^H f; \hat{\Theta})}_{\text{Initial reconstruction}} - u \right\|_2^2 + \lambda \underbrace{\|f - F_M u\|_2^2}_{\text{Subsampled Fourier Transform}}$$

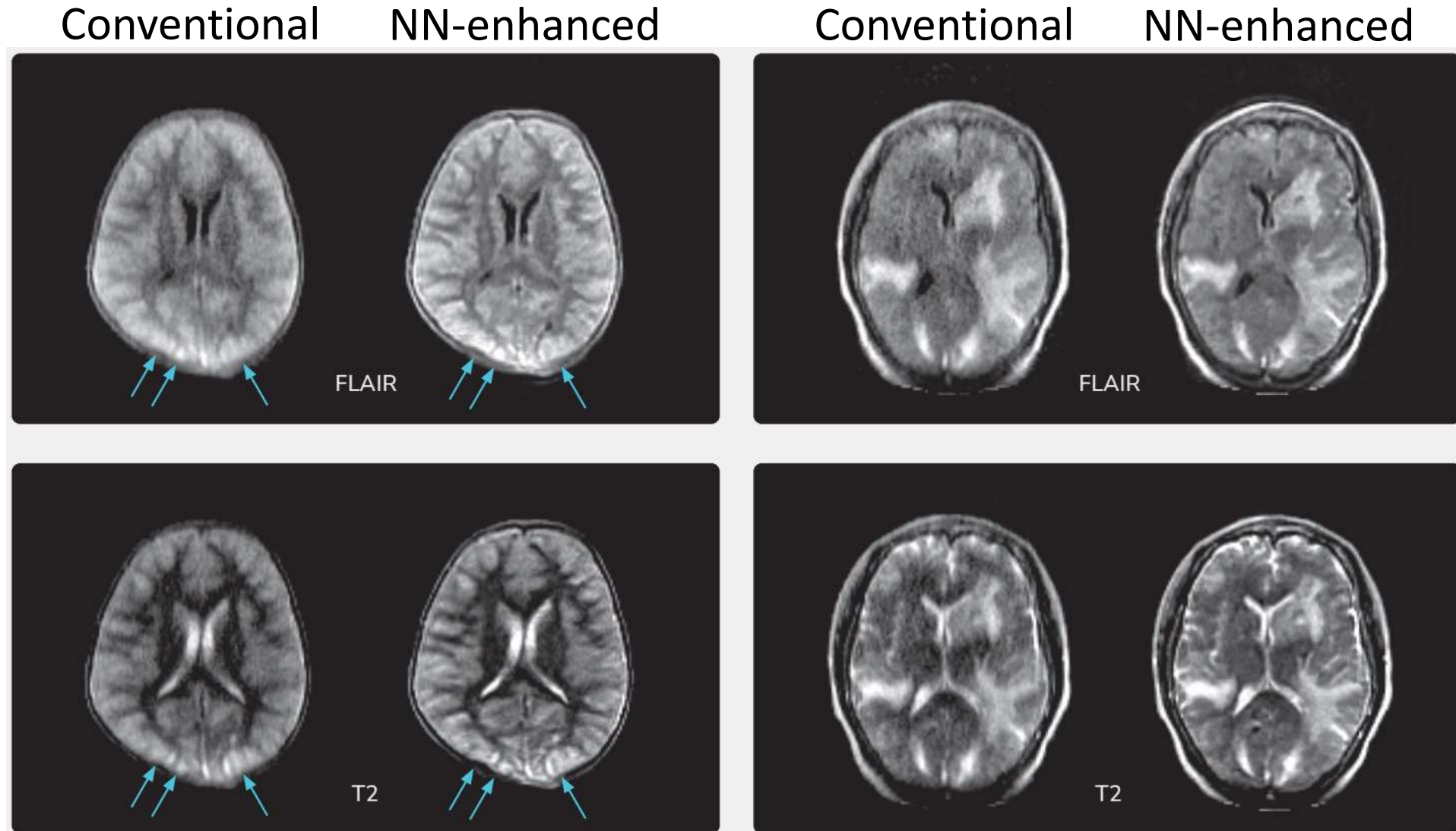
CNN with weights $\hat{\Theta}$ Measurements Data fidelity weight

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Image-to-Image (Wang et al.): Example Result



Example: Reconstruction in Ultra-Low Field MRI



5 year old with cortical cerebral malaria

62 year old with metastatic lesions

Summary: Neural Network based Reconstruction

- **Neural networks** can help maintain high quality in difficult circumstances (e.g., accelerated acquisition, ultra-low field)
 - Can learn a detailed and specific prior for (brain MR) images
 - Can learn to ignore noise and specific artifacts
- Introduced two specific approaches:
 - **AUTOMAP** learns reconstruction end-to-end
 - **Image-to-image** constrains classical reconstruction
- Based on **fundamental concepts**:
 - Artificial neurons and widely used activation functions
 - Feed-forward networks and fully connected layers
 - Convolution and transpose convolution layers

What to Take Away From This Lecture

- Basic principle of Nuclear Magnetic Resonance
 - Relevance of static vs. RF fields
 - Longitudinal vs. Transversal relaxation
- Basic echo sequences
 - Spin echo vs. Gradient echo
- MR image acquisition in k-space
 - Slice selection, frequency vs. phase encoding
- Different types of contrast
 - Impact of relaxation times (T_1 , T_2 , T_2^*) and inversion pulses
- Key ideas from compressed sensing and neural networks

References

- M. A. Flower (Ed): *Webb's Physics of Medical Imaging*, 2nd edition, CRC Press 2012
- *Magnets, Spins and Resonances*, Siemens Healthcare
- *Is Quantum Mechanics Necessary for Understanding Magnetic Resonance?*, L.G. Hanson, Conc. Magn. Res. A, Vol.32A(5), pp. 329-340, 2008
- Lustig et al., *Sparse MRI: The Application of Compressed Sensing for Rapid MR Imaging*, MRM 58:1182-1195, 2007
- Zhu et al., *Image reconstruction by domain-transform manifold learning*, Nature 555:487-492, 2018
- Wang et al., *Accelerated Magnetic Resonance Imaging via Deep Learning*, IEEE ISBI, pp. 514-517, 2016

Announcements

- Due to the large number of students, final exams will be in written form, dates will be announced as soon as I know them
- *Tentative* dates for the exercise classes are:
 - 2025: Mon Nov 3, Mon Nov 17, Mon Dec 1, Mon Dec 15
 - 2026: Thu Jan 15, Mon Jan 26
- Presenting your own solution strengthens your understanding and provides opportunity for feedback
 - Presenting at least once is mandatory

Exam-like Questions

1. What do the T_1 , T_2 , and T_2^* time constants in Magnetic Resonance Imaging have in common?
Also explain how each of them differs from the others.
2. Name the three mechanisms that are used to encode the three spatial dimensions in classical Magnetic Resonance Imaging. For each of them, briefly state at which point of the acquisition sequence it is employed.
3. State an important difference between a classical MR image acquisition and Echo Planar Imaging (EPI). Mention two practical consequences of that difference.